

Normalization by the Matching Number for Weighted Matching-Support Polynomials

Small ranks, unit and one-shore counterexamples, rank six, and eventual scaling

Research notes prepared for Vladimir Reshetnikov with OpenAI
(source 72: AI-assisted mathematical research)

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Abstract

For a finite bipartite graph G with positive vertex activities, the matching-support polynomial $p_G(t) = \sum_k a_k t^k$ counts each pair of endpoint sets of a k -edge matching once, weighted by the product of its activities. Its degree is the matching number $\nu(G)$. This report collects batch-72 research notes on the question of *preorder-root-polytopes* (Parts XI–XIII) whether p_G is ultra-log-concave of order $\nu(G)$. Part I (sources 74, 72, 76 and 75) proves it for arbitrary positive activities on both shores whenever some minimum vertex cover meets one shore in at most two vertices, through an explicit transversal-matroid lift; in particular for every graph with $\nu(G) \leq 5$. With the rank-six Hall counterexample of *preorder-root-polytopes*, the least matching number of a weighted failure is exactly six, and there are failures at every larger matching number. Equality in any one normalized inequality occurs only for balanced star forests; graphs with a vertex cover of two vertices on each shore have real-stable basis polynomials and real-rooted support polynomials; and the full Hall family is ultra-log-concave for all activities exactly when one core has at most two vertices. Part II (sources 71, 73, 55, 47 and 51) answers the unit-weight and one-shore questions of *preorder-root-polytopes* negatively: a connected all-unit graph with 6916 vertices and matching number 3450 fails, and so does a 125-vertex graph whose left activities are all one and whose right activities are one except for activity 10000 on its 42 right-core vertices; for one common activity on the right core of a full Hall graph the least failing matching number is exactly 38. Part III (sources 60, 10, 23, 13, 05, 69 and 63) treats matching number six with all left activities one: the second and the last of the five inequalities hold for every graph and every positive right field, all five hold for several core families, and an all-right-core conditional cubic can fail its own Newton inequalities. Nothing here is formalized in Lean or Rocq, and nothing is refereed.

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1 Guide to this report

1.1 Sources, status and plan

This report is a merge of research manuscripts delivered to ProveIt’s incoming-report intake on 1 October 2026 (batch 72, arrival commit 1512ef835, placement commit 8bb543f0f). The sources are numbered by their batch-72 manuscript numbers. Twenty-six archives and one manuscript delivered only inside archive 08 (called 08s here) are sources; a twenty-eighth archive (78) is an exact duplicate of the leaf-compression manuscript already printed as Part XIII of *preorder-root-polytopes* and is not a source. The full record, archive by archive, is Appendix A. At this stage of the write-up the report prints Parts I–III (sixteen sources); Part IV, built from the other eleven sources, is added in the following commit of the same write.

Every source is printed in full: statements, proofs, examples, remarks, limitations and verification notes. A result proved by several sources is printed once, under the source named in the Part’s introduction, and the other proofs follow as marked second routes. A result that *preorder-root-polytopes* already proves is cited by its number there and is not presented as new; where a source re-proves it, the source’s proof is either printed as a marked second route or replaced by a pointer, as recorded at the place and in Appendix A.

Status. All sources are AI-assisted research notes (their title pages read “Research note prepared for Vladimir Reshetnikov with OpenAI”, except source 72’s “AI-assisted mathematical research”). Their proofs are ordinary mathematical arguments supported by exact finite checks; they are not refereed, and nothing in them is formalized in Lean or Rocq. No source claims global priority. Text in square brackets marked *Merge note* was added in this merge; everything else is the sources’ text, with labels prefixed and section levels shifted (Section 1.5).

1.2 What *preorder-root-polytopes* already proves

The neighbouring report `SetTheory/Cardinals/docs/reports/enumerative-combinatorics/preorder-root-polytopes` ([8], cited here as *preorder-root-polytopes*) introduced the question and proves the following. They are cited by their printed numbers and labels in that report and are not reprinted here.

- Part V: Theorem 70.1 (`lor:thm:weightedULC`), the two-shore inequality: p_G is $\text{ULC}_{\min(|X|,|Y|)}$ for arbitrary positive activities (normalization by the smaller shore); its Research question 37 (`lor:q:first`) asks for normalization by the matching number.
- Part VI: the stability tools, Theorem 85.3 (`ade:mat:thm:stable-bridge`, the reciprocal bridge between the signed support polynomial and the basis polynomial), Theorem 91.2 (`ade:thm:expected`, independent-column expectations are real stable), Theorem 93.2 (`ade:mat:thm:vertex-stable`, stability under one-vertex sums), Theorem 94.1 (`ade:mat:thm:biclique`, complete bipartite blocks are stable) and Corollary 94.3 (`ade:mat:cor:block-family`).

- Part XI (Hall bottlenecks): the family $H_{s,n}$ and its exact coefficient formula, Theorem 175.3 (`hb:thm:coefficients`); the 30-vertex counterexample $H_{4,11}$ with weight 500 on both cores, Theorem 176.1 (`hb:thm:witness`); the complete classification of the core-three family $H_{3,n}$, Theorem 177.1 (`hb:thm:rank6phase`), whose last order-six margin fails at $n = 33$ (the rank-six counterexample $H_{3,33}$), with its asymptotic Corollary 177.2 (`hb:cor:asymptoticweight`); the endpoint threshold, Theorem 178.2 (`hb:thm:endpoint`); and the first-margin defect identity with its equality case, Theorem 180.2 (`hb:thm:firstidentity`) and Corollary 180.3 (`hb:cor:equality`): $(r-1)a_1^2 \geq 2ra_2$ for every positively weighted graph of matching number $r \geq 2$, with equality exactly for r disjoint stars of equal edge totals.
- Part XII (rank-three matching supports): the first rank inequality, Theorem 190.2 (`mr:thm:first`), and its unweighted equality case, Theorem 190.3 (`mr:thm:equality`); normalization by the matching number at unit weights for $\nu \leq 3$, Theorem 191.1 (`mr:thm:rankthree`); the explicit rank-three bimatroid obstruction, Theorem 192.1 (`mr:thm:matrix`); and a connected rank-four family, Theorem 196.1 (`mr:thm:family`).
- Part XIII (leaf compression): the weighted smaller-shore inequality, Lemma 205.1 (`lc:lem:ambient`); weighted leaf compression, Lemma 206.1 (`lc:lem:compression`); normalization by the matching number for every minimum cover with a singleton shore or on one shore, Theorem 206.2 (`lc:thm:singleton`); and weighted rank three, Corollary 206.3 (`lc:cor:rankthree`).

Archive 78 of this batch is byte-identical to the batch-71 archive printed as Part XIII; it is not printed again.

1.3 Notions and notation

The sources share one setting but not one vocabulary. The following conventions hold throughout; where a source uses another word or letter, it is listed here and kept in the source’s text.

Notion	Meaning in this report, and the readings to avoid
Support polynomial $p_G(t) = \sum_k a_k t^k$	$a_k = \sum u_I v_J$ over pairs $I \subseteq X, J \subseteq Y$ with $ I = J = k$ for which $G[I, J]$ has a perfect matching. Each pair of endpoint sets is counted <i>once</i> , not once per matching: this is not the matching (rook) polynomial. At unit activities it is the perfectly matchable set polynomial of Davis–Kohl [5]. Source 75 writes $P_H(t) = \sum p_k t^k$; <i>preorder-root-polytopes</i> writes $p_G(t; u, v)$ (Part XIII) and $A_H(t; u, v)$ (Part V).
Matching number $\nu(G)$, also r	The maximum size of a matching. The sources call it the <i>matching rank</i> ; <i>preorder-root-polytopes</i> calls it the <i>matching number</i> . Both words are used here with one meaning. For positive activities $\deg p_G = \nu(G)$ and every a_k with $0 \leq k \leq \nu(G)$ is positive.
ULC _D and the gaps $\Delta_k^{(D)}$	p is ULC _D if $a_k / \binom{D}{k}$ (padded by zeros) is log-concave without internal zeros; for support polynomials this is $\Delta_k^{(D)} = k(D-k)a_k^2 - (k+1)(D-k+1)a_{k-1}a_{k+1} \geq 0$ for $1 \leq k \leq D-1$ (display (1)). “Normalization by the matching number” means $D = \nu(G)$. The sources call Δ_k a <i>gap</i> , <i>margin</i> , <i>normalized gap</i> or <i>Newton gap</i> ; the <i>first</i> gap is $k = 1$, the <i>last</i> is $k = D-1$. <i>Strict</i> ULC _D means all $D-1$ gaps are positive.

Notion	Meaning in this report, and the readings to avoid
Weightings	<i>Two-shore</i> (or simply <i>weighted</i>): arbitrary positive activities u on X and v on Y (Part I). <i>Unit</i> : $u \equiv v \equiv 1$ (Research question 84 of <i>preorder-root-polytopes</i> ; Part II). <i>One-shore</i> : $u \equiv 1$ on the left shore, v arbitrary positive on the right shore (Research question 85 of <i>preorder-root-polytopes</i> ; Parts II–IV). <i>One common right-core activity</i> : activity w on the right core of a full Hall graph and 1 elsewhere, a special one-shore weighting (sources 73, 55, 47, 51).
Displayed cover $A \sqcup B$	A vertex cover with $A \subseteq X$ on the left and $B \subseteq Y$ on the right; $L = X \setminus A$ and $R = Y \setminus B$ are the exterior sets and there are no L – R edges; $G[A, B]$ is the <i>core</i> . Source 74 writes a $2 + q$ cover for $ A = 2$, $ B = q$; a cover is <i>genuine</i> when $\nu(G) = A + B $, that is, when it is minimum.
Full Hall graph $H_{a,b;n,m}$	Core A (a vertices) and B (b), exterior L (n) and R (m), edges $(A \times B) \cup (L \times B) \cup (A \times R)$ (sources 75, 73, 55, 47, 51; source 74 writes $H_{a,b;N,M}$, source 76 $H_{2,s;n,m}$). <i>preorder-root-polytopes</i> ’s $H_{s,n}$ is $H_{s,s;n,n}$ with weight w on <i>both</i> cores. The weighting differs by source: both cores (74, 75, <i>preorder-root-polytopes</i>), the right core only (73, 55, 47, 51).
Scalings	Two different scalings occur. Source 74 (Section 6) multiplies the activities of a <i>whole</i> minimum cover, on both shores, by a common factor; then failures persist at every matching number ≥ 6 . The sources of Part IV multiply only the <i>right</i> part of a fixed minimum cover, with unit left activities; there no failure occurs through matching number eight. These statements do not conflict.
Real stability, real-rootedness, Lorentzian	A polynomial is real stable if it has no zero with all variables in the upper half-plane $\mathbb{H}$; a real-stable univariate polynomial is real-rooted. A homogeneous bivariate polynomial is Lorentzian exactly when its coefficients are ULC of its degree. Real-rootedness of p_G implies $\text{ULC}_{\deg p_G}$ by Newton’s inequalities; the converse fails (source 75’s rank-five example, Section 16).
Conditioning	<i>Forcing</i> prescribed vertices into the support (sources 60, 21, 23, 63): the conditional polynomial is the coefficient of the product of the forced activities. It is not probabilistic conditioning.
Shifted and unshifted Newton inequalities	For a conditional sector $t^j C(t)$ of a degree-six polynomial, the <i>shifted</i> inequalities are the degree-six gaps restricted to the sector; the <i>unshifted</i> ones are the degree-three Newton inequalities of the cubic C itself (constant three). Source 23’s 25-vertex graph fails two unshifted inequalities while its shifted ones, and full ULC_6 , hold; it does not contradict source 21, which bounds shifted ones with smaller constants.

Letters local to a source	Letters are local to the source (often to the section) that defines them. In particular A, B are cover parts in source 74 but numerical gap factors in Section 3 of source 75 and <i>preorder-root-polytopes</i> 's core in Part XI; a, b, c are right-class activity sums in Section 5 but core sizes in Section 6; q is $ B $ in source 74 and $2^{-1/2}$ in source 76; $Q = ab + ac + bc + d$ in source 74; H is $G[L, B]$ in source 74, a full Hall graph in sources 75 and 76, and $H_{s,n}$ in <i>preorder-root-polytopes</i> ; h is a residual sector in source 74 and a population offset in source 75.
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Renamed symbols. No printed symbol of a source was renamed. Source 69 defines the macro `\rank` as ν ; it is printed as ν here, which is how source 69 renders it, so that the macro can keep source 60's meaning (rank).

1.4 Research questions of *preorder-root-polytopes* answered here

Question of <i>preorder-root-polytopes</i>	Status after this report
RQ 86 <code>hb:q:rank</code> (first possible weighted rank)	Answered: every weighted graph with $\nu \leq 5$ is ULC_ν (Corollary 2.2); Part XI's $H_{3,33}$ fails at $\nu = 6$. The least weighted failure rank is exactly six.
RQ 87 <code>hb:q:order</code> (a universal rank-only order)	In part: $\mathcal{D}(4) = 4$ and $\mathcal{D}(5) = 5$ (Corollary 2.2); the question beyond rank five is open.
RQ 93 <code>mr:q:rankfour</code> (finish rank four)	Answered affirmatively, for unit and arbitrary two-shore weights (Corollary 2.2; second route by real stability, Corollary 15.2).
RQ 89 <code>hb:q:asymmetry</code> (asymmetric bottlenecks)	In large part: full Hall graphs are ULC for all activities exactly when one core has at most two vertices (Theorem 16.1); explicit failures for all unequal cores $a, b \geq 3$ (Sections 6 and 16.5). Unequal core activities: Part II (one core at activity one).
RQ 90 <code>hb:q:defects</code> (higher-index defect identities)	Bears on it: under the two-vertex-shore cover hypothesis, equality in any one gap holds only for balanced star forests (Theorem 14.1); no higher defect identity is given.
RQ 84 <code>hb:q:unit</code> (unit weights)	Answered negatively (source 71, Theorem 17.1): a connected all-unit graph with $\nu = 3450$ is not ULC_ν ; infinitely many deficiency-one examples. The least matching number of a unit counterexample is open.
RQ 85 <code>hb:q:oneshore</code> (one-shore fields)	Answered negatively (source 73: $\nu = 50$, 125 vertices; $\nu = 38$; source 55: for one common right-core activity on full Hall graphs the least failing matching number is exactly 38). Positive pieces: $\nu \leq 5$ (Part I); at $\nu = 6$ the first, second and last gaps for every graph (Part XI of <i>preorder-root-polytopes</i> , sources 05 and 13) and all gaps for several families (Part III).
RQ 88 <code>hb:q:phase</code> (Hall phase diagrams, $s \geq 4$)	Not answered (it concerns weight on both cores). Its one-shore analogue is answered by sources 47 and 51: the last gap controls all gaps through $\nu = 38$ but not at $\nu = 49$.

Question of <i>preorder-root-polytopes</i>	Status after this report
RQ 94 mr:q:allrank (a structural all-rank proof)	Answered negatively for unit weights (source 71): no all-rank proof can exist.
RQ 101 mr:q:bimatroid (the bimatroid boundary)	In part: rank-two exterior blocks with ordinary columns are Lorentzian (source 60, Theorem 22.1).
RQ 37 lor:q:first (Part V)	Weighted: holds through $\nu = 5$, fails from $\nu = 6$ (Part I). Unit and one-shore: fail (Part II).

1.5 Labels and editorial marks

Every label carries the prefix **mrn:**. Labels written in this merge are **mrn:sec:...**, **mrn:src:NN** (the start of source NN) and **mrn:w:tab:...**; a delivered label of source NN carries a sub-prefix naming its Part (**w** for Part I, **u** for Part II, **s** for Part III, **e** for Part IV) and the source, for example **mrn:w:rf:thm:main** for source 74's **thm:main**. The table of sub-prefixes is in Appendix A. Source 74, the base, keeps its sections as sections of Part I, and its appendices are ordinary sections there; every other source is one section, whose sections are subsections. Section and theorem numbers therefore differ from the delivered PDFs, which are not shipped.

Part I

Weighted activities on both shores: normalization through matching number five

Contents of Part I. Part I prints four sources. Source 74, *Weighted Rank Five and Covers with a Two Vertex Shore*, is the base of the whole report; its sections are Sections 2–13, and its appendices are printed as ordinary sections (Sections 8–13). Source 72 (Section 14) classifies equality under the same cover hypothesis; source 76 (Section 15) proves real stability for covers with two vertices on each shore; source 75 (Section 16) classifies the full Hall family.

What is new here and what is cited. New relative to *preorder-root-polytopes*: the transversal lift of Theorem 2.1 and its consequences (normalization by the matching number for every minimum cover meeting one shore in at most two vertices, hence for $\nu \leq 5$, and the cover obstruction Corollary 5.1); the forced-sector and scaling refinements; the asymmetric negative shapes; the equality classification at every gap (source 72); real stability for $2 + 2$ covers, the $2 \times s$ stability threshold and the 17-vertex example (source 76); the full Hall classification, its strictness and real-rootedness refinements and the nonreal rank-five example (source 75). Cited, not new: the smaller-shore inequality and leaf compression (Parts V and XIII of *preorder-root-polytopes*; source 74’s Lemmas 3.1 and 3.2 and the corresponding steps of sources 72 and 76 are pointers); the first-gap inequality with its equality case (Part XI’s Theorem 180.2 and Corollary 180.3, Part XII’s Theorem 190.2; the Motzkin–Straus proofs of sources 74 and 72 are printed as second routes); the rank-six and rank-eight witnesses $H_{3,33}$ and $H_{4,11}$ (Part XI, restated by source 74); and the stability tools of source 76 (Part VI).

Same theorem in several sources. Source 74’s Corollary 2.2 contains source 76’s Corollary 15.2 ($\nu \leq 4$) and Part XIII’s Corollary 206.3 ($\nu \leq 3$); source 76’s real-stability route is printed as a second route. The positive half of source 75’s Theorem 16.1 is the full Hall case of source 74’s Theorem 2.1; source 75’s comparison-polynomial proof is printed as a second route. For equal cores and populations the negative half of Theorem 16.1 is Part XI’s endpoint theorem.

Non-claims of Part I. Source 74’s theorem asserts neither real-rootedness nor stability of the support polynomial (remark after Corollary 5.1); at higher rank it does not address the unrestricted unit-weight and one-shore problems (Section 13); finite computations are regression checks, not proofs (sources 72 and 74); source 76 does not assert real-rootedness for every graph of matching number four; source 75’s classification concerns the full Hall family only and its negative assertion says nothing about unit or one-shore weights.

Abstract of source 74, as delivered. Every bipartite matching-support polynomial with a minimum vertex cover meeting one shore in at most two vertices is ultra-log-concave after normalization by its matching rank, for arbitrary positive activities on both shores. The main construction replaces a displayed $2 + q$ cover by a rank- $(q + 2)$ represented matroid whose Lorentzian basis polynomial specializes exactly to the homogenized support polynomial. Consequently weighted rank normalization holds through rank five, and known rank-six Hall obstructions show that the first weighted failure rank is exactly six. We also prove a sharp all-core boundary inequality with constant three, give independent Lorentzian transfer proofs

for star cores, and derive quantitative eventual strictness under two natural activity scalings. Full proofs, finite exact coefficient certificates, sum-of-squares identities and independent verification scripts are included.

Scope and status. The rank-five weighted question posed in the inspected ProveIt draft is resolved here. At higher rank the theorem covers every minimum cover with a shore of size at most two; the unrestricted unit-weight and one-shore-weighted problems at higher rank are not addressed here. These are proofs in ordinary mathematics, not Lean-certified or independently refereed results. The work extends the inspected rank manuscript and the separate rank-four and full-Hall classification reports [10, 14, 13]; global priority is not claimed. [*Merge note:* The “inspected ProveIt draft” and “rank manuscript” [10] is the manuscript printed as Part XII of *preorder-root-polytopes* (pinned at [1b3960d8a](#)); the companions [14] and [13] are sources 76 and 75, printed in Sections 15 and 16; [11] and [9] are Parts XIII and XI of *preorder-root-polytopes*. The unit-weight and one-shore problems that this paragraph leaves aside are the subject of Parts II–IV (batch-72 sources 71, 73 and later).]

2 Definitions and main conclusions

For a finite bipartite graph $G = (X, Y; E)$ and positive vertex activities u_x, v_y , write

$$p_G(t) = \sum_{k=0}^r a_k t^k, \quad a_k = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|=k \\ G[I,J] \text{ perfectly matchable}}} u_I v_J, \quad r = \nu(G).$$

Each endpoint pair is counted once. In particular $p_{K_{2,2}}(t) = 1 + 4t + t^2$ at unit activities. The polynomial is ULC_D , for $D \geq \deg p_G$, if its zero-padded sequence $a_k / \binom{D}{k}$ is log-concave and has no internal zeros. Equivalently, for these support polynomials,

$$\Delta_k^{(D)} := k(D-k)a_k^2 - (k+1)(D-k+1)a_{k-1}a_{k+1} \geq 0. \quad (1)$$

Strict ULC_D means all $D-1$ displayed gaps are positive.

A displayed $2+q$ cover is $A \sqcup B$, where $A = \{A_1, A_2\} \subseteq X$ and $|B| = q$, $B \subseteq Y$. Put $L = X \setminus A$ and $R = Y \setminus B$. There are no L – R edges. The induced graph $G[A, B]$ is called the core. We call the displayed cover *genuine* when $\nu(G) = q+2$.

Theorem 2.1 (A two-vertex cover shore). *If G has a displayed $2+q$ cover, fix positive activities v on its right shore. The homogeneous left marginal*

$$\sum_{\substack{I \subseteq X \\ |I| \leq q+2}} \left(\sum_{\substack{J \subseteq Y \\ (I,J) \text{ feasible}}} v_J \right) z_I s^{q+2-|I|}$$

is Lorentzian. In particular, the degree- $(q+2)$ homogenization of its weighted matching-support polynomial is Lorentzian. Thus p_G is ULC_{q+2} for every positive activity assignment. If the displayed cover is minimum, the normalization is by the actual matching rank.

Corollary 2.2 (Weighted rank five and the first failure rank). *Every bipartite matching-support polynomial of matching rank $r \leq 5$ is ULC_r for arbitrary positive activities on both shores. The smallest rank admitting a weighted counterexample is exactly six. At every rank $r \geq 6$ there is a weighted counterexample.*

The positive conclusion follows from Theorem 2.1, the pure and singleton-side cover cases proved below, and König’s theorem. The negative construction is given in Section 6. For nonnegative activities, delete zero-activity vertices first; the same conclusion holds with r interpreted as the surviving degree. [Merge note: Corollary 2.2 contains Part XIII’s Corollary 206.3 of *preorder-root-polytopes* ($\nu \leq 3$, `lc:cor:rankthree`) and source 76’s Corollary 15.2 ($\nu \leq 4$), which is printed in Section 15 as a second route by real stability. The rank-six counterexample it relies on is Part XI’s $H_{3,33}$ (Theorem 177.1 of *preorder-root-polytopes*, restated in Section 6). It answers Research questions 86 and 93 of *preorder-root-polytopes* and gives $\mathcal{D}(4) = 4$, $\mathcal{D}(5) = 5$ in its Research question 87 (Section 1.4).]

Theorem 2.3 (Forced-sector and strict-scaling refinements). *For any displayed $2 + q$ cover with $q \geq 1$, the coefficient of the product of the two A activities in p_G is $t^2 C(t)$, where C is ULC_q . For a genuine cover, common scaling of only the two A activities makes p_G strictly ULC_{q+2} for all sufficiently large scale factors.*

For a rank-five $2 + 3$ cover whose core has matching rank two, common scaling of all five cover activities also makes all four gaps strictly positive eventually. An explicit coefficient bound gives a sufficient threshold for both scalings.

Every fixed minimum-cover scaling at rank at most five is therefore ULC at *every* positive scale factor. At each rank at least six, some fixed graph and minimum cover fail for every sufficiently large common cover scale, so the universal scaling threshold is sharp as well. [Merge note: Here the whole minimum cover is scaled, on both shores. Sources 08s, 08, 03, 02 and 01 (Part IV) scale only the right part of a minimum cover, with unit left activities, and there no failure occurs below matching number nine. See the entry “Scalings” in Section 1.3.]

3 Lorentzian input and singleton covers

We use the following published facts from Brändén–Huh [1]: matroid basis polynomials are Lorentzian (Theorem 3.10); partial derivatives, nonnegative linear substitutions, products and coefficientwise limits preserve this class (Corollary 2.11, Theorem 2.10, Corollary 2.32, and closedness); homogeneous stability preservers that preserve nonnegative coefficients preserve Lorentzian polynomials (Theorem 3.4); and multiaffine truncation and polarization preserve the class (Corollary 3.5 and Proposition 3.1). A homogeneous bivariate polynomial of degree D is Lorentzian exactly when its coefficients form a ULC_D sequence (Example 2.26). The zero polynomial is included by closure.

Lemma 3.1 (Weighted smaller-shore inequality). *Every bipartite support polynomial is $\text{ULC}_{\min(|X|, |Y|)}$.*

Proof (pointer). This is Theorem 70.1 of *preorder-root-polytopes* (Part V, `lor:thm:weighted ULC`), restated as Lemma 205.1 of its Part XIII (`lc:lem:ambient`). [Merge note: Source 74 proves it by the same specialization of the basis polynomial of the transversal matroid on the private copies $\bar{X} \sqcup Y$, at s/u_x and tv_y , followed by the bivariate Lorentzian characterization and transposition, and calls it the standard bimatroid specialization [7]. That proof is Part XIII’s proof of Lemma 205.1 and is not reprinted.] $\square$

Lemma 3.2 (Singleton-side minimum covers). *If a rank- r graph has a minimum cover entirely on one shore or split as $1 + (r - 1)$, its support polynomial is ULC_r for all positive activities.*

Proof (pointer). This is Theorem 206.2 of *preorder-root-polytopes* (Part XIII, `lc:thm:singleton`), proved there by weighted leaf compression, Lemma 206.1 (`lc:lem:compression`). [Merge note: Source 74’s proof is the same argument (delete isolated vertices for a pure cover;

for a cover $\{x_0\} \sqcup B$ merge the leaves at x_0 into one leaf carrying their total activity, which leaves p_G unchanged, and apply Lemma 3.1); source 74 credits it as “the exact leaf-compression argument of” [11], which is Part XIII. It is not reprinted.] $\square$

4 The exact two-row formula

Let the two A activities be x, y , and let $H = G[L, B]$. For a virtual left vertex with neighborhood $S \subseteq B$ and activity one, define g_S by

$$p_H = f, \quad p_{H+S} = f + tg_S.$$

If S_1, S_2 are the actual core-row neighborhoods, define h by

$$p_{H+A} = f + t(xg_1 + yg_2) + xyt^2h, \quad g_i = g_{S_i}, \quad g_U = g_{S_1 \cup S_2}.$$

The feasible support set for g_U is the union of those for g_1, g_2 : in a matching the virtual row uses one edge, and that edge belongs to at least one of the original neighborhoods.

For R , let a, b, c be the sums of activities in classes with neighborhoods $\{A_1\}$, $\{A_2\}$ and A , respectively. If the individual common-class activities are $r_1, \dots, r_m$, put

$$d = \sum_{i < j} r_i r_j, \quad Q = ab + ac + bc + d, \quad 0 \leq d \leq c^2/2.$$

Then the exact full polynomial is

$$\begin{aligned} p_G = f[1 + (x(a+c) + y(b+c))t + xyQt^2] + t(xg_1 + yg_2) \\ + xyt^2(h + ag_2 + bg_1 + cg_U). \end{aligned} \tag{2}$$

To prove this, condition on the selected A vertices and then on the number of selected R vertices. With both A selected, two selected R contribute Qf ; one singleton-class vertex forces the corresponding A and gives ag_2 or bg_1 ; one common-class vertex leaves the union-neighborhood alternative and gives cg_U ; no selected R gives h . These cases have disjoint endpoint sets. In particular the common term is cg_U , not $c(g_1 + g_2)$.

5 The full rank compression matroid lift

We prove Theorem 2.1. First suppose $Q > 0$. The same construction includes $q = 0$, with no original B slots. Form a transversal matroid matched into the $q + 2$ slots

$$B \sqcup \{e_1, e_2\}.$$

Its elements and allowed slots are as follows:

- a private element $\bar{b}$ for each $b \in B$, allowed only in slot b ;
- an element for each $l \in L$, allowed in its original neighborhood in B ;
- two augmented core elements $\hat{A}_i$, allowed in $S_i \cup \{e_i\}$;
- two plane elements P_1, P_2 , allowed only in e_1, e_2 , respectively;
- one common element Z_j for each common right-exterior vertex, allowed in both e_1, e_2 .

The rank is $q + 2$, since all private B elements together with P_1, P_2 form a basis. Every basis is counted once, regardless of how many injections into the slots it admits. We call P_1, P_2, Z_j the plane elements, as they use only the two additional slots.

Give the private B variables the values s/v_b , and the exterior-left variables tu_l . Keep core variables X, Y , and specialize the plane variables to

$$P_1 : \frac{bs}{Q}, \quad P_2 : \frac{as}{Q}, \quad Z_j : \frac{r_js}{Q}. \quad (3)$$

Multiply this basis specialization by $Q \prod_{b \in B} v_b$. It is Lorentzian by the matroid and nonnegative-substitution theorems.

We now compute it exactly. Put

$$f_h = s^q f(t/s), \quad G_i = s^{q-1} g_i(t/s), \quad G_U = s^{q-1} g_U(t/s), \quad H_h = s^{q-2} h(t/s),$$

where $G_i = G_U = 0$ is defined directly for $q = 0$, and $H_h = 0$ is defined directly for $q < 2$; the displayed homogenizing formulas are used only for nonnegative exponents. Classify a basis by the number k of core elements and ℓ of plane elements. Other elements use only the q original slots, so $k + \ell \geq 2$ and $0 \leq k, \ell \leq 2$. There are six cases.

1. $(k, \ell) = (0, 2)$. Any pair of distinct plane elements can fill the two extra slots, and the other q elements fill the B slots. The sum of plane-pair activities is

$$\frac{s^2}{Q^2} \left(ab + (a + b)c + \sum_{i < j} r_i r_j \right) = \frac{s^2}{Q}.$$

After the exterior factor Q , the contribution is $s^2 f_h$.

2. $(k, \ell) = (1, 1)$. With core $\hat{A}_1$, the plane element must be able to occupy e_2 , hence is P_2 or a Z_j . Its total activity is $(a + c)s/Q$. The remaining q elements fill the B slots, giving $Xs(a + c)f_h$. With $\hat{A}_2$, the contribution is $Ys(b + c)f_h$.
3. $(k, \ell) = (1, 2)$. The plane pair fills both extra slots, forcing the core element into B . The remaining support uses row S_i and $q - 1$ ordinary elements, with profile G_i . The contribution is $Xs^2 G_1 + Ys^2 G_2$.
4. $(k, \ell) = (2, 0)$. The two core elements occupy their dedicated extra slots. The other q elements fill B , contributing $QXY f_h$.
5. $(k, \ell) = (2, 1)$. The element P_1 occupies e_1 , forcing $\hat{A}_2$ into e_2 and $\hat{A}_1$ into B ; hence its profile is G_1 and its activity coefficient is b . Similarly P_2 gives G_2 with coefficient a . A common Z_j can occupy either extra slot, so feasibility holds exactly when either core row completes the remaining B -slot support. This is the union profile G_U , counted once even if both alternatives work. Thus the contribution is $XYs(bG_1 + aG_2 + cG_U)$.
6. $(k, \ell) = (2, 2)$. The two plane elements fill the extra slots, forcing both core elements into B together with $q - 2$ other elements. The contribution is $XYs^2 H_h$; this case is absent for $q < 2$.

Consequently the constructed Lorentzian polynomial is

$$\begin{aligned} \mathcal{L}(s, t, X, Y) &= s^2 f_h + sX[(a + c)f_h + sG_1] + sY[(b + c)f_h + sG_2] \\ &\quad + XY[Qf_h + s(bG_1 + aG_2 + cG_U) + s^2 H_h]. \end{aligned} \quad (4)$$

Set $X = xt$ and $Y = yt$. Equation (2) identifies (4) with $s^{q+2}p_G(t/s)$, proving the bivariate assertion for $Q > 0$. More generally, keep each exterior-left element variable as its own z_l rather than tu_l , and set $X = z_{A_1}$, $Y = z_{A_2}$. The same six basis cases give exactly the homogeneous left marginal in Theorem 2.1. Every plane and private variable remains a nonnegative multiple of s , so this full marginal is Lorentzian. No analogous opposite-marginal assertion is inferred when $q > 2$.

The six cases are disjoint classes of matroid bases. In particular the union case does not count injections or matching allocations. The construction also has a generic matrix representation obtained by giving each allowed slot incidence its own indeterminate. Its six block-determinant expansions provide a second interpretation of the same identity, including dependent or zero original core-row vectors.

If $Q = 0$, add a singleton right vertex at each A_i with activity $\varepsilon > 0$. The perturbed graph still has a displayed $2 + q$ cover, its Q is positive, and its coefficients converge to those of the original graph as $\varepsilon \downarrow 0$. Closedness gives the result. Other zero activities are handled by continuity at fixed order $q + 2$; private reciprocals are used only before taking the positive-activity limit.

Corollary 5.1. *Any failure of weighted rank-normalized ultra-log-concavity must have every minimum vertex cover meet both shores in at least three vertices.*

Proof. A pure or singleton-side minimum cover is covered by Lemma 3.2; a cover meeting one shore in exactly two vertices is covered by Theorem 2.1. $\square$

Remark 5.2. The new rank- $(q + 2)$ transversal matroid is used through its Lorentzian basis polynomial. The theorem asserts neither real-rootedness of p_G nor stability of the original complemented-row polynomial. The support identity, and not the values of nonzero determinants, determines all coefficients.

6 Exact failure at rank six and at every higher rank

We include the elementary obstruction calculation to make the sharpness conclusion self-contained. It is the negative construction from the Hall reports [13, 9]. [*Merge note:* [13] is source 75 (Section 16.5) and [9] is Part XI of *preorder-root-polytopes*. For $a = b$ and $N = M$ the graph $H_{a,b;N,M}$ with weight w on both cores is Part XI's $H_{s,n}$, and formula (5) is Part XI's Theorem 175.3 (`hb:thm:coefficients`) extended to unequal cores and populations. The endpoint quadratic below is the asymmetric form of Part XI's endpoint analysis (Theorem 178.2, `hb:thm:endpoint`).]

Let $H_{a,b;N,M}$ have a complete $a \times b$ core, N exterior-left vertices adjacent to every right-core vertex, and M exterior-right vertices adjacent to every left-core vertex. Give core vertices activity w and exterior vertices activity one. If $N \geq b$ and $M \geq a$, the core is a minimum cover of size $r = a + b$: match its two shores independently into the opposite exterior sets.

A size- k endpoint support with i selected left-core and j selected right-core vertices is feasible exactly when $i + j \geq k$, subject to the cardinality bounds. Match all exterior vertices to the core first, then use the complete core for the remainder. Thus

$$a_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{b}{j} \binom{N}{k-i} \binom{M}{k-j} w^{i+j}. \quad (5)$$

Put $A_i = \binom{N}{b-i}$, $B_i = \binom{M}{a-i}$ for $i = 0, 1, 2$, and

$$X = aA_0B_1 + bA_1B_0, \quad Y = aA_1B_2 + bA_2B_1, \quad Z = \binom{a}{2}A_0B_2 + abA_1B_1 + \binom{b}{2}A_2B_0.$$

Counting at most two omitted core vertices gives

$$a_r = A_0B_0w^r, \quad a_{r-1} = A_1B_1w^r + Xw^{r-1}, \quad a_{r-2} = A_2B_2w^r + Yw^{r-1} + Zw^{r-2}.$$

Hence

$$\Delta_{r-1}^{(r)} = w^{2r-2}(\alpha w^2 + \beta w + \gamma), \tag{6}$$

where

$$\begin{aligned} \alpha &= (r-1)A_1^2B_1^2 - 2rA_0A_2B_0B_2, \\ \beta &= 2(r-1)A_1B_1X - 2rA_0B_0Y, \\ \gamma &= (r-1)X^2 - 2rA_0B_0Z. \end{aligned}$$

6.1 A concrete rank-six witness

Take $a = b = 3$ and $N = M = 33$. Here $A_0 = B_0 = 5456$, $A_1 = B_1 = 528$, $A_2 = B_2 = 33$, and exact simplification of (6) gives

$$\Delta_5^{(6)} = 404794368 w^{10}(-w^2 + 26784w + 522784).$$

At $w = 26804$, the parenthesis equals -13296 . This is an exact positive-integer-activity counterexample on 72 vertices and 207 edges, with minimum cover and matching rank six. Combined with Corollary 2.2, the first weighted failure rank is exactly six. [*Merge note*: This witness is Part XI's $H_{3,33}$: Theorem 177.1 of *preorder-root-polytopes* (`hb:thm:rank6phase`) gives the last order-six margin of $H_{3,n}$ at $n = 33$ as 16 times the same quadratic $-w^2 + 26784w + 522784$. Source 74 restates and verifies it (its bibliography says so); it is not new here.]

6.2 Permanent scaling failure at every rank at least six

Let $a, b \geq 3$. Write

$$K = (r-1)ab, \quad J = (r+1)ab - 2r(r-1).$$

Because $ab \geq 3(r-3)$ and $r \geq 6$,

$$J \geq r^2 - 4r - 9 > 0.$$

Choose $N = b + 30$, $M = a + 30$, so that

$$\frac{A_1^2B_1^2}{A_0A_2B_0B_2} = \frac{ab}{(a-1)(b-1)} \left(\frac{32}{31} \right)^2.$$

The inequality $\alpha < 0$ is equivalent to

$$K(2 \cdot 31 + 1) < J \cdot 31^2.$$

Indeed

$$15J - K = (14r + 16)ab - 30r(r-1) \geq 12(r-6)(r+2) \geq 0,$$

and $63/961 < 1/15 \leq J/K$. Thus (6) is negative for every sufficiently large w . An explicit sufficient integer is

$$w \geq 1 + \left\lfloor \frac{|\beta| + |\gamma|}{-\alpha} \right\rfloor,$$

since for $w \geq 1$ the bracket is at most $w(\alpha w + |\beta| + |\gamma|) < 0$. Taking $a = 3$, $b = r - 3$ supplies every $r \geq 6$. In contrast, every graph of rank at most five remains ULC at every positive minimum-cover scale by Corollary 2.2. This proves the sharp universal rank threshold for minimum-cover scaling.

6.3 The shapes recorded by source 74's verifier

[*Merge note:* This subsection was written in the merge from source 74's recorded output `data/74-weighted-rank-five-negative_shapes_verification.json`; its numbers were recomputed independently from (5). Source 74's text does not print them.] Source 74's negative-family verifier records seven exact witnesses, listed in Table 3. Each has exterior activity one and one common integer activity w on all $a + b$ core vertices; the populations are $N = b - 1 + h$ and $M = a - 1 + h$ with source 75's offset $h = 1 + \lceil 2K/H \rceil$ (Section 16.5), and w is the least integer at which the last gap $\Delta_{r-1}^{(r)}$ is negative for these populations: the primitive endpoint quadratic (the bracket of (6) divided by its content) is negative at w and positive at $w - 1$. The (4, 4) row is Part XI's 30-vertex counterexample (Theorem 176.1 of

(a, b)	r	(N, M)	vertices	edges	w	primitive quadratic	in <i>preorder-root-polytopes</i>
(3, 3)	6	(33, 33)	72	207	26804	$-w^2 + 26784w + 522784$	Part XI, $H_{3,33}$
(3, 4)	7	(16, 15)	38	121	2012	$-3w^2 + 6006w + 55601$	new
(3, 5)	8	(15, 13)	36	129	418	$-115w^2 + 46860w + 469964$	new
(4, 4)	8	(11, 11)	30	104	485	$-w^2 + 480w + 2304$	Part XI, $H_{4,11}$
(4, 5)	9	(11, 10)	30	115	203	$-215w^2 + 42560w + 192472$	new
(5, 5)	10	(10, 10)	30	125	131	$-11w^2 + 1400w + 4928$	new
(3, 8)	11	(16, 11)	38	185	235	$-158w^2 + 34860w + 518355$	new

Table 3: Exact last-gap failures of $H_{a,b,N,M}$ recorded by source 74's verifier (merge table).

preorder-root-polytopes, `hb:thm:witness`, which uses $w = 500$); the (3, 3) row is its rank-six witness. The other five rows are asymmetric or larger-core shapes, which bear on Research question 89 of *preorder-root-polytopes*; none has fewer than 30 vertices, and none has matching number below six, as Corollary 2.2 requires.

7 An established geometric interpretation

For unit activities, p_G is the *perfectly matchable set polynomial* of Davis–Kohl [5]. It is distinct from the ordinary matching or rook polynomial, which counts matchings with multiplicity. Their Theorem 3.10, restating results of Ohsugi–Tsuchiya, gives the following precise Ehrhart interpretation.

Given bipartite $G = (X, Y; E)$, form $\hat{G}$ by adjoining x_0 on the left and y_0 on the right, with edges from x_0 to every old right vertex, from y_0 to every old left vertex, and $x_0 y_0$. For the edge polytope

$$Q_{\hat{G}} = \text{conv}\{e_x + e_y : xy \in E(\hat{G})\},$$

one has

$$h^*(Q_{\hat{G}}; t) = p_G(t) = h^*(\text{STAB}(\overline{G}); t).$$

The second identity is their Lemma 3.11; $\text{STAB}(\overline{G})$ is the stable-set polytope of the complement.

Corollary 7.1. *If G has matching rank $r \leq 5$, or a minimum vertex cover meeting one shore in at most two vertices, these two h^* -polynomials are ULC_r , normalized by their degree $r = \nu(G)$.*

Only the unit-activity interpretation is used here. No assertion is made about arbitrary Ehrhart h^* -polynomials or about the h^* -polynomial of an unrelated preorder polytope. [*Merge note:* At unit activities without the cover or rank hypothesis the conclusion fails: source 71 (Part II) gives a connected graph with $\nu = 3450$ whose two h^* -polynomials are not ULC_{3450} .] [*Merge note:* Sections 8–13 were the appendices of source 74; they are printed here as ordinary sections of Part I.]

8 A generic-line matroid model for the forced-left sector

By (2), the polynomial in Theorem 2.3 is

$$C = Qf + ag_2 + bg_1 + cg_U + h. \quad (7)$$

Write $f_h = s^q f(t/s)$, $G_i = s^{q-1} g_i(t/s)$, $G_U = s^{q-1} g_U(t/s)$ and $H_h = s^{q-2} h(t/s)$. For $q = 1$ the last term is zero. The right-complement transversal matroid of $H + A$ gives a degree- q basis specialization

$$F = f_h + z_1 G_1 + z_2 G_2 + z_1 z_2 H_h.$$

The private B variables are s/v_b , the exterior-left variables are tu_l , the two distinguished elements have variables z_1, z_2 , and we multiply by $\prod_b v_b$.

Represent this matroid by a generic matrix: use an independent indeterminate for each allowed adjacency and private standard basis columns. A square determinant is nonzero exactly when its support admits a matching, since different matchings give distinct monomials. Denote the two distinguished columns by v_1, v_2 . Adjoin columns

$$U_j = \alpha_j v_1 + \beta_j v_2 \quad (1 \leq j \leq m),$$

with generic coefficients, algebraically independent over the original matrix field. For a basis containing just one U_j from this distinguished set, its determinant is $\alpha_j D_1 + \beta_j D_2$ and is nonzero iff at least one of D_1, D_2 is nonzero. Thus its remaining-element profile is exactly G_U . Any pair of distinct distinguished columns has determinant a nonzero scalar multiple of the determinant using v_1, v_2 , hence profile H_h . Three distinguished columns are dependent. These statements also hold when v_1, v_2 are dependent or zero: pair profiles then vanish, and the single profile remains the union.

The resulting matroid basis specialization is therefore

$$\begin{aligned} F_{\text{ext}} = f_h + z_1 G_1 + z_2 G_2 + \left(\sum_j Z_j \right) G_U \\ + \left[z_1 z_2 + (z_1 + z_2) \sum_j Z_j + \sum_{i < j} Z_i Z_j \right] H_h. \end{aligned} \quad (8)$$

It is Lorentzian. For $Q > 0$, substitute

$$z_1 = bs/Q, \quad z_2 = as/Q, \quad Z_j = r_j s/Q.$$

The bracket becomes s^2/Q , and

$$QF_{\text{ext}} = Qf_h + s(ag_2 + bG_1 + cG_U) + s^2 H_h = s^q C(t/s).$$

This proves the first part of Theorem 2.3. For $Q = 0$, add two singleton right vertices with positive activities tending to zero and use coefficientwise closedness.

Corollary 8.1 (Scaling only the two cover vertices). *For a genuine $2 + q$ cover with $q \geq 1$, scaling only the two A activities by a common sufficiently large factor makes every ULC_{q+2} gap strictly positive.*

Proof. Every matching of size $q + 2$ uses exactly one cover vertex per edge. Thus R matches both A vertices and L matches all B , so $Q > 0$ and $f_q > 0$. Consequently all coefficients $C_0, \dots, C_q$ are positive. Write the two activities as $w\alpha, w\beta$. With E the weighted edge sum incident with A at relative activities α, β ,

$$a_1(w) = wE + O(1), \quad a_k(w) = w^2\alpha\beta C_{k-2} + O(w) \quad (k \geq 2).$$

The graph induced by A and the opposite shore has matching rank two, so $E^2 \geq 4\alpha\beta C_0$. The first two leading gaps are respectively

$$(q+1)E^2 - 2(q+2)\alpha\beta C_0 \geq 2q\alpha\beta C_0 > 0, \quad 2q(\alpha\beta C_0)^2 > 0,$$

at powers w^2, w^4 . For $k = i + 2$, $1 \leq i < q$, the leading coefficient at w^4 is at least

$$\frac{2(q-i+1)}{i}(\alpha\beta)^2 C_{i-1} C_{i+1} > 0,$$

by substituting the ULC_q inequality for C_i^2 into (1). Positive leading coefficients prove the claim. $\square$

9 The sharp all-core inequality at rank five

The full lift gives a short general boundary theorem before the independent rank-five certificate below.

Theorem 9.1 (Sharp all-core constant for a two-vertex shore). *For a $2 + q$ cover with $q \geq 2$, factor the product of all core activities from the all-core sector and write it as $t^q(c_0 + c_1 t + c_2 t^2)$. Then*

$$c_1^2 \geq \frac{2q}{q-1} c_0 c_2,$$

and the constant is optimal for every q .

Proof. In the lifted matroid basis polynomial, set all private B variables to zero and differentiate once in each augmented core variable X, Y . These operations preserve the Lorentzian property, giving a degree- q polynomial. Under the remaining specializations its three plane-count sectors give, up to the fixed positive scaling factor,

$$c_0 s^2 t^{q-2} + c_1 s t^{q-1} + c_2 t^q.$$

The bivariate ULC_q inequality at index $q - 1$ is the desired bound. Zero activities and $Q = 0$ follow by continuity as before.

For the complete $2 \times q$ core with $N \geq q$ unit common-left vertices and $M \geq 2$ unit common-right vertices,

$$c_0 = \binom{N}{q-2}, \quad c_1 = M \binom{N}{q-1}, \quad c_2 = \binom{M}{2} \binom{N}{q}.$$

Their ratio is

$$\frac{c_1^2}{c_0 c_2} = \frac{2q}{q-1} \frac{M}{M-1} \frac{N-q+2}{N-q+1} \longrightarrow \frac{2q}{q-1}.$$

$\square$

The following argument independently proves the $q = 3$ case using only a finite quartic coefficient test and explicit sums of squares. It does not rely on the full lift.

Now fix a $2 + 3$ cover. Strip off the product of the five core activities from the sector in which all five core vertices are selected. Its polynomial is

$$t^3(c_0 + c_1t + c_2t^2).$$

Theorem 9.2 (Sharp constant three). *For arbitrary nonnegative exterior activities and every $2 + 3$ core,*

$$c_1^2 \geq 3c_0c_2.$$

The constant 3 cannot be increased uniformly.

Let T be the weight of exterior-left triples matching all of B . Let C be the weight of single exterior-left vertices which, together with the two core rows, match B . Let P_i be the weight of exterior-left pairs completable with core row S_i , and let P_U use the union row. Set

$$O = P_1 + P_2 - P_U.$$

The union interpretation gives $0 \leq O \leq \min(P_1, P_2)$. The central left-side inequality is

$$4P_1P_2 - 2O^2 \geq 3CT. \quad (9)$$

9.1 A finite exact coefficient certificate

Encode a nonempty neighborhood in B by its bit mask $1, \dots, 7$. For three nonempty masks, let $M(r, s, t)$ be one exactly when they admit a matching to all three columns. Hall's condition gives the elementary rule: their bitwise union is 7, and no singleton mask $1, 2, 4$ appears twice. For general tuples including an empty mask, use ordinary Boolean matchability.

Index left exterior vertices by i , with masks t_i and weights z_i . Put

$$\begin{aligned} c_i &= M(S_1, S_2, t_i), & p_{ij} &= M(S_1, t_i, t_j), & q_{ij} &= M(S_2, t_i, t_j), \\ o_{ij} &= p_{ij}q_{ij}, & \tau_{ijk} &= M(t_i, t_j, t_k). \end{aligned}$$

Then $C = \sum c_i z_i$, $P_1 = \sum_{i < j} p_{ij} z_i z_j$, and similarly for P_2, O, T . In the quartic

$$I = 4P_1P_2 - 2O^2 - 3CT$$

every exponent is at most two. Its coefficient of $z_i^2 z_j^2$ is $2o_{ij} \geq 0$. Its coefficient of $z_i^2 z_j z_k$ for distinct indices is

$$4(p_{ij}q_{ik} + p_{ik}q_{ij}) - 4o_{ij}o_{ik} - 3c_i\tau_{ijk}. \quad (10)$$

Its coefficient of $z_1 z_2 z_3 z_4$ is

$$4 \sum_{ij|kl} (p_{ij}q_{kl} + p_{kl}q_{ij}) - 4 \sum_{ij|kl} o_{ij}o_{kl} - 3 \sum_i c_i \tau_{[4] \setminus \{i\}}, \quad (11)$$

where each of the three unordered pair partitions is used once.

If the core has matching rank at most one, $C = 0$ and (9) follows from $O^2 \leq P_1P_2$. Otherwise row exchange and column permutation give exactly the eight representatives in Table 4. The table records the complete range of possible coefficients in (10) and (11). It is obtained by the stated three-row Boolean rule, over $7^3 = 343$ ordered triples and $\binom{10}{4} = 210$ neighborhood multisets per core. The included exact verifier implements an independent permutation matcher and checks all 4424 substitutions, as well as all 46 labeled matching-rank-two cores.

Core masks	Repeated-exponent coefficients	Squarefree coefficients
(1, 2)	0, 1, 4	0, 2, 3
(1, 3)	0, 1, 4	0, 2, 3, 6, 8, 12
(1, 6)	0, 1, 4	-4, 0, 2, 3
(1, 7)	0, 1, 4	-4, 0, 2, 3, 8, 12
(3, 3)	0, 1, 4	0, 2, 3, 6, 8, 12
(3, 5)	0, 1, 4	0, 2, 3, 8, 12
(3, 7)	0, 1, 4	0, 2, 3, 8, 12
(7, 7)	0, 1, 4	0, 2, 3, 8, 12

Table 4: The complete finite coefficient test. In each exceptional row the only four negative neighborhood multisets are listed in (12).

Only cores (1, 6) and (1, 7) have negative coefficients. In both cases the negative neighborhood multisets are exactly

$$(2, 3, 4, 5), \quad (2, 3, 5, 5), \quad (3, 3, 4, 5), \quad (3, 3, 5, 5), \quad (12)$$

and each coefficient is -4 . Thus every term involving another neighborhood class is nonnegative. It remains to treat the four classes 2, 3, 4, 5 in these two cores.

9.2 The four-class sum-of-squares certificate

For core (1, 6) let a, b, c, d now denote the activity sums in classes 2, 3, 4, 5, and let B_2, D_2 be the second elementary moments of classes 3, 5. Put $u = a + b$, $v = c + d$. Exact support counting yields

$$\begin{aligned} C &= u + v, & P_1 &= uv, \\ P_2 &= (a + c)(b + d) + bd + B_2 + D_2, & O &= ad + bc + bd, \\ T &= v(ab + B_2) + u(cd + D_2). \end{aligned}$$

Since $0 \leq B_2 \leq b^2/2$ and $0 \leq D_2 \leq d^2/2$, and I is affine in both moments, it suffices to check the four rectangle corners. Write $I_{00}, I_{01}, I_{10}, I_{11}$ for the lower/lower, lower/upper, upper/lower and upper/upper choices. The following exact identities have only nonnegative summands:

$$\begin{aligned} I_{00} &= 2(ad - bc)^2 + uv(ab + cd) + (ac + bd)(ad + bc) + 2b^2d^2, \\ I_{01} &= \frac{1}{2}d^2(a - b)^2 + \frac{1}{2}ad(b - d)^2 + ac(b - d)^2 + bc(a - d)^2 + R_{01}, \\ R_{01} &= a^2(bd + cd) + a\left(\frac{1}{2}b^2d + bc^2 + c^2d + \frac{1}{2}cd^2\right) \\ &\quad + b^2(2c^2 + cd) + b\left(c^2d + \frac{1}{2}cd^2 + \frac{1}{2}d^3\right), \\ I_{11} &= \frac{1}{2}bd(u - v)^2 + \frac{1}{2}(ad + bc + 3ac)(b - d)^2 \\ &\quad + (b + d)\left(ac(a + c) + \frac{1}{2}(a^2d + bc^2)\right). \end{aligned}$$

The expression for I_{10} is obtained from I_{01} by interchanging (a, b) with (c, d) . For core (1, 7), the restricted P_2, O both increase by ac while C, T, P_1 are unchanged. Since $P_1 - O = ac$, its quartic is exactly the preceding $I + 2a^2c^2$. This proves (9) in every case. The included symbolic checker verifies all five identities exactly.

9.3 Passing to the right exterior and proving sharpness

Return to the right-side notation a, b, c, d of Section 4 [*Merge note*: source 74 writes “Section 3”]. Selecting all core vertices gives

$$c_0 = C, \quad c_1 = aP_2 + bP_1 + cP_U, \quad c_2 = TQ.$$

Let $A = a + c$, $B = b + c$, $p = P_1$, $q = P_2$. Then

$$c_1 = qA + pB - Oc, \quad Q \leq AB - c^2/2.$$

For $pq > 0$, the exact quadratic identity

$$\begin{aligned} & (qA + pB - Oc)^2 - (4pq - 2O^2)(AB - c^2/2) \\ &= \left(1 - \frac{O^2}{2pq}\right) (qA - pB)^2 + \frac{[O(qA + pB) - 2pqc]^2}{2pq} \geq 0 \end{aligned}$$

uses only $O^2 \leq pq$. Together with (9) it implies $c_1^2 \geq 3CTQ = 3c_0c_2$. If $pq = 0$, the same assertion follows by continuity, or directly from $O = 0$ and $CT = 0$.

For sharpness, take the complete 2×3 core, N common left-exterior vertices and M common right-exterior vertices, all with unit exterior activities. Then

$$c_0 = N, \quad c_1 = \binom{N}{2}M, \quad c_2 = \binom{N}{3}\binom{M}{2}, \quad \frac{c_1^2}{c_0c_2} = 3\frac{N-1}{N-2}\frac{M}{M-1} \rightarrow 3.$$

This completes Theorem 9.2.

10 Lorentzian transfer across a star core

Theorem 10.1 (Star-core transfer). *For a displayed $2+q$ cover with matching-rank-at-most-one core, both root orientations admit direct Lorentzian transfer proofs of ULC_{q+2} .*

This is now a special case of Theorem 2.1. We retain the independent proofs because the transfer operators and their interlacing certificates are useful beyond the full-lift construction.

10.1 A support-exact vertex-sum identity

Suppose G_1, G_2 intersect in just one vertex v of activity u , and write

$$p_1 = A(t) + utB(t), \quad p_2 = C(t) + utD(t).$$

Then

$$p_{G_1 \cup G_2} = AC + ut(BC + AD). \tag{13}$$

If v is absent, each component has balanced selected shores. If v is selected, exactly one component uses it, and that component is determined by its selected-vertex imbalance away from v : its opposite shore has one extra selected vertex. Thus the two root-using terms cannot count the same endpoint support, even if that support has several perfect matchings.

10.2 Stars centered on the two-vertex shore

Suppose every core edge is incident with $v = A_1$. Let G_1 have shores $L \cup \{v\}$ and B , and let G_2 have shores $\{v, A_2\}$ and R . Their intersection is just v . The right-complement basis construction for G_1 gives a Lorentzian polynomial

$$F(s, t, z) = A_h(s, t) + zB_h(s, t), \quad A_h = s^q A(t/s), \quad B_h = s^{q-1} B(t/s).$$

Here z is the root element variable; setting $z = ut$ gives the degree- q homogenization of p_1 .

For G_2 , write $C = 1 + dt$, $D = e + ft$. If A_2 has activity y , and the exterior class sums are a, b, c with common second moment δ , then

$$d = y(b + c), \quad e = a + c, \quad f = y(ab + ac + bc + \delta), \quad de - f = y(c^2 - \delta) \geq 0.$$

The homogeneous quadratic

$$K(s, t, z) = s^2 + dst + esz + ftz$$

is Lorentzian. Its Hessian is

$$\begin{pmatrix} 2 & d & e \\ d & 0 & f \\ e & f & 0 \end{pmatrix}, \quad \det = 2f(de - f) \geq 0.$$

For $f > 0$, its t, z principal submatrix has one positive and one negative eigenvalue; the determinant and eigenvalue interlacing exclude two positive eigenvalues of the full Hessian. For $f = 0$, $K = s(s + dt + ez)$ is a product of nonnegative linear forms.

The product FK is Lorentzian. Retain only its terms of degree at most one in z , then set $z = ut$. The output is

$$A_h(s^2 + dst) + ut[B_h(s^2 + dst) + A_h(es + ft)],$$

which is the degree- $(q + 2)$ homogenization of (13). Partial truncation preserves Lorentzian polynomials: polarize s, t , take the full multiaffine part (only powers z^2 are then removed), and identify the polarized variables again. Thus the output is Lorentzian and p_G is ULC_{q+2} .

10.3 Stars centered on the opposite shore

Suppose every core edge is incident with $v = B_1$. Let $G_1 = G[L, B]$ and let G_2 have shores A and $R \cup \{v\}$. For G_1 , the private-root variable now gives

$$F(s, t, z) = zA_h + tB_h, \quad A_h = s^{q-1} A(t/s), \quad B_h = s^{q-1} B(t/s).$$

The other private variables are s/v_b , the left variables are tu_l , and the basis polynomial is multiplied by $\prod_{b \neq v} v_b$. Hence F is Lorentzian and $uF(s, t, s/u) = s^q p_1(t/s)$.

For G_2 put $C_h = s^2 + c_1 st + c_2 t^2$ and $D_h = es + ft$. Give its two left vertices fixed activities α, β ; these are constants, distinct from the private-root indeterminate z . With exterior class sums a, b, c and second moment δ ,

$$c_1 = \alpha(a + c) + \beta(b + c), \quad c_2 = \alpha\beta(ab + ac + bc + \delta),$$

so

$$c_1^2 - 4c_2 = [\alpha(a + c) - \beta(b + c)]^2 + 4\alpha\beta(c^2 - \delta) \geq 0.$$

The rooted interlacing inequality is

$$c_1 e f - f^2 - c_2 e^2 \geq 0. \quad (14)$$

If v is adjacent only to the first left vertex, $e = \alpha$, $f = \alpha\beta(b + c)$ and the expression is $\alpha^3\beta(c^2 - \delta)$. The other singleton case gives $\alpha\beta^3(c^2 - \delta)$. If v is adjacent to both, $e = \alpha + \beta$, $f = \alpha\beta(a + b + c)$ and it is exactly

$$\alpha\beta \left[(a\alpha - b\beta + \frac{c}{2}(\alpha - \beta))^2 + \left(\frac{3c^2}{4} - \delta \right) (\alpha + \beta)^2 \right] \geq 0.$$

An isolated root has $e = f = 0$.

Factor $C_h = (s + \lambda_1 t)(s + \lambda_2 t)$, with $0 \leq \lambda_1 \leq \lambda_2$. If $e > 0$, (14) gives $\lambda_1 \leq f/e \leq \lambda_2$. Therefore

$$h(s, t) = \frac{stD_h}{C_h}$$

is a nonnegative linear combination of $st/(s + \lambda_i t)$ with positive total coefficient. Each such fraction maps $\mathbb{H}^2$ into $\mathbb{H}$, since its reciprocal is $1/t + \lambda_i/s$, of strictly negative imaginary part. Repeated λ_i give the same conclusion directly.

Consider the homogeneous degree-two operator

$$\mathcal{T}(P) = C_h P + stD_h \partial_z P.$$

It preserves real stability. At upper-half-plane values, a stable nonzero P has $\text{Im}(\partial_z P/P) \leq 0$, by its univariate factorization in z . A zero of $\mathcal{T}(P)$ with $D_h \neq 0$ would require this quotient to equal $-1/h$, which has strictly positive imaginary part. If $D_h = 0$, stable multiplication suffices.

The operator also preserves nonnegative coefficients. Restrict it to any finite coordinatewise degree box containing F ; its output lies in a larger degree box. Brändén–Huh’s Theorem 3.4 applies, so $\mathcal{T}(F)$ is Lorentzian. Finally

$$u\mathcal{T}(F)|_{z=s/u} = sA_h C_h + utB_h C_h + ustA_h D_h$$

is precisely the degree- $(q + 2)$ homogenization of (13). This proves the column-star case.

Every nonempty bipartite graph of matching rank one has all its edges incident with a common vertex. The two orientations therefore exhaust such cores. An empty core gives a product of ULC_q and ULC_2 polynomials, which is ULC_{q+2} by Lorentzian product closure. Theorem 10.1 follows.

11 Exact equality in the first Newton gap

[*Merge note:* The theorem of this section is *preorder-root-polytopes*’s Theorem 180.2 with Corollary 180.3 (Part XI, `hb:thm:firstidentity` and `hb:cor:equality`), proved there by an exact defect identity over a minimum vertex cover; the inequality is also Part XII’s Theorem 190.2 (`mr:thm:first`), proved there by the same Motzkin–Straus route, and its unweighted equality case is Part XII’s Theorem 190.3. Source 74’s proof is printed as a second route; no novelty is claimed. Source 72 proves the same statement by the same route (Theorem 14.6).] The following gives a useful structural interpretation of non-strictness. Its ingredient is the classical Motzkin–Straus bound [6]; we include the short argument in the form needed here.

Theorem 11.1. *For a positive weighted bipartite graph of matching rank $r \geq 2$,*

$$(r-1)a_1^2 - 2ra_2 = 0$$

if and only if every nontrivial connected component is a star and their weighted edge totals are all equal. In that case $p_G(t) = (1 + Et)^r$ for their common edge total E ; otherwise the first gap is strict.

Proof. Form the compatibility graph K whose vertices are the edges of G , with two vertices adjacent when the corresponding edges are disjoint. Give an edge xy weight $w_{xy} = u_x v_y$. Put $S = \sum w_e = a_1$ and

$$m_2 = \sum_{\{e,f\} \in E(K)} w_e w_f.$$

The clique number of K is r , and $m_2 \geq a_2$ because m_2 counts two-edge matchings rather than endpoint supports.

For completeness, the Motzkin–Straus bound is

$$m_2 \leq \frac{r-1}{2r} S^2.$$

To see it, merge the weights of two nonadjacent vertices onto whichever has the larger weighted neighbor sum. This preserves S and does not decrease m_2 . Continue until the positive support is a clique of size at most r , where the bound follows from Cauchy–Schwarz.

Equality in the asserted first gap forces equality in both preceding bounds. Since the original compatibility weights are all positive, they give an interior maximum of m_2 on the simplex of fixed sum. If K were not complete multipartite, it would contain an induced edge ik and a vertex j nonadjacent to both. The sum-preserving direction $e_i - 2e_j + e_k$ has strictly positive second derivative for m_2 , contradicting a local maximum. Hence K is complete multipartite. Its complement, the line graph of G , is a disjoint union of cliques. In a bipartite graph a nontrivial connected component whose edges are pairwise intersecting is a star. There are exactly r such components. The compatibility bound is then equality precisely when the total weights in the r parts are equal. Conversely, for stars support and matching counts coincide and the polynomial is the stated product. $\square$

12 Quantitative eventual strictness under common cover scaling

Fix a genuine $2 + 3$ cover whose core has matching rank two. Write every cover activity as w times its fixed positive relative activity. Let e_1, e_2 be the first two support coefficients of the induced core at those relative activities, and let ζ be the product of all five relative activities. For the all-core coefficients of Theorem 9.2, set $\gamma_i = \zeta c_i$.

Every size-five matching uses exactly one cover vertex per edge; a core edge would consume two cover vertices and preclude five edges. Thus the exterior-left graph matches all three B vertices and the exterior-right graph matches both A vertices, giving $c_2 > 0$. A core two-edge matching leaves one B vertex unmatched; that vertex has an exterior-left neighbor, so adding it gives $c_0 > 0$. Also $e_1, e_2 > 0$ and $e_1^2 \geq 4e_2$ by Lemma 3.1 for the core.

Counting selected cover vertices gives

$$\begin{aligned} a_0 &= 1, & a_1 &= e_1 w^2 + O(w), & a_2 &= e_2 w^4 + O(w^3), \\ a_3 &= \gamma_0 w^5 + O(w^4), & a_4 &= \gamma_1 w^5 + O(w^4), & a_5 &= \gamma_2 w^5. \end{aligned}$$

Consequently the leading gap limits are

$$\begin{aligned}\Delta_1^{(5)}/w^4 &\longrightarrow 4e_1^2 - 10e_2 \geq 6e_2 > 0, \\ \Delta_2^{(5)}/w^8 &\longrightarrow 6e_2^2 > 0, \\ \Delta_3^{(5)}/w^{10} &\longrightarrow 6\gamma_0^2 > 0, \\ \Delta_4^{(5)}/w^{10} &\longrightarrow 4\gamma_1^2 - 10\gamma_0\gamma_2 \geq 2\gamma_0\gamma_2 > 0.\end{aligned}$$

The last inequality uses the sharp constant three from Theorem 9.2. All four gaps are therefore strictly positive eventually. This is stronger than the already proved non-strict ULC_5 property at every $w > 0$.

For either this scaling or Corollary 8.1, write each gap as

$$\Delta_k(w) = L_k w^{d_k} + \sum_{j < d_k} b_{kj} w^j, \quad L_k > 0.$$

A directly computable sufficient threshold is

$$W = 1 + \max_k \frac{\sum_{j < d_k} \max(-b_{kj}, 0)}{L_k}. \quad (15)$$

For $w \geq W$, every negative lower term is bounded in magnitude by its coefficient times w^{d_k-1} , so every gap is positive. This threshold depends on the graph and its fixed field; no uniform bound is asserted.

For a rank-five minimum cover of split $0 + 5$ or $1 + 4$, Lemma 3.2 already gives ULC_5 at every scale. For a $2 + 3$ cover, Theorem 2.1 does likewise; the calculation above adds strictness when its core has matching rank two. Thus all choices of minimum cover are covered, with the relative activities fixed before scaling.

13 Exact verification and scope of the conclusions

The proofs above are universal arguments. Their exact checks are independently implementable and do not use numerical root searches or numerical optimization as evidence for a theorem.

- The full transversal lift is checked by Boolean basis enumeration on 180 weighted graphs with $q = 0, 1, 2, 3, 4, 5$. It tests 140450 candidate basis subsets and compares the complete left-subset marginal, before diagonalizing the left variables.
- An independent finite-field representation check covers all 64 labeled 2×3 cores and 73441 basis determinants, comparing the full weighted support polynomial exactly. A separate generic-line check covers 192 graphs and 23553 determinants, including 54 rank-deficient cores.
- The union formula is checked on 1352 weighted graphs with $q = 1, 2, 3, 4$, including overlapping, nested and empty core neighborhoods.
- The rank-five quartic certificate checks all 4424 stated coefficient cases, all 46 labeled rank-two cores, the four moment-corner identities, the exceptional correction, and the right-side quadratic identity.
- The star-core and scaling checks cover 820 weighted star-core graphs for $q = 1, \dots, 5$ and 230 rank-five common-scaling instances. All displayed interlacing sums of squares, grading identities and computed strictness thresholds are verified exactly.

- The negative-family verifier checks complete endpoint support counts for seven core shapes and the universal exterior population choice for all $3 \leq a, b \leq 20$. The rank-six witness and its exact negative gap are included.

The package contains the Python sources, fixed seeds and JSON outputs. Only SymPy is needed beyond the Python standard library. Run `python3 checks/run_all.py`; the JSON outputs are regenerated in `data/`. The LaTeX sources compile with ordinary pdfLaTeX; `build_local.sh` also handles the local TeX configuration used to generate this PDF. [*Merge note:* In this report the scripts are shipped as `code/74-weighted-rank-five-*.py` and the outputs as `data/74-weighted-rank-five-*.json`; the delivered layout (`checks/`) is not kept, so rerun on a copy of the delivered layout as the README explains. `build_local.sh` is not shipped, and the eight section files are inlined into this report's `article.tex`.]

Röhrle–Ulirsch's bimatroid theorem [7] normalizes by the smaller ambient shore size. The new lift replaces that order by $q + 2$, and hence by matching rank when the cover is minimum. The polynomial is the perfectly matchable set polynomial of [5], not a matching count with multiplicities. A bounded primary-literature comparison located no prior universal $2 + q$ -cover or weighted rank-five theorem; global priority remains unclaimed.

Higher-rank unrestricted unit-weight and one-shore-only questions are outside this report. [*Merge note:* Both are answered negatively by later sources of this batch (sources 71 and 73, Part II).] The proof uses Lorentzian matroid bases; it does not assume that transversal matroids have the half-plane property. The cover obstruction in Corollary 5.1 is sharp.

14 Strictness and equality for weighted matching-support polynomials (source 72)

Abstract of source 72, as delivered. For positively weighted bipartite graphs having a minimum vertex cover that meets one shore in at most two vertices, we characterize equality in every rank-normalized ultra-log-concavity inequality. Equality at any one index occurs precisely when all nontrivial connected components are stars with equal total edge activities. In that case the polynomial is $(1 + Et)^r$; otherwise every inequality is strict. In particular, this characterizes equality for every bipartite graph of matching rank at most five and gives strictness for connected graphs in that range with rank at least two. The proof combines the recently obtained rank-compression lift with a direct weighted version of the Chan–Pak contraction criterion and explicit rank-two contraction witnesses. We also give a short Motzkin–Straus proof of the same equality characterization for the first inequality at every rank, without a cover restriction.

14.1 Definitions and conclusions

Let $G = (X, Y; E)$ be a finite simple bipartite graph, with positive vertex activities u_x, v_y . Its matching-support polynomial is

$$p_G(t) = \sum_{k=0}^r a_k t^k, \quad a_k = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|=k \\ G[I, J] \text{ perfectly matchable}}} u_I v_J, \quad r = \nu(G).$$

An endpoint support is counted once, regardless of how many perfect matchings it has. Positive activities and a matching of size r imply $a_k > 0$ for every $0 \leq k \leq r$. The order- r normalized gap is

$$\Delta_k = k(r - k)a_k^2 - (k + 1)(r - k + 1)a_{k-1}a_{k+1}, \quad 1 \leq k \leq r - 1. \quad (16)$$

The condition $\Delta_k \geq 0$ for all k is ULC_r . A star includes a single edge; isolated vertices are disregarded. The *edge total* of a star is the sum of $u_x v_y$ over its edges, equivalently its center activity times the sum of its leaf activities.

Theorem 14.1 (Equality classification). *Suppose $r \geq 2$ and G has a minimum vertex cover meeting one shore in at most two vertices. The following are equivalent:*

1. $\Delta_k = 0$ for at least one $k \in \{1, \dots, r-1\}$;
2. every nontrivial component of G is a star and all its edge totals are equal to one common $E > 0$;
3. $p_G(t) = (1 + Et)^r$ for some $E > 0$.

If these conditions fail, then $\Delta_k > 0$ for every $1 \leq k \leq r-1$.

Corollary 14.2. *Theorem 14.1 applies to every bipartite graph with $2 \leq r \leq 5$. Every connected graph in that range with rank at least two is strictly ULC_r , for every positive activity assignment.*

Indeed a minimum cover has size r by König's theorem, so its smaller shore contribution is at most two when $r \leq 5$.

This is a companion to the rank-compression theorem [12], which proves ULC_r under the cover hypothesis. [Merge note: [12] is source 74: Theorem 2.1 and Corollary 2.2.] The lift is recalled precisely below. The unweighted matroid equality characterization is due to Chan–Pak [3, Theorem 1.3]. Their Theorem 1.11 shows that single-index equality need not imply total equality in general matroids; our conclusion uses the special graph lift. We prove the needed real-weight version directly, rather than infer strictness by a limit from rational weights. The arguments are ordinary mathematical proofs, not externally refereed or proof-assistant formalized. No exhaustive publication-priority claim is made.

14.2 A weighted matroid equality criterion

Let M be a rank- r matroid on a ground set $E = \mathcal{R} \sqcup \mathcal{B}$, with positive element weights ω_e . Define

$$P_j = \binom{r}{j}^{-1} \sum_{\substack{S \text{ basis of } M \\ |S \cap \mathcal{R}|=j}} \prod_{e \in S} \omega_e.$$

Loops are allowed. Parallel classes below refer only to nonloops.

Theorem 14.3 (Weighted contraction criterion). *Let $1 \leq a \leq r-1$ and $P_a > 0$. Then $P_a^2 = P_{a-1}P_{a+1}$ if and only if there is a common $\lambda > 0$ such that, for every independent I with*

$$|I| = r-2, \quad |I \cap \mathcal{R}| = a-1,$$

every nonloop parallel class J of M/I satisfies

$$\sum_{e \in J \cap \mathcal{R}} \omega_e = \lambda \sum_{e \in J \cap \mathcal{B}} \omega_e. \tag{17}$$

In particular, one such contraction with a monochromatic nonloop parallel class proves strictness at index a .

We use two established Lorentzian facts from Brändén–Huh [1]: a matroid basis polynomial is Lorentzian (Theorem 3.10), and every nonnegative directional derivative preserves the class (Corollary 2.11). A quadratic Lorentzian polynomial has a Hessian with at most one positive eigenvalue.

Lemma 14.4 (Kernel form of equality). *If a real symmetric matrix H has at most one positive eigenvalue and $y^T H y > 0$, then its quadratic form is negative semidefinite on $\{z : z^T H y = 0\}$. If also $z^T H z = 0$, then $H z = 0$.*

Proof. Otherwise the span of y and a positive vector in its H -orthogonal complement would have two positive directions. On that complement, zero quadratic value for a negative semidefinite form implies orthogonality to the whole complement. Together with orthogonality to y , it implies $H z = 0$. $\square$

Proof of Theorem 14.3: necessity. Let F be the unweighted basis generating polynomial. Let $f_e = \omega_e$ on $\mathcal{R}$ and zero on $\mathcal{B}$, and let $g_e = \omega_e$ on $\mathcal{B}$ and zero on $\mathcal{R}$. Write D_f, D_g for directional differentiation. The matrix

$$C = \text{Hess} (D_f^{a-1} D_g^{r-a-1} F) \quad (18)$$

has at most one positive eigenvalue. Direct differentiation gives

$$f^T C f = r! P_{a+1}, \quad f^T C g = r! P_a, \quad g^T C g = r! P_{a-1}. \quad (19)$$

At equality both neighbors are positive. Set

$$\lambda = P_a / P_{a-1} = P_{a+1} / P_a, \quad z = f - \lambda g.$$

Then $z^T C g = z^T C z = 0$, so Lemma 14.4 gives $C z = 0$.

Suppose first $r \geq 3$ and $a \geq 2$. For each red nonloop i , put

$$M_i = \text{Hess} (D_f^{a-2} D_g^{r-a-1} \partial_i F). \quad (20)$$

The exponents are nonnegative, and the differentiated polynomial has degree two. Commuting derivatives yields

$$C = \sum_{i \in \mathcal{R}} \omega_i M_i, \quad (C z)_i = z^T M_i f. \quad (21)$$

Loop terms are zero.

Every mixed value $f^T M_i g$ is strictly positive. Here is the required combinatorial verification. Equality supplies bases with red counts $a - 1$ and $a + 1$. Insert the nonloop i into each by one basis exchange, or keep it if already present. The resulting bases containing i have red counts at most a and at least $a + 1$, respectively. Bases of M/i are connected by exchanges, and each exchange changes the red count by at most one. Consequently M/i has a basis with exactly $a - 1$ red elements. Its positive weight contributes to $f^T M_i g$.

Equations (21) and $C z = 0$ now give

$$f^T M_i f = \lambda g^T M_i f > 0.$$

Each M_i has at most one positive eigenvalue, so Lemma 14.4 implies $z^T M_i z \leq 0$. Their positive-weighted sum is $z^T C z = 0$; hence every term is zero. The same lemma gives $M_i z = 0$. By (19) applied to M/i , equality propagates to index $a - 1$ of that contraction with the same ratio λ .

Contract red elements until the index is one. Then reverse colors and contract blue elements until rank two. Color reversal replaces λ by $1/\lambda$, and reversing back restores λ . This covers

$a = 1$ by starting with blue contractions and $a = r - 1$ by red contractions alone. Any independent I of the specified color counts can be ordered with red elements first and blue elements second; all prefixes remain independent, so each element contracted is a nonloop. The result therefore applies to every required M/I .

At rank two the Hessian has entry one on independent pairs and zero elsewhere. Its kernel equation for z says that the sum of z over the nonloops outside each parallel class is zero. Summing over the at least two classes makes the total over nonloops zero, and hence every class sum is zero. Loops have zero rows and impose no condition. This is precisely (17). $\square$

Proof of Theorem 14.3: sufficiency. For each specified independent I , let H_I be the Hessian of the basis polynomial of M/I . Condition (17) means $H_I z = 0$. Expansion of the directional derivatives gives

$$C = (a - 1)!(r - a - 1)! \sum_I \omega_I H_I.$$

Thus $Cz = 0$. Since $f^T Cg = r!P_a > 0$ and $\lambda > 0$, the three values in (19) are in geometric progression with ratio λ . This proves equality. $\square$

This proof establishes the criterion for arbitrary positive real weights. Parallel cloning proves its rational-weight version, but continuity alone would not justify the necessity or strictness conclusions for real weights.

14.3 The rank compression lift and its two bases

[Merge note: The lift recalled here is source 74's, Section 5; its six basis classes are the six cases there.] Suppose a minimum cover is $A \sqcup B$, where $A = \{A_1, A_2\}$, $|B| = q$ and $r = q + 2$. Put $L = X \setminus A$, $R = Y \setminus B$; there are no L - R edges. A size- r matching uses exactly one cover vertex per edge. Thus L matches all B and R matches both A vertices.

Partition R by neighborhoods $\{A_1\}, \{A_2\}, A$. Let their activity sums be a, b, c , and let the common-class individual activities be r_j . Put

$$d = \sum_{i < j} r_i r_j, \quad Q = ab + ac + bc + d > 0.$$

The lift of [12] is a rank- r transversal matroid with slots $B \sqcup \{e_1, e_2\}$. It has private columns for B , ordinary columns for L on their original neighborhoods, augmented core columns on $S_i \cup \{e_i\}$, private plane columns on e_1, e_2 , and common plane columns on both extra slots. Its red elements are the ordinary and augmented core columns; its blue elements are private and plane columns.

Give the red elements their original left activities, the private blue element at b weight $1/v_b$, and the plane elements weights

$$b/Q, \quad a/Q, \quad r_j/Q,$$

respectively. Delete zero-weight plane elements before applying Theorem 14.3. The positive value of Q guarantees a positive-weight independent pair of plane elements remains. The colored basis polynomial, multiplied by the harmless positive constant $Q \prod_b v_b$, is exactly

$$\sum_{k=0}^r a_k s^{r-k} t^k. \tag{22}$$

The common positive prefactor does not affect equality or strictness of the binomial-normalized coefficients. This identity counts bases once, not matching injections. It follows

by classifying selected augmented core and plane columns: their six possible counts are $(0, 2), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)$. The one-plane/two-core case uses the union of the two possible core-row completion profiles. The independent plane-pair activity sum is $1/Q$, while the eligible single-plane sums for A_1, A_2 are $(a + c)/Q, (b + c)/Q$. These are exactly the support alternatives with zero, one or two selected vertices in R .

We use the following generic representation, which is the usual transversal representation after harmless column normalization. Let V have dimension q and let E_0 have basis e_1, e_2 . Private B columns are coordinate vectors of V . Ordinary L columns have independent generic entries on their neighborhoods in V and zero projection to E_0 . The core columns are

$$(v_1, e_1), \quad (v_2, e_2),$$

with independent generic top entries on S_1, S_2 . Plane columns are $(0, e_1), (0, e_2)$ and $(0, \alpha_j e_1 + \beta_j e_2)$, with generic independent coefficients. A permitted square submatrix has nonzero determinant exactly when it admits a matching; different matching monomials do not cancel.

Choose two positive-weight independent plane columns P, Q_0 . Together with the q private B columns they form a blue basis. Order them so that the coefficient of e_1 along P and of e_2 along Q_0 are nonzero. Such an ordering exists because the two-by-two change-of-basis matrix is invertible.

Choose q ordinary L columns matching all B ; together with the two core columns they form a red basis. Assign each ordinary column to its matching B coordinate and the two core columns to P, Q_0 . For every subset of assigned positions, the corresponding principal row-column submatrix is nonsingular over the generic field: the ordinary block contains its matching monomial, and the bottom block is empty, one of the selected nonzero diagonal entries, or the full invertible two-by-two block. The bottom-left block is zero.

14.4 Contractions witnessing strictness

We apply Theorem 14.3 at an arbitrary index $k \in \{1, \dots, r - 1\}$. The following construction gives an independent I with $r - 2$ elements and $k - 1$ red elements. Leave two assigned positions i, j uncontracted. Select any $k - 1$ of the other $r - 2$ red basis elements, and contract them together with the blue basis elements at every remaining position except i, j . The principal-minor observation above shows that I is independent and the two uncontracted blue coordinate images are independent in M/I .

14.4.1 An ordinary column with two neighbors

Suppose some $u \in L$ has two B neighbors. Since the ordinary-column transversal matroid has rank q , its nonloop u extends to a basis. Choose a matching of this basis assigning u to i , and choose another neighbor j of u . Leave i, j uncontracted in the preceding construction, excluding both u and the ordinary red basis column assigned to j from I .

The contraction set and all blue columns are independent of every incidence parameter in column u . Its two entries at i, j are independent parameters multiplying independent quotient coordinate vectors. For any nonloop blue element b , the determinant of the images of u, b is a nonzero polynomial in those two parameters: its two coefficients cannot both vanish. Hence u is not parallel to any nonloop blue element in the generic matroid. Its parallel class is monochromatic red. Theorem 14.3 proves strictness at k . The construction works for every k .

14.4.2 A nonempty core

Suppose a core edge $A_1 B_i$ exists, and let $u = (v_1, e_1)$ be the augmented core column. Choose an ordinary basis matching all B . Leave the coordinate B_i and the bottom coordinate P

uncontracted; exclude u and the ordinary column matched to B_i from the red contraction choices.

The contraction set and all blue elements are independent of the top entries of u . Its entry at B_i is an independent parameter multiplying the surviving B_i coordinate. Moreover the image of $u_0 = (0, e_1)$ is independent of that coordinate. Indeed the $q - 1$ contracted top columns together with B_i span V . The one contracted bottom projection is either Q_0 , not parallel to e_1 by the chosen ordering, or e_2 from the other core column. In either case adding e_1 gives a nonzero full determinant.

For any nonloop blue element b , if its determinant with the B_i coordinate is nonzero, the corresponding top parameter makes the determinant with u nonzero. Otherwise b is parallel to B_i , and its determinant with u_0 is nonzero. Thus the determinant of u, b is not identically zero in either case. The class of u is monochromatic red, again proving strictness at every k . A core edge incident with A_2 is handled by interchanging labels.

These arguments include empty singleton plane classes: only a positive-weight plane basis was used, not the presence of both private plane elements. They concern generic representation parameters, independent of the arbitrary positive activity values.

14.5 Completing the graph classification

Suppose neither witness case occurs. The core is empty, and every ordinary left vertex has at most one B neighbor. Since L matches all B , these components are stars with positive edge totals $E_1, \dots, E_q$. The remaining graph is on A_1, A_2 and R . If the two left activities are ξ, η , its support polynomial is

$$1 + Et + \xi\eta Qt^2, \quad E = \xi(a + c) + \eta(b + c).$$

Its discriminant equals

$$[\xi(a + c) - \eta(b + c)]^2 + 4\xi\eta(c^2 - d) \geq 0. \quad (23)$$

Thus it factors as $(1 + \lambda_1 t)(1 + \lambda_2 t)$ with positive parameters. If $c > 0$, then $c^2 - d \geq c^2/2 > 0$, so the two parameters differ. If $c = 0$, the whole graph is a disjoint union of stars, with additional edge totals $\xi a, \eta b$.

Lemma 14.5. *For $r \geq 2$ and $p(t) = \prod_{i=1}^r (1 + \lambda_i t)$ with all $\lambda_i > 0$, equality in any one order- r normalized gap occurs if and only if all λ_i are equal.*

Proof. Use a direct sum of r rank-one matroids, each with a red element of weight λ_i and a blue element of weight one. For any two classes and any index k , contract one element from each other class, choosing exactly $k - 1$ red elements. Theorem 14.3 forces the two remaining class ratios to agree. Thus all λ_i agree. The converse follows from $(1 + \lambda t)^r$. $\square$

Equation (23) and Lemma 14.5 complete the classification for genuine $2 + q$ covers. They also handle $q = 0$ directly.

For a graph whose active right shore has size $r = \nu(G)$, use its ordinary right-complement transversal matroid with a blue coordinate basis. The first witness construction applies whenever a left vertex has two neighbors; otherwise the graph is a union of stars and Lemma 14.5 applies. This proves the same classification for pure minimum covers.

For a minimum cover split $1 + (r - 1)$, every nonisolated right vertex outside the cover is a leaf at its single left-cover vertex. Replace all these leaves by one with activity equal to their sum. [*Merge note:* This is the weighted leaf compression of *preorder-root-polytopes*, Lemma 206.1 (`lc:lem:compression`) of Part XIII.] There is at least one such leaf, since

otherwise the rank would be at most $r - 1$. The support polynomial is unchanged and the compressed graph has an active right shore of size r . It is a union of stars exactly when the original graph is, and expansion of the compressed leaf preserves its component edge total. The pure-shore result therefore transfers. Shore reversal and König’s theorem prove Theorem 14.1 and Corollary 14.2.

14.6 The first inequality at every rank

The following result requires no restriction on the cover and does not depend on the lift or on higher-gap ultra-log-concavity. [*Merge note:* Theorem 14.6 is *preorder-root-polytopes*’s Theorem 180.2 with Corollary 180.3 (Part XI, `hb:thm:firstidentity`, `hb:cor:equality`), and the inequality is Part XII’s Theorem 190.2; it is also printed above as source 74’s Section 11. Source 72’s Motzkin–Straus proof is printed as a second route; as source 72 itself says below, it is not a new optimization theorem.]

Theorem 14.6. *For every positively weighted bipartite graph of matching rank $r \geq 2$,*

$$(r - 1)a_1^2 \geq 2ra_2.$$

Equality occurs exactly for disjoint stars with equal edge totals.

Proof. Give edge xy weight $w_{xy} = u_x v_y$ and put $S = \sum_e w_e = a_1$. Let m_2 count pairs of disjoint edges with their weight products. Every feasible two-edge endpoint support has at least one such matching, with the same product, so $a_2 \leq m_2$.

Form the edge-compatibility graph: its vertices are edges of G , and its adjacency means disjointness. Its clique number is r . The Motzkin–Straus theorem [6] gives $m_2 \leq (r - 1)S^2/(2r)$. The upper bound also has a short proof: merge the masses of a nonadjacent pair onto the endpoint of larger weighted neighbor sum, without decreasing the objective, until the positive support is a clique; then apply Cauchy–Schwarz to the masses on that clique.

If the support inequality is equality, the original strictly positive edge-weight vector is an interior global maximizer of the compatibility objective. If that graph were not complete multipartite, it would contain an edge ik and a vertex j nonadjacent to both. The fixed-sum direction $e_i - 2e_j + e_k$ has positive second variation, contradicting an interior maximum. Thus the complement, which is the line graph of G , is a disjoint union of cliques. Each nontrivial component of G has pairwise intersecting edges; in a bipartite graph such a component is a star. If their totals are $E_1, \dots, E_r$, equality in $m_2 = (S^2 - \sum E_i^2)/2$ requires all totals equal. The converse follows from $p_G(t) = \prod_i (1 + E_i t)$. $\square$

Theorem 14.6 is a graph-support consequence of the classical Motzkin–Straus theorem, not a new quadratic optimization theorem. It should not be used to infer higher-gap strictness by coefficient comparison.

14.7 Verification and scope

The accompanying exact script [*Merge note:* shipped as `code/72-strictness-verify.py` and `code/72-strictness-verify_witnesses.py`, with outputs `data/72-strictness-*`] enumerates all 512 simple 3×3 bipartite graphs under three positive integer activity assignments. Excluding ranks zero and one, it verifies 1434 weighted graph cases and 2175 individual gaps: 42 balanced-star cases have all gaps zero, and 1392 cases have all gaps positive. An independent Boolean-transversal test examines 375 genuine cover graphs with $q = 0, \dots, 5$ and verifies 1589 monochromatic contraction witnesses; four star cases have no such structural witness. These are finite regression tests, not proof of the universal statements.

The novelty boundary is the separate rank-compression report [12]: the present companion adds equality classification and strictness to its nonnegative-gap theorem. The unweighted matroid characterization is credited to [3]; the weighted criterion above is proved directly, and no broad bibliographic priority claim is made for that extension. The full graph classification uses the particular generic contraction witnesses, not a general claim about all transversal or represented matroids.

All activity assumptions in the strictness theorems are positive. If some activities vanish, delete those vertices and reconsider the surviving graph and its actual matching rank. A limit of strict inequalities need not be strict. In particular no unspecified continuity argument is used for equality cases.

15 Stability of two-by-two covers and weighted rank four (source 76)

Abstract of source 76, as delivered. Every bipartite graph with a vertex cover consisting of two vertices on each shore has a real stable complemented-row matching-support polynomial. In particular, its matching-support polynomial is real-rooted under arbitrary positive vertex activities. The proof reduces to six four-vertex core types. Two independent-column covariance constructions treat the matching and three-edge cores; a two-block polarization argument treats the complete core. Together with the smaller-shore inequality and exact leaf compression, this settles rank-normalized ultra-log-concavity for every bipartite matching-support polynomial of rank at most four, including arbitrary weights on both shores. All finite certificates are exact, and the universal argument is given in full.

Status and provenance. This note answers the incoming ProveIt manuscript’s question “Finish rank four” [10], strengthening that question to arbitrary positive vertex activities. It follows the separate leaf-compression note [11], which settled weighted rank three and all singleton-side minimum covers. The conclusions are proofs in ordinary mathematics, not Lean-certified results or independently refereed theorems. The stability machinery is classical [2]; novelty is asserted only relative to the inspected ProveIt drafts, not as a claim of worldwide priority. [*Merge note:* [10] is the manuscript printed as Part XII of *preorder-root-polytopes*, and “Finish rank four” is its Research question 93 (**mr:q:rankfour**); [11] is Part XIII. Source 74 (Corollary 2.2) answers the same question through matching number five, so Corollary 15.2 below is printed as a second route. The real-stability Theorem 15.1 is not implied by source 74, which asserts no real-rootedness. The tools of Section 15.2 are Part VI’s (see the notes there).]

15.1 Definitions and the conclusions

Let $G = (X, Y; E)$ be a finite simple bipartite graph. A pair (I, J) , with $I \subseteq X$, $J \subseteq Y$ and $|I| = |J|$, is *feasible* when $G[I, J]$ has a perfect matching. For positive activities u_x, v_y , define

$$p_G(t; u, v) = \sum_{(I, J) \text{ feasible}} u_I v_J t^{|I|} = \sum_{k=0}^r a_k t^k, \quad r = \nu(G).$$

Each pair is counted once, regardless of the number of perfect matchings. Thus $p_{K_{2,2}}(t; \mathbf{1}, \mathbf{1}) = 1 + 4t + t^2$.

A degree- r polynomial with nonnegative coefficients and no internal zeros is ULC_r when $a_k / \binom{r}{k}$ is log-concave. Equivalently,

$$k(r-k)a_k^2 \geq (k+1)(r-k+1)a_{k-1}a_{k+1} \quad (1 \leq k < r). \quad (24)$$

Write $\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$. A nonzero real polynomial is *real stable* if it has no zero when every variable belongs to $\mathbb{H}$.

Introduce a private copy $\bar{x}$ of each $x \in X$ and put

$$F_G(z; y) = \sum_{(I, J) \text{ feasible}} z^{\overline{X \setminus I}} y^J. \quad (25)$$

This is the basis polynomial of the transversal matroid in which $\bar{x}$ is adjacent only to x and each $y \in Y$ is adjacent to its neighbours in G . It is homogeneous of degree $|X|$ and multiaffine.

Theorem 15.1 (Two vertices on each shore). *If G has a vertex cover $A \sqcup B$ with $A \subseteq X$, $B \subseteq Y$ and $|A| = |B| = 2$, then F_G is real stable. Consequently $p_G(t; u, v)$ has only real negative zeros for every positive activity assignment and is $\text{ULC}_{\deg p_G}$.*

Corollary 15.2 (Weighted rank four). *Every bipartite graph with matching rank $r \leq 4$ has ULC_r matching-support polynomial for arbitrary positive vertex activities. For nonnegative activities the same assertion holds with the actual surviving degree as normalization order. In rank four, the three inequalities are*

$$3a_1^2 \geq 8a_2, \quad 4a_2^2 \geq 9a_1a_3, \quad 3a_3^2 \geq 8a_2a_4.$$

The corollary does not assert real-rootedness for every rank-four graph. Its real-rootedness input concerns the two-plus-two cover class; other cover types use ultra-log-concavity directly. [Merge note: Corollary 15.2 is the case $\nu \leq 4$ of source 74's Corollary 2.2.]

15.2 Stability tools and the six cores

For a graph, also define its signed unmatched-support polynomial

$$\Phi_G(z) = \sum_{S \text{ perfectly matchable}} (-1)^{|S|/2} \prod_{v \notin S} z_v.$$

For bipartite G , the exact inversion identity is

$$\Phi_G(z) = \left(\prod_{y \in Y} z_y \right) F_G((z_x)_{x \in X}; (-1/z_y)_{y \in Y}). \quad (26)$$

The map $z \mapsto -1/z$ preserves $\mathbb{H}$, and the inverse substitution has the same form. Thus F_G is stable if and only if Φ_G is stable. Stability here concerns these polynomials, not the raw matched-support generating function. [Merge note: Identity (26) and this equivalence are *preorder-root-polytopes*'s Theorem 85.3 (Part VI, `ade:mat:thm:stable-bridge`).]

If F_G is stable, substitute $z_x = 1/u_x$ and $y_y = tv_y$ in (25). Boundary specialization and positive scaling preserve stability, possibly giving zero; here the constant term excludes zero. Therefore

$$p_G(t; u, v) = \left(\prod_{x \in X} u_x \right) F_G((1/u_x)_x; (tv_y)_y) \quad (27)$$

is a real stable univariate polynomial, so it is real-rooted. Its nonnegative coefficients and constant term one force all roots to be negative. Newton's inequalities give (24) with its actual degree.

Lemma 15.3 (Leaf attachment and vertex sums). *Attaching a leaf to a graph with stable Φ preserves stability. More generally, if two graphs with stable Φ intersect in one vertex and have no other connecting edges, their union has stable Φ .*

Proof (pointer). The vertex-sum statement is *preorder-root-polytopes*'s Theorem 93.2 (Part VI, `ade:mat:thm:vertex-stable`), proved there from the identity $\Phi_G = z_v A_1 A_2 + B_1 A_2 + A_1 B_2$ and $\text{Im}(B_i/A_i) \geq 0$. Attaching a leaf is the one-vertex sum with an edge, whose signed polynomial $z_v z_w - 1$ is stable. [*Merge note:* Source 76 gives the same two arguments (for the leaf, $\Phi_{G+w} = z_w \Phi_G - \partial_v \Phi_G$ and the sign of the logarithmic derivative); they are not reprinted.] $\square$

Delete isolated vertices and all exterior vertices whose neighbourhood in the opposite part of the cover is a singleton. These are pendant leaves and can be reattached by Lemma 15.3. The remaining exterior vertices form sets $L = \{L_1, \dots, L_n\} \subseteq X$ and $R = \{R_1, \dots, R_m\} \subseteq Y$, where every L_i is adjacent to both B_1, B_2 and every R_j to both A_1, A_2 . There are no edges between L and R .

Up to row and column permutations and transposition, the four-vertex core $G[A, B]$ has six types: empty, one edge, two adjacent edges, two disjoint edges, three edges, or four edges. For the first three types the graph is a vertex sum of complete bipartite graphs $K_{2,k}$, bridges and isolated vertices. The basis polynomial of $K_{2,k}$, with the two-vertex shore on the left, is the elementary symmetric polynomial e_2 on $k+2$ variables, hence is stable. Equation (26) removes dependence on orientation. Lemma 15.3 handles these three cases. [*Merge note:* Stability of $K_{2,k}$ is *preorder-root-polytopes*'s Theorem 94.1 (`ade:mat:thm:biclique`), and these three core types are covered by its Corollary 94.3 (`ade:mat:cor:block-family`). Only the matching, three-edge and complete cores treated below are new.]

15.3 Independent columns and Boolean minor weights

Lemma 15.4 (Expected determinant). *Let the columns K_j of a real random matrix be independent and have finite second moments. Then*

$$\mathbb{E} \det(\text{diag}(z) + K \text{diag}(y) K^T)$$

is real stable.

Proof (pointer). This is *preorder-root-polytopes*'s Theorem 91.2 (Part VI, `ade:thm:expected`). [*Merge note:* Source 76's proof is the one given there: stability of $\det(\text{diag}(z) + \sum_j t_j C_j)$ with $C_j = \mathbb{E}(K_j K_j^T)$, the operators $1 + y_j \partial_{t_j}$, specialization $t_j = 0$, and independence of the columns. It is not reprinted.] $\square$

This is standard stability machinery, also used in the repository's even-multi-theta argument [8]; the covariance choices below are the particular constructions needed here.

Cauchy–Binet expands the expected determinant as

$$\sum_{|I|=|J|} \mathbb{E} \det(K[I, J])^2 z^{X \setminus I} y^J. \quad (28)$$

It therefore equals F_G if the squared-minor expectation is one on every feasible support and zero on every infeasible support.

Set

$$q = 2^{-1/2}, \quad \rho = \sqrt{q} = 2^{-1/4}.$$

Rows are indexed by $A_1, A_2, L_1, \dots, L_n$. Every R -column has covariance supported on the two A rows, with matrix

$$\begin{pmatrix} 1 & q \\ q & 1 \end{pmatrix}. \quad (29)$$

All columns are independent. Within either B -column, the covariance on the L rows is $(1 - q)I + qJ$, so diagonal entries are one and distinct-row entries are q .

15.3.1 The matching core

Use core edges A_2B_1, A_1B_2 . In the B_1 column, the A_1 entry is zero, the A_2 variance is one, and its covariance with each L_i is ρ . In the B_2 column, the A_2 entry is zero, the A_1 variance is one, and its covariance with each L_i is $-\rho$.

These matrices are positive semidefinite in every dimension. For independent centred variance-one variables $Z, Z_1, \dots, Z_n$, take the nonzero A entry to be Z or $-Z$, respectively, and take

$$K_{L_i} = \rho Z + \sqrt{1-q} Z_i.$$

Use independent copies for different columns. Matrix (29) is positive definite, so it too is a covariance matrix.

15.3.2 The three-edge core

Use edges A_1B_1, A_2B_1, A_1B_2 . In the B_1 column, the A -block is (29), and the covariance vector from (A_1, A_2) to any L_i is $(\rho/\sqrt{2}, 0)$. For independent centred variance-one Z, W, Z_i , realize it by

$$K_{A_1} = (Z + W)/\sqrt{2}, \quad K_{A_2} = W, \quad K_{L_i} = \rho Z + \sqrt{1-q} Z_i.$$

In the B_2 column, take $K_{A_1} = Z$, $K_{A_2} = 0$, and the same formula for K_{L_i} , using a fresh independent family. This realizes variance one at A_1 and covariance ρ from A_1 to every L_i .

15.3.3 Verification of all minor types

Both constructions have the graph's zero pattern. Therefore unsupported minors vanish identically. Supported minors use at most two L rows and two R columns, and have size at most four.

Size-one supported minors have expectation one. A size-two supported minor has either one matching, giving one by column independence and unit entry variances, or two matchings around a four-cycle. In the latter case the expectation is $2 - 2q^2 = 1$, except for the path core's $\{A_1, L_i\} \times \{B_1, B_2\}$ minor, where it is

$$2 - 2(\rho/\sqrt{2})\rho = 1.$$

For size three, the only case without a forced row or column uses rows A_1, A_2, L_i and columns B_1, B_2, R_j . For the matching core, the two determinant terms have diagonal contributions one each, and their cross term is $-2\rho^2q$. Thus

$$\mathbb{E} \det(K[I, J])^2 = 2 - 2\rho^2q = 1.$$

For the path core there are three matching terms. Their three pairwise cross terms are

$$-2(\rho/\sqrt{2})\rho, \quad -2q^2, \quad 0,$$

so the expectation is again one. In every other supported size-three case a forced entry lies in a column independent of the remaining minor; its variance is one and the remaining size-two minor has squared expectation one.

Every supported size-four minor uses both A rows, both B columns, two L rows and two R columns. Up to sign its determinant factors as

$$\det K[\{L_i, L_j\}, \{B_1, B_2\}] \det K[\{A_1, A_2\}, \{R_s, R_t\}].$$

The factors depend on disjoint independent column families, and each has squared expectation one. This proves the Boolean identity required in (28) for every minor size. Lemma 15.4 proves stability for both core types.

15.4 The complete core by two-block polarization

Now suppose all four edges between A and B are present. Partition the ground set of the fundamental transversal matroid into

$$\mathcal{U} = \bar{A} \sqcup R, \quad \mathcal{V} = \bar{L} \sqcup B, \quad M = m + 2, \quad N = n + 2.$$

Its bases are exactly the N -element subsets containing at most two elements of $\mathcal{U}$. Indeed, let a support select i vertices of A , j of B , ℓ of L and s of R . Size balance is $i + \ell = j + s$. Feasibility requires $s \leq i$ and $\ell \leq j$, and these conditions are equivalent under balance. They suffice: match the exterior vertices to the selected opposite core, then match the remaining equally numerous core vertices across the complete core. The corresponding basis contains $2 - i + s$ elements of $\mathcal{U}$, so feasibility is exactly the bound of two.

Consequently the basis polynomial is

$$F_G = e_N(\mathcal{V}) + e_1(\mathcal{U})e_{N-1}(\mathcal{V}) + e_2(\mathcal{U})e_{N-2}(\mathcal{V}). \quad (30)$$

Its separate block diagonal, with $\mathcal{U}$ variables set to a and $\mathcal{V}$ variables to b , is

$$b^{N-2} \left(b^2 + MNab + \binom{M}{2} \binom{N}{2} a^2 \right).$$

The quadratic discriminant is

$$M^2N^2 - 4 \binom{M}{2} \binom{N}{2} = MN(M + N - 1) > 0. \quad (31)$$

Since $M, N \geq 2$, its factors are $(b + \lambda_+ a)(b + \lambda_- a)$ with both $\lambda_{\pm} > 0$. The block diagonal is therefore stable.

Polynomial (30) is exactly its separate multiaffine polarization in M variables and N variables. The Grace–Walsh–Szegő theorem, in the convex upper half-plane, says that polarization preserves stability [2]. Thus F_G is stable. This completes all six core cases; leaf reattachment proves Theorem 15.1.

15.5 From stability to the rank-four inequalities

For completeness we recall the two inputs from the preceding note [11]. [*Merge note:* They are *preorder-root-polytopes*'s Theorem 70.1 (Part V) or Lemma 205.1 (Part XIII), and Lemma 206.1 with Theorem 206.2 (Part XIII); source 74 states them as Lemmas 3.1 and 3.2.] For arbitrary positive activities, p_G is $\text{ULC}_{\min(|X|, |Y|)}$: matroid basis polynomials are Lorentzian [1], and the nonnegative linear substitution

$$\left(\prod_{x \in X} u_x \right) F_G((s/u_x)_x; (tv_y)_y) = \sum_k a_k s^{|X|-k} t^k$$

preserves the Lorentzian property. The bivariate characterization gives $\text{ULC}_{|X|}$; transpose the graph for $\text{ULC}_{|Y|}$. This is the weighted form of the established smaller-shore input; see also [7].

If a minimum cover of size r lies wholly on one shore, remove isolated vertices and apply that input. If a minimum cover is $\{x_0\} \sqcup B$ with $|B| = r - 1$, every active right vertex outside B is a leaf at x_0 . A feasible support contains at most one such leaf, so replacing the class by one leaf of activity equal to the sum of its activities preserves the entire polynomial. The compressed right shore has size r , giving ULC_r . The transposed case is identical.

By König's theorem a rank-four graph has a cover of size four. Its split is $0 + 4$, $1 + 3$, $2 + 2$, or a transpose. The preceding paragraph treats the first two types, and Theorem 15.1 treats the third by Newton's inequalities. Covers of size at most three have a pure or singleton-side split, proving the smaller ranks too. For zero activities, delete their vertices and apply the result with the surviving degree. This proves Corollary 15.2.

15.6 A sharp boundary for complete mixed cores

For integers $s \geq 2$ and $n, m \geq 0$, let $H_{2,s;n,m}$ have left core A of size two, right core B of size s , n exterior left vertices L , and m exterior right vertices R . Its edges are all pairs in

$$(A \times B) \cup (L \times B) \cup (A \times R),$$

with no edges between L and R . When $n \geq s$ and $m \geq 2$, its matching rank is $s + 2$.

Theorem 15.5 (Sharp stability criterion). *The signed support polynomial of $H_{2,s;n,m}$ is real stable if and only if*

$$(m + 2)(n + 2)s \geq 2(m + 1)(n + 1)(s - 1). \quad (32)$$

In particular every two by two Hall core is stable. For a two by three core, stability is equivalent to

$$(n - 2)(m - 2) \leq 12. \quad (33)$$

Proof. In the extended transversal matroid put $F = \bar{A} \sqcup R$ and $O = \bar{L} \sqcup B$, with

$$M = |F| = m + 2, \quad N = |O| = n + s, \quad r = n + 2.$$

Exactly as in the complete two by two core, the bases are all r -element subsets containing at most two members of F . To check sufficiency directly, a selected set with $k \leq 2$ elements of F and j elements of B uses $r - k - j$ private L -elements. Match the F -elements into A , use $2 - k$ selected B -elements on the remaining positions of A , and use the other $k + j - 2$ selected B -elements on the vacant L -positions. All counts are nonnegative by the size of the selected subset.

Thus its basis polynomial is

$$\mathcal{B} = \sum_{k=0}^2 e_k(F) e_{r-k}(O).$$

Its two-block diagonal is

$$y^n \left[\binom{N}{r} y^2 + M \binom{N}{r-1} xy + \binom{M}{2} \binom{N}{r-2} x^2 \right].$$

All three bracketed coefficients are positive. The bracket is stable exactly when its discriminant is nonnegative, equivalently

$$M^2 \binom{N}{r-1}^2 \geq 4 \binom{N}{r} \binom{M}{2} \binom{N}{r-2}.$$

Using

$$\frac{\binom{N}{r-1}^2}{\binom{N}{r} \binom{N}{r-2}} = \frac{r}{r-1} \frac{N-r+2}{N-r+1}$$

reduces this inequality to (32). Necessity for full stability follows by identifying all variables within each block. Sufficiency follows by polarization, using M slots for x and N slots for y . The signed support polynomial is stable exactly when the extended basis polynomial is stable, by the invertible upper-half-plane substitution $z \mapsto -1/z$ on one shore. For $s = 3$, expansion of (32) gives (33). $\square$

A smallest unstable member of the rank-five family. Among members with $s = 3$, $n \geq 3$ and $m \geq 2$, instability requires $(n - 2)(m - 2) > 12$. If $(n - 2) + (m - 2) \leq 7$, the product is at most $\lfloor 7^2/4 \rfloor = 12$. Thus every unstable member has at least $n + m + 5 = 17$ vertices; $n = m = 6$ attains this bound. Its diagonal factor is

$$y^6(9y^2 + 288xy + 2352x^2),$$

whose quadratic discriminant is -1728 .

This does not refute positive-field real-rootedness or rank-normalized ULC. Indeed the same graph with unit vertex weights has

$$p(t) = 1 + 36t + 333t^2 + 974t^3 + 1005t^4 + 300t^5,$$

with five distinct negative roots. An exact certificate consists of its alternating signs at

t	-2	-1	$-\frac{1}{2}$	$-\frac{1}{5}$	$-\frac{1}{20}$	0
$p(t)$	-51	29	$-\frac{33}{16}$	$\frac{21}{25}$	$-\frac{1329}{16000}$	1

The intermediate value theorem supplies a root in each of the five disjoint intervals. Consequently the rank-four proof by universal signed stability has a genuine rank-five boundary, even though the rank-five ULC problem remains unresolved here. [*Merge note:* The rank-five problem is settled affirmatively by source 74 (Corollary 2.2); this graph has the $2 + 3$ minimum cover $A \sqcup B$, so Theorem 2.1 gives ULC_5 for every positive activity assignment.]

15.7 Scope and reproducibility

The rank-four question in [10] is therefore resolved for unit weights, one-shore weights and arbitrary two-shore weights. Combined with the rank-six weighted counterexamples in [9], the first possible failure rank for arbitrary two-shore rank normalization lies in $\{5, 6\}$. General unit-weight and one-shore rank normalization are not settled by this note. [*Merge note:* Superseded within the batch: the first failure rank is exactly six by source 74 (Corollary 2.2). The shipped `76-rank-four-PROOF_STATUS.md` says likewise that “the smallest failure rank is reduced to five or six”; it is kept as delivered. The unit-weight and one-shore questions are answered negatively in Part II (sources 71 and 73).]

The expected-determinant lemma, polarization theorem, uniform-matroid stability and gluing principles are established tools; they are not claimed as new. This note’s contributions relative to the inspected drafts are the two covariance realizations, the complete-core two-block application, the resulting two-plus-two cover theorem, and the weighted rank-four conclusion. A targeted search of the stability literature did not certify global priority, which is not claimed.

The accompanying scripts supplement the proof:

- [*Merge note:* The scripts are shipped as `code/76-rank-four-verify_*.py` and their outputs as `data/76-rank-four-*_verification.json`; the build wrapper is not shipped.]
- `verify_covariance.py` checks all 70 square minors of the four-by-four representative for each covariance construction, including infeasible and empty minors. All 140 exact identities pass in $\mathbb{Q}(2^{1/4})$. It also checks every principal covariance minor in those representatives.
- `verify_complete_core.py` independently checks the support-to-basis characterization and block counts for all 49 pairs $0 \leq n, m \leq 6$, together with the discriminant identity.

- `verify_boundary.py` checks 75 complete mixed cores against the two-block basis description, the exact stability threshold, and the rank-five unit-weight polynomial and rational root certificate.
- The six-core proof and the classification of all possible supported minor types establish the universal assertions; the finite scripts do not replace those arguments.

Run the scripts with Python 3 and SymPy, and compile this source twice with `pdflatex`. The package includes a build wrapper for the preparation environment and recorded verification outputs.

16 A sharp weighted classification of full Hall bottleneck graphs (source 75)

Abstract of source 75, as delivered. A full Hall-bottleneck graph has two complete core blocks, complete connections from each exterior block to the opposite core, and no exterior-to-exterior edges. We prove that this family satisfies matching-rank-normalized ultra-log-concavity for every positive vertex-activity assignment and every exterior population if and only if one core has size at most two. The new positive argument treats a two-by- s core uniformly for every s : deleting the core edges gives a real-rooted comparison polynomial, and restoring them increases every Newton margin. A twelve-case coefficient calculation proves this using only binomial ratios and Newton inequalities for exterior elementary symmetric functions. When both cores have size at least three, we give explicit finite populations and an integer activity producing a failed last margin. The positive inequalities are strict in the genuine two-by- s rank regime. An exact weighted rank-five example has nonreal roots despite strict ultra-log-concavity. The classification concerns this full Hall family; arbitrary graphs with a two-plus- s minimum cover remain outside its scope.

Status. The results have ordinary mathematical proofs and reproducible exact algebraic checks. They are not formally verified in Lean or externally refereed. The contribution is compared with the inspected ProveIt Hall-bottleneck report; global priority is not asserted. [*Merge note:* The inspected Hall-bottleneck report [9] is Part XI of *preorder-root-polytopes* (pinned at 4a3b6991d). The positive half of Theorem 16.1 is the full Hall case of source 74’s Theorem 2.1: for $n \geq b$, $m \geq a$ and $\min(a, b) \leq 2$ the core $A \sqcup B$ is a minimum cover with at most two vertices on one shore (for $\min(a, b) = 1$ it is also Part XIII’s Theorem 206.2). Source 75’s comparison-polynomial proof is printed as a second route, with what it adds: strictness when $\min(a, b) = 2$, real-rootedness when $\min(a, b) = 1$, and the nonreal rank-five example. For $a = b$ and $n = m$ the negative half is Part XI’s endpoint theorem (Theorem 178.2 of *preorder-root-polytopes*, `hb:thm:endpoint`), and the population offset h below gives $n = 33$ and $n = 11$ at $s = 3, 4$, Part XI’s minimal outer sizes; it is new for $a \neq b$ (Research question 89 of *preorder-root-polytopes*).]

16.1 The family and the classification

For positive integers a, b , define $H_{a,b;n,m}$ by the disjoint shores

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = a, \quad |B| = b, \quad |L| = n, \quad |R| = m,$$

and the exact edge set

$$(A \times B) \cup (L \times B) \cup (A \times R). \tag{34}$$

Thus every exterior-left vertex is adjacent to *all* of B , every exterior-right vertex to *all* of A , the core $A \times B$ is complete, and no exterior pair is adjacent. Assign an arbitrary positive activity to each vertex. Its support polynomial is

$$P_H(t) = \sum_{(I,J) \text{ feasible}} u_I v_J t^{|I|} = \sum_k p_k t^k,$$

where $I \subseteq X$, $J \subseteq Y$, $|I| = |J|$, and feasibility means that the induced bipartite graph admits a perfect matching. A pair of endpoint sets contributes once, regardless of how many matchings realize it.

For a nonnegative degree- d coefficient sequence with no internal zeros, write ULC_d for log-concavity of $p_k / \binom{d}{k}$. Equivalently,

$$\Delta_k^{(d)}(P) := k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1} \geq 0 \quad (1 \leq k < d). \quad (35)$$

For positive activities the degree equals the maximum matching size. If $n \geq b$ and $m \geq a$, this rank is $d = a + b$: the core is a cover of that size, attained by matching each core into the opposite exterior block.

Theorem 16.1 (Sharp full-Hall classification). *Fix positive core sizes a, b . The following are equivalent.*

1. *For all $n \geq b$, $m \geq a$ and every positive activity assignment, $P_{H_{a,b,n,m}}$ is ULC_{a+b} .*
2. $\min(a, b) \leq 2$.

If $\min(a, b) = 2$, every internal inequality is strict. If $a, b \geq 3$, the failure of the last inequality can be realized with activity one at every exterior vertex and one common positive integer activity at every core vertex; explicit parameters are given in Section 16.5.

The positive assertion also extends to smaller exterior populations with normalization by the actual matching rank, as explained in Section 16.4. Zero activities are handled there separately. Nothing in the negative assertion concerns unit weights or activities on just one shore. [Merge note: Those cases are treated by sources 71, 73, 55, 47 and 51 in Part II.]

16.2 A uniform formula for a two by s core

Assume $a = 2$, $b = s$, $n \geq s$ and $m \geq 2$, and put $d = s + 2$. Normalize both exterior activity sums to one by dividing every left activity by $e_1(L)$ and every right activity by $e_1(R)$. This only replaces $P_H(t)$ by $P_H(ct)$ for a positive c , and multiplies its k th margin by c^{2k} .

Let u_1, u_2 be the normalized left-core activities, $v_1, \dots, v_s$ the normalized right-core activities, and define

$$U_1 = u_1 + u_2, \quad U_2 = u_1 u_2, \quad V_j = e_j(v_1, \dots, v_s), \quad E_j = e_j(L), \quad z = e_2(R).$$

Here $E_0 = E_1 = 1$, $0 \leq z \leq 1/2$, and every out-of-range elementary symmetric function is zero.

A size- k support selecting i left-core and j right-core vertices is feasible exactly when $i + j \geq k$. Necessity follows because its $k - i$ exterior-left vertices must be matched into its j selected right-core vertices. Conversely match those exterior vertices first, then the selected exterior-right vertices into the left core, and finally the remaining core vertices to one another. Therefore

$$p_k = E_k V_k + U_1 E_{k-1} (V_k + V_{k-1}) + U_2 E_{k-2} (V_k + V_{k-1} + z V_{k-2}). \quad (36)$$

Deleting the $2s$ core edges gives two disjoint complete bipartite graphs. Their support polynomial Q has coefficients

$$q_k = E_k V_k + U_1 E_{k-1} V_{k-1} + U_2 E_{k-2} z V_{k-2}. \quad (37)$$

The proof will establish the stronger comparison

$$\Delta_k^{(d)}(P_H) - \Delta_k^{(d)}(Q) \geq 0 \quad (1 \leq k < d). \quad (38)$$

Lemma 16.2 (The comparison polynomial). *The polynomial Q has degree d and only negative real zeros. Its order- d Newton inequalities are strict.*

Proof. For a complete bipartite graph with left activities $\alpha_1, \dots, \alpha_p$ and right activities $\beta_1, \dots, \beta_q$, its support polynomial is

$$\left(\prod_i \alpha_i \right) e_p(\alpha_1^{-1}, \dots, \alpha_p^{-1}, t\beta_1, \dots, t\beta_q).$$

Elementary symmetric polynomials are real stable: differentiate $\prod_i (w + x_i)$ in w the required number of times and specialize $w = 0$. Differentiation, positive scaling, and real specialization preserve stability or give zero [4]. The displayed univariate polynomial is nonzero with positive constant term and nonnegative coefficients, so its zeros are negative and real.

Apply this to $K_{2,m}$ and $K_{n,s}$. Their product has degree $2 + s = d$. The first factor is

$$1 + U_1 t + z U_2 t^2.$$

Its discriminant obeys

$$U_1^2 - 4zU_2 \geq (u_1 - u_2)^2 + 2u_1 u_2 > 0,$$

so Q has at least two distinct negative zeros. Newton's inequalities are strict at every internal index unless all roots coincide. This proves the lemma. $\square$

16.3 Twelve coefficient families

Every core variable has exponent at most two in the difference in (38). The possible nonzero left exponent patterns, up to swapping u_1, u_2 , are

$$10, \quad 20, \quad 11, \quad 21, \quad 22.$$

Fix one representative monomial with left pattern λ . Suppose its right-core exponents have p twos and q ones, and put

$$a = k - p, \quad j = 2a - q, \quad \ell = d - k, \quad A = k\ell, \quad B = (k+1)(\ell+1).$$

The letters A, B in this section are numerical gap factors, not vertex blocks. A nonzero increment coefficient has

$$0 \leq j < |\lambda|, \quad k \geq a, \quad a \geq \lceil j/2 \rceil, \quad \ell \geq a - j + 2. \quad (39)$$

Indeed the total right-core degree is $2k - j$; a core-edge contribution raises total core degree above $2k$, giving $j < |\lambda|$. Also the number of right-core variables occurring is

$$p + q = k + a - j \leq s = d - 2,$$

which gives the last inequality. These restrictions leave exactly twelve pattern- j families.

Divide the coefficient by the following positive central binomial factor:

$$C_0 = \binom{2a}{a}, \quad C_1 = \binom{2a-1}{a-1}, \quad C_2 = \binom{2a-2}{a-1}, \quad C_3 = \binom{2a-3}{a-2}.$$

Each is used only when its indices are allowed by (39). To derive the formulas below, the coefficient of a monomial with p twos and q ones in $V_i V_h$ is

$$\binom{q}{i-p} \quad (i+h=2p+q), \quad (40)$$

with the usual zero convention. The repeated variables occur in both subsets; one chooses which of the q single-occurrence variables belong to the first subset. Thus all twelve formulas follow by direct substitution of (36) and (37) into (35), followed by elementary ratios of binomial coefficients. No matching multiplicities are introduced.

We use these consequences of Newton's inequalities:

$$\begin{aligned} E_{k-2}E_{k+1} &\leq \frac{k-1}{k+1}E_{k-1}E_k, \\ E_{k-2}E_k &\leq \frac{k-1}{k}E_{k-1}^2, & E_{k-3}E_{k+1} &\leq \frac{k-2}{k+1}E_{k-2}E_k, \\ E_{k-3}E_k &\leq \frac{k-2}{k}E_{k-2}E_{k-1}, & E_{k-3}E_{k-1} &\leq \frac{k-2}{k-1}E_{k-2}^2. \end{aligned} \quad (41)$$

For $k \geq 2$ these also cover boundary cases by the zero convention. They follow, for example, from log-concavity of $j!E_j$. The case $k=1$ is checked at the end of the argument.

16.3.1 Pattern 10

Only $j=0$ occurs. The normalized coefficient is

$$2AE_kE_{k-1} - B\frac{a}{a+1}(E_kE_{k-1} + E_{k-2}E_{k+1}).$$

Using the first inequality in (41), it is at least

$$\frac{2k(\ell-a)}{a+1}E_kE_{k-1} \geq 0,$$

since $\ell \geq a+2$.

16.3.2 Pattern 20

Put $X = E_{k-1}^2$ and $Y = E_{k-2}E_k$. For $j=0$ the normalized coefficient is

$$AX - B\frac{a}{a+1}Y;$$

for $j=1$ it is twice this expression. The bound $Y \leq (k-1)X/k$ reduces its sign to

$$k(a+1)\left(A - B\frac{a}{a+1}\frac{k-1}{k}\right) = k^2(\ell-a) + a(\ell+1) \geq 0.$$

Here $\ell \geq a+1$ in both cases.

16.3.3 Pattern 11

Keep X, Y and put $Z = E_{k-3}E_{k+1}$. The two normalized coefficients are

$$\begin{aligned} j = 0 : \quad & 2A(X + Y) - B \frac{a}{a+1}(X + 2Y + Z), \\ j = 1 : \quad & 4AX + 2AY - B \left(X + \frac{4a}{a+1}Y + \frac{a-1}{a+1}Z \right). \end{aligned}$$

Use $Z \leq (k-2)Y/(k+1)$, and write the resulting lower bound as

$$\alpha \left(X - \frac{k}{k-1}Y \right) + \beta Y. \quad (42)$$

For $j = 0$ the two factors are

$$\alpha = 2A - \frac{Ba}{a+1} \geq 0, \quad \beta = \frac{2k(2k-1)(\ell-a)}{(k-1)(a+1)} \geq 0.$$

The sign of α is immediate for $a = 0$; for $a \geq 1$, use $k \geq a$ and $\ell \geq a+2$ in $A/B = k\ell/((k+1)(\ell+1))$.

For $j = 1$ one has $a \geq 1$, $\ell \geq a+1$, and

$$\alpha = 4A - B \geq 0,$$

while

$$\begin{aligned} \frac{(k-1)(a+1)}{2} \beta &= \ell(3k^2 - 3k + 1 + a) \\ &\quad - a(3k^2 - k - 1) - 2k + 1. \end{aligned}$$

The right side increases with ℓ . At $\ell = a+1$ it becomes

$$(a-k+1)(a-k+2) + 2k(k-1) \geq 0,$$

because the first product consists of consecutive integers. Equation (42) and Newton's bound now prove both cases.

16.3.4 Pattern 21

Put $M = E_{k-2}E_{k-1}$ and $N = E_{k-3}E_k$. The normalized coefficient is

$$AMS_j - B(MH_j + NJ_j),$$

where

j	S_j	H_j	J_j
0	2	$\frac{H_j}{a}$	$\frac{J_j}{a}$
1	4	$\frac{a+1}{2a}$	$\frac{a+1}{2a}$
2	$2 + \frac{2z(a-1)}{a}$	$z + \frac{a-1}{a}$	$\frac{a-1}{a} + \frac{z(a-1)(a-2)}{a(a+1)}$

Newton gives $N \leq \nu M$, where $\nu = (k-2)/k$. For $j = 0, 1$,

$$\frac{H_j + \nu J_j}{S_j} = \frac{a}{a+1} \frac{k-1}{k} \leq \frac{a}{a+1} \frac{k}{k+1} \leq \frac{A}{B},$$

using $\ell \geq a$. These two coefficients are nonnegative.

For $j = 2$, $a \geq 1$ and $\ell \geq a$. Define

$$D(\ell, z) = k\ell S_2 - (k+1)(\ell+1)(H_2 + \nu J_2).$$

It is affine in each of ℓ, z . Direct simplification gives

$$D(a, 0) = \frac{2(a^2 + k^2 - 1)}{ak} \geq 0, \quad D(a, \tfrac{1}{2}) = \frac{(k-a)^2 + (k-a) + 2a(a-1)}{ak} \geq 0.$$

Thus $D(a, z) \geq 0$ on $0 \leq z \leq 1/2$. Since $H_2 + \nu J_2 \geq 0$,

$$\frac{\partial D}{\partial \ell} = \frac{D(a, z) + (k+1)(H_2 + \nu J_2)}{a} \geq 0.$$

Therefore $D(\ell, z) \geq 0$ for all $\ell \geq a$, proving the last case.

16.3.5 Pattern 22

Put $U = E_{k-2}^2$ and $W = E_{k-3}E_{k-1}$. The normalized coefficient is

$$AUS_j - BWT_j,$$

with

j	S_j	T_j
0	1	$\frac{1}{a+1}$
1	2	$\frac{a+1}{2a}$
2	$1 + \frac{2z(a-1)}{a}$	$\frac{a-1}{a} + \frac{2z(a^2 - a + 1)}{a(a+1)}$
3	$2z$	$\frac{2z(a-1)}{a}$

For $j \leq 2$, $T_j/S_j \leq a/(a+1)$. In the $j = 2$ row this follows by checking $z = 0, 1/2$ in the affine cross-multiplied inequality; the endpoint ratios are $(a-1)/a$ and $a/(a+1)$. For $j = 3$ and $z > 0$ the ratio is $(a-1)/a$; at $z = 0$ the row is zero.

Now

$$W \leq \frac{k-2}{k-1}U, \quad \frac{k-2}{k-1} \leq \frac{k}{k+1}.$$

When $j \leq 2$, feasibility gives $\ell \geq a$, so A/B dominates the product of the moment ratio and $a/(a+1)$. When $j = 3$, use $\ell \geq a-1$ and the ratio $(a-1)/a$ instead. All four cases follow.

16.3.6 The small index boundary and completeness

At $k = 2$, all displayed arguments remain valid with $E_{-1} = 0$. At $k = 1$, patterns 21 and 22 vanish. For patterns 10 and 11 with $j = 0$ the normalized coefficient is $2A - Ba/(a+1)$; here $a = 0$ or 1 and $\ell \geq a+2$, making it nonnegative. Pattern 20 gives A or $2A$, and pattern 11 with $j = 1$ gives $4A - B \geq 0$. Thus the boundary is covered without division by $k-1$.

The twelve cases exhaust every nonzero monomial coefficient of the increment in the independent positive core activities. Every coefficient is nonnegative, proving (38). Lemma 16.2 then proves strict ULC_{s+2} for every full two-by- s Hall graph in the genuine-rank regime.

16.4 Singleton cores and deficient exterior populations

A singleton-core case has an elementary stable-matroid description. Suppose $a = 1$. In the extended transversal matroid, let

$$F = \bar{A} \sqcup R, \quad O = \bar{L} \sqcup B.$$

Its rank is $n + 1$, and its bases are exactly the $(n + 1)$ -element subsets containing at most one element of F . Indeed F can only be matched to the one core vertex, and every selected element of B is available for any remaining left position. Its basis polynomial is

$$\mathcal{B} = e_{n+1}(O) + e_1(F)e_n(O).$$

Its two-block diagonal is $y^n(c_0y + c_1x)$ for positive constants c_0, c_1 . This is real stable, and polarization separately in the two blocks recovers $\mathcal{B}$, preserving stability [4]. Setting each private variable to the inverse left activity and each right variable to t times its activity recovers P_H , up to a positive factor. Thus P_H is real-rooted and satisfies Newton's inequalities. Exchanging shores handles $b = 1$.

For $\min(a, b) \leq 2$ but insufficient exterior populations, the matching rank is

$$\min\{a + b, n + a, m + b\}.$$

The support criterion proves this formula directly: for every k below all three bounds, choose $i = \min(a, k)$ and $j = \min(b, k)$ selected core vertices. If $n < b$ or $m < a$, the formula equals the smaller shore size. The established weighted smaller-shore ULC theorem therefore applies [7, 1]. [*Merge note:* This is *preorder-root-polytopes*'s Theorem 70.1 (Part V), source 74's Lemma 3.1.] Its weighted form follows by specializing the basis polynomial of the extended transversal matroid at private variables s/u_x and right variables tv_y , then using the bivariate Lorentzian criterion. Hence Theorem 16.1's positive direction holds for every exterior population when the actual rank is used.

For nonnegative activities, delete the zero-activity vertices. The graph remains a full Hall graph with cores of size at most their original sizes. An empty core leaves a complete bipartite graph and isolates; singleton cores and all other cases are covered above. Thus the positive direction remains true with normalization by the surviving degree, not merely by its original value.

16.5 Explicit counterexamples when both cores are large

[*Merge note:* Source 74 (Section 6) prints the same family with the populations $N = b + 30$, $M = a + 30$; source 75's offset h is a second choice, and Table 3 records its first failing integer activities for seven shapes.] Now let $a, b \geq 3$, and set

$$r = a + b, \quad K = (r - 1)ab, \quad H = (r + 1)ab - 2r(r - 1).$$

Since $ab \geq 3(r - 3)$ and $r \geq 6$,

$$H \geq r^2 - 4r - 9 > 0.$$

Choose the explicit integer

$$h = 1 + \left\lceil \frac{2K}{H} \right\rceil, \quad n = b - 1 + h, \quad m = a - 1 + h.$$

Give every exterior vertex activity one and every core vertex activity $w > 0$. Put

$$A_i = \binom{n}{b-i}, \quad B_i = \binom{m}{a-i} \quad (i = 0, 1, 2),$$

and define

$$\begin{aligned} X &= aA_0B_1 + bA_1B_0, \\ Y &= aA_1B_2 + bA_2B_1, \\ Z &= \binom{a}{2}A_0B_2 + abA_1B_1 + \binom{b}{2}A_2B_0. \end{aligned}$$

The top three support coefficients are exactly

$$\begin{aligned} p_r &= A_0B_0w^r, \\ p_{r-1} &= A_1B_1w^r + Xw^{r-1}, \\ p_{r-2} &= A_2B_2w^r + Yw^{r-1} + Zw^{r-2}. \end{aligned}$$

They are obtained by omitting zero, one or two core vertices in the general support formula. Consequently

$$\Delta_{r-1}^{(r)}(P_H) = w^{2r-2}(\alpha w^2 + \beta w + \gamma), \quad (43)$$

where

$$\begin{aligned} \alpha &= (r-1)A_1^2B_1^2 - 2rA_0A_2B_0B_2, \\ \beta &= 2(r-1)A_1B_1X - 2rA_0B_0Y, \\ \gamma &= (r-1)X^2 - 2rA_0B_0Z. \end{aligned}$$

The leading coefficient is strictly negative. Indeed

$$\frac{A_1^2B_1^2}{A_0A_2B_0B_2} = \frac{ab}{(a-1)(b-1)} \left(1 + \frac{1}{h}\right)^2,$$

while the choice of h gives

$$\frac{H}{K} \geq \frac{2}{h-1} > \frac{2}{h} + \frac{1}{h^2}.$$

Therefore this coefficient ratio is strictly below $2r/(r-1)$, exactly the threshold for $\alpha < 0$.

For a completely finite activity choice, take

$$w = 1 + \left\lfloor \frac{|\beta| + |\gamma|}{-\alpha} \right\rfloor.$$

Then $w \geq 1$ and

$$\alpha w^2 + \beta w + \gamma \leq \alpha w^2 + (|\beta| + |\gamma|)w < 0.$$

This gives an explicit positive-integer two-shore activity counterexample for every $a, b \geq 3$, completing the negative direction of Theorem 16.1. The activity bound is conservative and is not claimed minimal.

16.6 A positive theorem does not imply real rootedness

The two-by- s conclusion is genuinely an ultra-log-concavity theorem. For $s = 3$, $n = m = 9$, activity 100 at all five core vertices and activity one at every exterior vertex, replacing t by $t/100$ gives

$$F(t) = 1 + 645t + 38730t^2 + 124500t^3 + 37800t^4 + 3024t^5.$$

Its discriminant is

$$-4431992565218959594687680000000 < 0,$$

so not all roots are real. Exact root counting gives three real roots. Its four order-five margins are nevertheless

$$1276800, \quad 8036447400, \quad 75433572000, \quad 1950480000,$$

all strictly positive. The program `verify_separation.py` [*Merge note:* shipped as `code/75-full-hall-verify_separation.py`] regenerates these integers from the graph parameters.

16.7 Verification and the remaining questions

The accompanying exact programs generate the twelve normalized coefficient families directly from (36)–(37). A separate check proves their signs by substituting the Newton bounds, reducing to corners of multilinear boxes, and verifying nonnegative polynomial coefficients in the free integer slacks. Denominator positivity and the $k = 1, 2$ boundary are checked explicitly. The generic extraction also reproduces all 49 orbit coefficients of the two-by-three case and all 80 of the two-by-four case exactly.

The readable argument above is uniform in s ; it is not an inference from finitely many core sizes. Exact scripts audit algebraic identities and boundary bookkeeping. They do not constitute a Lean formalization, and numerical root approximations are not used in any proof.

The existing Hall-bottleneck report [9] already disproves universal weighted rank normalization using a three-by-three core and studies symmetric higher cores. Theorem 16.1 gives a sharp boundary for the *full* Hall-core shapes, including all unequal cores and arbitrary independent positive activities in the positive range. Arbitrary graphs with a two-plus-three minimum cover are still not classified here: missing core edges and proper exterior neighbourhoods need different arguments. In particular, this report does not settle whether the first weighted counterexample among all bipartite graphs has rank five or six, and gives no negative result for unit weights or weights confined to one shore. [*Merge note:* Settled within the batch: the first weighted counterexample rank is exactly six (source 74, Corollary 2.2); unit-weight and one-shore counterexamples are sources 71 and 73 (Part II). Source 75’s delivered README, which is not shipped, likewise says that general weighted rank five remains open “here”. Arbitrary graphs with a $2 + 3$ minimum cover, which source 75 leaves unclassified, are covered by source 74’s Theorem 2.1.]

Part II

Unit weights and one weighted shore: counterexamples and the rank-38 threshold

Contents of Part II. Part II prints five sources. Source 71 (Section 17) gives a connected all-unit counterexample, answering Research question 84 of *preorder-root-polytopes* negatively. Source 73 (Section 18) gives one-shore counterexamples (all left activities one), answering Research question 85 there negatively. Sources 55, 47 and 51 (Sections 19, 20 and 21) study the full Hall graphs with *one common activity on the right core* and one elsewhere: the last gap first fails at matching number 38 (55), so does full ULC (47), and at matching number 49 the last gap no longer controls the others (51).

Which weighting. Every section title of this Part states its weighting. Source 71 is unit weighted; sources 73, 55, 47 and 51 are one-shore weighted, and the last three only with one common activity on the right core. None of them answers Research question 88 of *preorder-root-polytopes*, which concerns the two-shore family $H_{s,n}$ with weight on both cores; sources 47 and 51 answer its one-shore analogue.

Same theorem in several sources. Source 71's unit pendant-arm identity is the case $T = 1$ of source 73's weighted identity; it is printed once, in the weighted form, in source 73's appendix. Source 47 recalls source 55's endpoint theorem without proof (kept as a recall, with pointers) and re-proves the first-gap inequality by the Motzkin–Straus route (replaced by a pointer to Part XI of *preorder-root-polytopes*; its strictness step is kept). Source 55 extends source 73: the top-three formula and the rank-38 construction of 73 reappear in 55 with an exact threshold. The smaller-shore steps of sources 55 and 47 restate *preorder-root-polytopes*'s Theorem 70.1.

Non-claims of Part II. Source 71 does not determine the least matching number, vertex count or edge count of a unit counterexample, and keeps the smaller-shore bound and ordinary log-concavity; source 73 makes no minimal-rank claim; sources 55, 47 and 51 concern one common right-core activity on full Hall graphs only, and not arbitrary one-shore activities; source 55 concerns the last gap only.

17 Unit weight failure of matching rank normalization (source 71)

Abstract of source 71, as delivered. We give a connected bipartite graph whose unweighted matching-support polynomial fails ultra-log-concavity normalized by its maximum matching size. The explicit graph has 6916 vertices, 174228 edges and matching rank 3450. Its last Newton margin is a strictly negative integer, certified by a short exact formula for its top three coefficients. Every vertex activity is one. A second example has matching rank exactly one below the size of the smaller shore, showing that even this smallest possible discrepancy can obstruct rank normalization. The construction uses pendant length-two arms and admits exact support counting without enumerating the large graphs' matchings.

Status and scope. The counting identities and the counterexamples below are established in ordinary mathematics and independently checked in exact arithmetic. This is an unrefereed

research note, not a Lean-certified result. Its claim of resolving a question is relative to the inspected ProveIt manuscript *Hall Bottlenecks and the Limits of Rank Normalization* [9]; worldwide priority is not asserted. The examples do not determine the smallest possible matching rank or number of vertices of a counterexample.

17.1 The question and the counterexample

For a finite simple bipartite graph $G = (X, Y; E)$, define

$$p_G(t) = \sum_{k \geq 0} p_k t^k,$$

where p_k is the number of pairs $I \subseteq X$, $J \subseteq Y$ with $|I| = |J| = k$ for which $G[I, J]$ has a perfect matching. A feasible pair is counted once, regardless of its number of perfect matchings. This is the bipartite form of the perfectly matchable set polynomial; see [5] for earlier work on this polynomial. In particular $p_{K_{2,2}}(t) = 1 + 4t + t^2$.

For a polynomial of degree R with positive coefficients, rank-normalized ultra-log-concavity means that $p_k / \binom{R}{k}$ is log-concave. Equivalently,

$$k(R - k)p_k^2 \geq (k + 1)(R - k + 1)p_{k-1}p_{k+1} \quad (1 \leq k < R). \quad (44)$$

Here the proposed normalization is $R = \nu(G)$, the maximum matching size. The smaller-shore normalization is known to hold [7, Theorem A]; replacing the smaller shore by $\nu(G)$ is a genuinely stronger assertion. The unit-weight version was explicitly left open in [9, question “Unit weights”]. [*Merge note:* That question is Research question 84 of *preorder-root-polytopes* (`hb:q:unit`); the smaller-shore normalization is its Theorem 70.1 (Part V). The positive unit-weight theorems there, Theorem 191.1 ($\nu \leq 3$), Theorem 196.1 (a rank-four family) and Theorem 206.2 (singleton covers), and Theorem 2.1 of Part I, do not apply to the graph below.]

Theorem 17.1. *There is a connected simple bipartite graph with all vertex activities equal to one, 6916 vertices, 174228 edges and matching rank $R = 3450$, for which (44) fails at $k = 3449$.*

The complete graph definition, the rank certificate, and the exact negative margin are given below. No external field or limiting activity is used in this theorem. [*Merge note:* Theorem 17.1 answers Research question 84 of *preorder-root-polytopes* negatively, and with it the unit-weight half of its Research question 37 and its Research question 94 (no all-rank proof at unit weights can exist). By Corollary 5.1, every minimum cover of this graph meets both shores in at least three vertices; the cover of Lemma 17.2 has $12 + 3040 = 3052$ vertices on the left and 398 on the right.]

17.2 A graph family with pendant arms

Let $a, b, N, M \geq 1$, with $N \geq b$ and $M \geq a$, and let $W \geq 2$ be an integer. Start with left classes A, L of sizes a, N and right classes B, C of sizes b, M . Include all edges in

$$A \times B, \quad A \times C, \quad L \times B,$$

and no edges in $L \times C$. At each $c_j \in C$, attach $W - 1$ new length-two arms

$$c_j - z_{j,h} - \ell_{j,h} \quad (1 \leq h < W),$$

where $z_{j,h}$ is a new left vertex and $\ell_{j,h}$ is a new right leaf. There are no further edges. Write

$$K = M(W - 1), \quad r = a + b, \quad R = r + K.$$

Denote the resulting graph by $G(a, b, N, M, W)$. *Every activity is one; W specifies an arm count and is not a vertex weight.*

Lemma 17.2 (Size and rank). *The two shore sizes of $G(a, b, N, M, W)$ are $a + N + K$ and $b + M + K$. Its edge count is $ab + aM + bN + 2K$, and its matching rank is $R = a + b + K$.*

Proof. Match every vertex of A to a distinct vertex of C , every vertex of B to a distinct vertex of L , and every new center $z_{j,h}$ to its leaf $\ell_{j,h}$. This matching has $a + b + K$ edges. The set consisting of A , B , and all new centers is a vertex cover of that same size. The size and edge formulas follow directly from the construction. Connectivity holds because the original three complete blocks form a connected graph, and all new vertices lie on attached arms. $\square$

17.3 Exact support counting

Lemma 17.3 (The unit pendant-arm identity). *Let a vertex x of a bipartite graph have $k \geq 1$ new pendant length-two arms attached to it, all with unit activities. Write $p_G = f + g$, where f counts original supports omitting x and g counts original supports containing x . Then the augmented support polynomial is*

$$(1+t)^{k-1}[(1+(k+1)t)f + (1+t)g]. \quad (45)$$

Proof (pointer). This is the case $T = 1$ of source 73's weighted pendant-arm identity, Lemma 18.6 (Section 18.5), with the two shores exchanged: there the arms hang at a left vertex and their centers carry activity T . [Merge note: The same theorem in two sources: source 71's proof is source 73's proof at $T = 1$ (supports containing x contribute $(1+t)^k g$; for supports omitting x , choosing j centers gives $j+1$ distinct endpoint sets, whence $\sum_j \binom{k}{j} (j+1)t^j f = (1+t)^{k-1}(1+(k+1)t)f$; no matching multiplicities are counted). It is printed once, in the weighted form.] $\square$

For the original three-block Hall graph, mark the number ρ of selected vertices of C . Its corresponding support polynomial is

$$P_\rho(t) = \binom{M}{\rho} \sum_{\substack{0 \leq i \leq a, \\ 0 \leq j \leq b, \\ i \geq \rho}} \binom{a}{i} \binom{b}{j} \binom{N}{j+\rho-i} t^{j+\rho}, \quad (46)$$

where out-of-range binomial coefficients are zero. Indeed, i and j count selected core vertices in A and B ; balance requires $j + \rho - i$ selected vertices in L . Feasibility is precisely $i \geq \rho$: match the exterior vertices first, then match the remaining core vertices across the complete core. Applying Lemma 17.3 at all M vertices of C gives

$$p_{G(a,b,N,M,W)}(t) = \sum_{\rho=0}^a P_\rho(t) (1+t)^{K-M+\rho} (1+Wt)^{M-\rho}. \quad (47)$$

The factor $1 + Wt$ records W different endpoint sets; it does not assign an activity to a vertex.

17.3.1 A short formula for the top three coefficients

For $a, b \geq 2$ and $\delta \in \{0, 1, 2\}$, let e be the deficit of the original support size from $r = a + b$. If u left-core and v right-core vertices are omitted, then

$$u + v \leq e, \quad \ell = b - e + u, \quad \rho = a - e + v$$

are the exterior-left and exterior-right support sizes. Put $A_\rho = K - M + \rho$ and $B_\rho = M - \rho$. Extracting the top coefficients in (47) yields the entirely integer formula

$$p_{R-\delta} = \sum_{e=0}^{\delta} \sum_{u=0}^e \sum_{v=0}^{e-u} \binom{a}{u} \binom{b}{v} \binom{N}{b-e+u} \binom{M}{a-e+v} \cdot \sum_{z=0}^{\min(B_\rho, \delta-e)} \binom{A_\rho}{\delta-e-z} \binom{B_\rho}{z} W^{B_\rho-z}. \quad (48)$$

To see the last line, the pendant factors have degree K . Selecting their coefficient of degree $K - (\delta - e)$ means omitting $\delta - e$ of their linear terms, with z omissions from the B_ρ factors of slope W . Thus (48) counts supports exactly and requires only a few summands.

17.4 The finite certificate

Take

$$(a, b, N, M, W) = (12, 398, 410, 16, 191).$$

Thus $K = 3040$, the shore sizes are 3462 and 3454, and Lemma 17.2 gives $R = 3450$ and 174228 edges. The graph has 6916 vertices in total.

Define the positive integer

$$D = 160160547610973088925200.$$

Formula (48) gives the compact exact certificate

$$\begin{aligned} p_{3450} &= 455D \, 191^4, \\ p_{3449} &= 3020455956D \, 191^3, \\ p_{3448} &= 10022638278784267D \, 191^2. \end{aligned} \quad (49)$$

The required endpoint inequality is $3449p_{3449}^2 \geq 6900p_{3448}p_{3450}$. Its margin is

$$\begin{aligned} &3449p_{3449}^2 - 6900p_{3448}p_{3450} \\ &= D^2 \, 191^6 (3449 \cdot 3020455956^2 - 6900 \cdot 455 \cdot 10022638278784267) \\ &= -314102056577041236 \, D^2 \, 191^6 < 0. \end{aligned}$$

This proves Theorem 17.1. The exact coefficient formula, together with the displayed finite integer substitution, is the proof of the counterexample; the search that led to the parameters is not used as a premise.

17.5 Failure when the smaller shore exceeds rank by one

The phenomenon does not require a large gap between the smaller shore and the matching rank.

Theorem 17.4. *There is a connected unit-weight bipartite graph whose smaller shore has size 125509, whose matching rank is 125508, and whose support polynomial is not ULC_{125508} .*

Proof. Take $G(8, 100, 101, 30, 4181)$. Then $K = 125400$, $R = 125508$, the shore sizes are 125509 and 125530, and the graph has 251039 vertices and 261940 edges. Put $D_1 = 8567325$. The same formula (48) gives

$$\begin{aligned} p_R &= 69D_1 \, 4181^{22}, \\ p_{R-1} &= 61945839672D_1 \, 4181^{21}, \\ p_{R-2} &= 27806230344578726426D_1 \, 4181^{20}. \end{aligned}$$

Consequently

$$(R-1)p_{R-1}^2 - 2Rp_{R-2}p_R = -415296714996000603216 D_1^2 4181^{42} < 0.$$

The rank and the two shore sizes follow from Lemma 17.2. $\square$

The smaller-shore theorem [7] still guarantees ULC_{125509} for this graph. The conclusion is failure of the stronger order 125508, not failure of that published bound.

17.5.1 Why deficiency one is not an isolated accident

Fix a, b, N, M and let W grow, still keeping every vertex activity one. Write

$$h = N - b + 1, \quad \theta = \frac{b}{h}, \quad \zeta = \frac{b(b-1)}{h(h+1)}, \quad \mu = \frac{a}{M-a+1}, \quad \eta = \frac{a(a-1)}{(M-a+1)(M-a+2)}.$$

Expanding (48), with $R = a + b + M(W-1)$, gives

$$\frac{(R-1)p_{R-1}^2 - 2Rp_{R-2}p_R}{p_R^2 W^3} \longrightarrow M \left[\mu^2(a+\theta)^2 - 2\eta \left(\binom{a}{2} + a\theta + \zeta \right) \right]. \quad (50)$$

For example, the leading terms of p_{R-1}/p_R and p_{R-2}/p_R are respectively

$$W(M + \mu(a+\theta)), \quad W^2 \left[\frac{M^2}{2} + M\mu(a+\theta) + \eta \left(\binom{a}{2} + a\theta + \zeta \right) \right],$$

which imply (50) directly. Set $h = 2$, so $N = b + 1$. After division by μ^2 , the bracket in (50) tends as $M \rightarrow \infty$ to

$$2a - 1 + \frac{4a-1}{3a}b - \frac{a-4}{12a}b^2.$$

For any fixed $a \geq 5$, choose b sufficiently large, then choose a finite $M \geq a + 2$ sufficiently large. The bracket is negative, so all sufficiently large integers W give counterexamples. Their smaller shore is always $R + 1$. This proves existence of infinitely many deficiency-one examples; Theorem 17.4 supplies one explicit finite instance.

17.6 Consequences for Ehrhart h^* -polynomials

For either constructed graph $G = (X, Y; E)$, form $\widehat{G}$ by adding a new left vertex v , a new right vertex w , and the edges vy for $y \in Y$ and xw for $x \in X \cup \{v\}$. Let

$$Q_{\widehat{G}} = \text{conv}\{e_i + e_j : ij \in E(\widehat{G})\},$$

and let $\text{STAB}(\overline{G})$ be the stable set polytope of the complement of G . Davis–Kohl [5, Theorem 3.10 and Lemma 3.11] give

$$h^*(Q_{\widehat{G}}; t) = p_G(t) = h^*(\text{STAB}(\overline{G}); t).$$

Thus these two specific Ehrhart h^* -polynomials have degree $R = \nu(G)$ and fail ULC_R . [*Merge note:* Compare source 74's corollary in Section 7, which gives ULC_r for these polynomials when $\nu \leq 5$ or a minimum cover meets one shore in at most two vertices.] They also have nonreal zeros: otherwise their nonnegative coefficients would satisfy the degree-normalized Newton inequalities. Their coefficients still satisfy ordinary log-concavity and the smaller-shore ULC bound by [7].

17.7 Verification and remaining questions

The package contains standard-library Python scripts [*Merge note:* shipped as `code/71-unit-weight-verify*.py`, with outputs `data/71-unit-weight-*.json`] checking the pendant identity by independent endpoint-set enumeration on 150 small graphs, including nonuniform activities as an extra regression test. For both displayed examples, they evaluate (48) by integer binomial sums and independently recompute the tails from normalized reciprocal sums of the pendant factors. They also check the exact negative margins, explicit matching and cover constructions, edge counts and coverage. All checks pass with fixed seeds. They count supports without matching multiplicity and supplement the written proof.

The minimum matching rank, vertex count and edge count of an all-unit counterexample remain undetermined. [*Merge note:* By Part I (Corollary 2.2) such a counterexample has matching number at least six, so the least unit failure rank lies in $[6, 3450]$.] Rank 3450 is the best found in a bounded parameter search, not a proved minimum. Smaller deficiency-one examples and structural classes retaining rank-normalized ultra-log-concavity remain natural questions. The published smaller-shore normalization and ordinary log-concavity still hold.

18 One-shore activities can destroy matching-rank ULC (source 73)

Abstract of source 73, as delivered. The restriction to activities on one shore does not restore matching-rank-normalized ultra-log-concavity of perfectly matchable endpoint supports. We give an explicit full Hall-bottleneck graph on 125 vertices, with matching rank 50, whose left activities are all one and whose right activities take only the values one and 10000. Its last Newton margin is a positive factor times the negative integer -39998827 . A second example has matching rank 38. Both certificates follow from a closed formula for the top three coefficients and a quadratic in the common right-core activity. Thus no numerical root calculation or limiting argument is needed to certify either finite counterexample. We also give a general criterion for constructing such examples and record a separate pendant-arm transfer construction.

Status and scope. These are ordinary mathematical proofs with reproducible exact-integer checks, not formal Lean verification or external peer review. The result resolves the one-shore-activity question in the inspected ProveIt Hall-bottleneck report negatively. It does not produce a unit-weight counterexample, identify the smallest possible rank, or assert exhaustive literature priority. [*Merge note:* The question is Research question 85 of *preorder-root-polytopes* (`hb:q:oneshore`), with exactly these hypotheses: all left activities one, right activities arbitrary positive. The weighting here, activity w on the right core B and one elsewhere, is the “one common right-core activity” of Section 1.3. The unit-weight counterexample is source 71’s (Section 17).]

18.1 The polynomial and two explicit counterexamples

For a bipartite graph $G = (X, Y; E)$ with positive vertex activities u_x, v_y , define

$$P_G(t) = \sum_{(I, J) \text{ feasible}} u_I v_J t^{|I|} = \sum_{k=0}^d p_k t^k.$$

Here $I \subseteq X$, $J \subseteq Y$, $|I| = |J|$, and feasibility means that $G[I, J]$ admits a perfect matching. Each pair of endpoint sets is counted once, regardless of the number of realizing matchings.

The degree d is the maximum matching size. This is the vertex-weighted perfectly matchable set polynomial, with t recording half the support size.

The property ULC_d requires log-concavity of $p_k/\binom{d}{k}$ and no internal zeros. For these positive-activity support polynomials there are no internal zeros, so the inequalities are

$$\Delta_k^{(d)} = k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1} \geq 0. \quad (51)$$

In particular, the last inequality is $(d-1)p_{d-1}^2 \geq 2dp_{d-2}p_d$.

Write $H_{a,b;n,m}$ for the graph with shores $X = A \sqcup L$, $Y = B \sqcup R$, block sizes a, n, b, m , and exactly the edges

$$(A \times B) \cup (L \times B) \cup (A \times R). \quad (52)$$

All exterior-left vertices are adjacent to all of B , all exterior-right vertices to all of A , and there are no exterior-to-exterior edges. Give every vertex activity one except the b vertices of B , each of which has activity $w > 0$. Thus the weighting is confined to one shore. If $n \geq b$ and $m \geq a$, the matching rank is $d = a + b$: the core is a cover of size d , attained by disjoint matchings $A \rightarrow R$ and $L \rightarrow B$.

Theorem 18.1 (A counterexample on 125 vertices). *Let $G = H_{8,42;44,31}$, with activity w on B and one elsewhere. Its matching rank is 50, and*

$$\Delta_{49}^{(50)} = 98 \left[\binom{44}{42} \binom{31}{7} \right]^2 w^{82} (-w^2 + 5958w + 421173). \quad (53)$$

At $w = 10000$ the parenthesized integer is $-39998827 < 0$. The first positive integer w for which this particular margin fails is 6028, where the integer is -787 .

This graph has 125 vertices and 2432 edges. Only 42 activities differ from one. The theorem is an exact certificate against universal one-shore ULC_d .

Theorem 18.2 (A counterexample of matching rank 38). *Let $G = H_{8,30;32,1000}$ with the same one-shore weighting. Put*

$$Q(w) = -2404w^2 + 4878807600w + 10888224221475.$$

Then

$$\Delta_{37}^{(38)} = \frac{3}{2272} \left[\binom{32}{30} \binom{1000}{7} \right]^2 w^{58} Q(w). \quad (54)$$

At $w = 3000000$, one has $Q(w) = -6988688975778525 < 0$. The first positive integer giving a negative value of this particular margin is 2031684.

No minimal-rank claim is made. The rank-38 example has 1070 vertices; the rank-50 example is a substantially smaller graph. [Merge note: Within full Hall graphs with one common right-core activity, source 55 (Section 19) proves that 38 is the least matching number at which the last gap can fail, with 864 vertices the fewest at that rank, and source 47 (Section 20) extends this to every gap.]

18.2 Endpoint counting gives the complete certificate

A size- k support selecting i vertices of A and j vertices of B is feasible exactly when $i + j \geq k$. Necessity follows because the $k - i$ selected exterior-left vertices must match into the j selected

vertices of B . For sufficiency, first match the exterior-left vertices into B , then the exterior-right vertices into A , and finally the remaining core vertices to each other. Consequently

$$p_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{n}{k-i} \binom{b}{j} \binom{m}{k-j} w^j, \quad (55)$$

with out-of-range binomial coefficients set to zero.

Lemma 18.3 (The top three coefficients). *Let $a, b \geq 2$, $d = a + b$, $n \geq b$, $m \geq a$, and set*

$$X_i = \binom{n}{b-i}, \quad Y_i = \binom{m}{a-i} \quad (i = 0, 1, 2).$$

Then

$$p_d = X_0 Y_0 w^b, \quad (56)$$

$$p_{d-1} = (X_1 + aX_0)Y_1 w^b + bX_1 Y_0 w^{b-1}, \quad (57)$$

$$\begin{aligned} p_{d-2} = & \left(X_2 + aX_1 + \binom{a}{2} X_0 \right) Y_2 w^b \\ & + b(X_2 + aX_1)Y_1 w^{b-1} + \binom{b}{2} X_2 Y_0 w^{b-2}. \end{aligned} \quad (58)$$

Proof. At $k = d$, the condition $i + j \geq d$ forces $(i, j) = (a, b)$. At $k = d - 1$, the possible deficits $(a - i, b - j)$ are $(0, 0), (1, 0), (0, 1)$. At $k = d - 2$, these and $(2, 0), (1, 1), (0, 2)$ are possible. Substitution in (55) gives the formulas. $\square$

For a compact parametrization write

$$h = n - b + 1, \quad t = \frac{b}{h}, \quad z = \frac{b(b-1)}{h(h+1)}, \quad R = \frac{m-a+1}{a}, \quad \rho = \frac{a-1}{m-a+2}, \quad J = \binom{a}{2} + at + z. \quad (59)$$

Thus $t = X_1/X_0$, $z = X_2/X_0$, $R = Y_0/Y_1$ and $\rho = Y_2/Y_1$. Put $D = X_0 Y_1 > 0$.

Proposition 18.4 (Quadratic endpoint criterion). *For $a, b \geq 2$ and the above parameters,*

$$\Delta_{d-1}^{(d)} = D^2 w^{2b-2} (\alpha w^2 + \beta w + \gamma), \quad (60)$$

where

$$\alpha = (d-1)(a+t)^2 - 2dR\rho J, \quad (61)$$

$$\beta = 2bR((d-1)t(a+t) - d(at+z)), \quad (62)$$

$$\gamma = bR^2(b(d-1)t^2 - d(b-1)z). \quad (63)$$

Proof. Divide the three identities in Lemma 18.3 by D and substitute them into $(d-1)p_{d-1}^2 - 2dp_{d-2}p_d$. Grouping the three powers $w^{2b}, w^{2b-1}, w^{2b-2}$ yields the displayed coefficients. $\square$

For Theorem 18.1, formulas (61)–(63) give

$$(\alpha, \beta, \gamma) = 98(-1, 5958, 421173).$$

For Theorem 18.2, they give

$$(\alpha, \beta, \gamma) = \frac{3}{2272}(-2404, 4878807600, 10888224221475).$$

This proves both exact factorizations. Each primitive quadratic has a negative leading coefficient and a positive constant term, so it has precisely one positive zero. Direct substitution at the stated first integer and the preceding integer proves the threshold claims; the preceding values are respectively 5310 and 1792188719.

18.3 A general construction criterion

For fixed a, b, h the leading coefficient (61) tends, as $m \rightarrow \infty$, to

$$\alpha_\infty = (d-1)(a+t)^2 - \frac{2d(a-1)}{a} \left(\binom{a}{2} + at + z \right). \quad (64)$$

This is only a device for finding examples; the preceding theorems use finite exact identities.

Proposition 18.5 (From a negative limiting coefficient to a finite graph). *If $\alpha_\infty < 0$, set*

$$C = \frac{2d(a-1)}{a} \left(\binom{a}{2} + at + z \right), \quad m = a - 1 + \left\lfloor \frac{C}{-\alpha_\infty} \right\rfloor.$$

Then $m \geq a$ and $\alpha < 0$. With β, γ defined by the finite parameters, the positive integer

$$w = 1 + \left\lfloor \frac{|\beta| + |\gamma|}{-\alpha} \right\rfloor \quad (65)$$

necessarily makes the last margin negative.

Proof. One has $\alpha = \alpha_\infty + C/(m - a + 2)$. The chosen denominator is $1 + \lfloor C/(-\alpha_\infty) \rfloor > C/(-\alpha_\infty)$, so $\alpha < 0$. Also $C > -\alpha_\infty$, since their difference is $(d-1)(a+t)^2 > 0$, proving $m \geq a$. For (65), $(-\alpha)w > |\beta| + |\gamma|$ and $w \geq 1$, hence

$$\alpha w^2 + \beta w + \gamma \leq -(-\alpha)w^2 + |\beta|w + |\gamma| < 0.$$

□

For $a = 8, b = 30, h = 3$ (thus $n = 32$),

$$\alpha_\infty = -\frac{61}{4}, \quad \alpha = \frac{48379 - 61m}{4(m-6)}.$$

The leading coefficient is negative exactly when $m \geq 794$. Choosing $m = 1000$ gives Theorem 18.2. The alternative $a = 7, b = 31, h = 3$ also has rank 38 and gives $\alpha_\infty = -764/63$. These are construction examples, not a proof of any global optimum.

18.4 What this settles and what remains open

The inspected ProveIt Hall-bottleneck report [9] distinguishes the arbitrary-two-shore weighted failure from its unit-weight and one-shore variants. Theorems 18.1 and 18.2 supply a negative answer to the one-shore variant, even inside the full two-step Ferrers family (52). Since a degree- d polynomial with positive coefficients and only real zeros satisfies Newton's inequalities with normalization d , these examples also fail positive-field real-rootedness.

There is no conflict with the ambient-shore normalization in the bimatroid theorem of Röhrle and Ulirsch [7]. Nor does the current non-nesting-rook result of Alexandersson and Jal settle the stronger normalization used here: Corollary 4.1 of version 3 explicitly normalizes by the board's row count [15]. Our two examples have smaller shore sizes 52 and 40, respectively, while their matching ranks are 50 and 38. The older version's statement should not be substituted for that current row-normalized theorem.

The full-Hall constructions here have nonunit activities. Further concrete questions include the smallest rank admitting one-shore failure, the smallest graph admitting such failure, and which fixed full-Hall core shapes remain universally rank-ULC under one-shore activities. [Merge note: Re-scoped by the batch: for one common right-core activity on full Hall graphs the smallest failing rank is exactly 38 (sources 55 and 47); among all one-shore graphs it lies in [6, 38], since Part I gives ULC_ν for $\nu \leq 5$.] The present calculation also suggests studying the sign of (64) systematically; its negativity is sufficient for a counterexample, not asserted necessary for every one-shore failure.

Reproducibility. The accompanying script `verify_full_hall.py` [*Merge note:* shipped as `code/73-one-shore-verify_full_hall.py`, output `data/73-one-shore-full_hall_verification.json`] evaluates the complete feasibility sum (55), checks it independently against Boolean matching on six small graphs, and compares the top-three formula on 100 deterministic parameter samples. It then verifies both factorizations at four stated integer activities using exact integers and rational numbers. For all four instances, the only negative order- d Newton margin is the last one. These finite checks supplement the proof; they are not its logical basis.

18.5 Appendix: A pendant gadget that transfers fields between shores

The direct Hall counterexamples above give much smaller matching rank. The following independent construction gives a one-shore counterexample whose only nonunit activity is the modest integer 21. It also explains how a weighted obstruction can survive a transfer of fields between shores.

[*Merge note:* Lemma 18.6 contains source 71’s Lemma 17.3 as the case $T = 1$ (shores exchanged); it is printed here once for both sources.]

Lemma 18.6 (Exact pendant-arm identity). *Let x be a left vertex of activity one in a bipartite graph, and write $p_G(t) = f(t) + g(t)$, where f counts supports omitting x and g counts supports containing x . Attach $k \geq 1$ disjoint length-two arms $x - z_i - \ell_i$. Give the new right vertices z_i activity $T > 0$ and the new left leaves ℓ_i activity one. Then*

$$p_{G'}(t) = (1 + Tt)^{k-1}[(1 + (k + 1)Tt)f(t) + (1 + Tt)g(t)]. \quad (66)$$

Moreover, $\nu(G') = \nu(G) + k$.

Proof. Fix a support counted by g , so x must match within the original graph. Each selected new center then forces its own leaf, contributing $(1 + Tt)^k g$. For a support counted by f , choose j new centers. Either all their leaves are selected and x is absent, or x replaces exactly one of those j leaves. The latter support forces x to that leaf’s center, so it cannot also use x in the original matching. These $j + 1$ endpoint sets are distinct. Their total contribution is

$$\sum_{j=0}^k \binom{k}{j} (j + 1)(Tt)^j f = (1 + Tt)^{k-1} (1 + (k + 1)Tt) f.$$

This counts supports, not allocations or perfect matchings. An original maximum matching together with all k center–leaf edges has size $\nu(G) + k$. An original minimum vertex cover together with the k centers covers the augmented graph, giving the reverse inequality. $\square$

The top power of T in (66) is

$$T^k t^k ((k + 1)f + g).$$

Thus the forced-center limit simulates activity $1/(k + 1)$ at x , up to a positive scalar and the degree shift k . For gadgets at several left vertices, these factors multiply. The rank shift must be accounted for: a polynomial $t^K p(t)$ of degree $r + K$ has last normalized inequality

$$(r + K - 1)p_{r-1}^2 \geq 2(r + K)p_{r-2}p_r.$$

In particular, a top-three ratio strictly below 2 survives every such shift, and continuity then gives a finite positive center activity. This observation is a sufficient criterion; the finite example below is checked by its exact endpoint polynomial.

18.5.1 A rank 310 example with activity 21

[*Merge note:* This example is one-shore weighted, not unit weighted: its ten right-core vertices have activity 21. Its rank is superseded by the rank-50 and rank-38 examples above; it is kept for the transfer construction.] Start with a full 20×10 Hall core, with exterior populations $N = 14$ on the left and $M = 30$ on the right. Attach 20 pendant arms to each exterior-left vertex. Give all left vertices activity one. Give the ten original right-core vertices activity 21, and every other right vertex activity one, including every pendant center. There are $K = 280$ arms, so the graph has

$$634 \text{ vertices,} \quad 1500 \text{ edges,} \quad \nu(G) = 30 + 280 = 310.$$

For completeness, the original 30-edge matching pairs the twenty left-core vertices to distinct exterior-right vertices and the ten right-core vertices to distinct exterior-left vertices. Add all 280 center–leaf edges. The thirty original core vertices together with the 280 centers form a cover of the same size.

Here is an exact count of the endpoint coefficients. More generally let a, b, N, M be Hall parameters, put $W = k + 1$, $K = N(W - 1)$, and let $P_\ell(t)$ count original supports using exactly ℓ exterior-left vertices when only the right core has activity W . Then

$$P_\ell(t) = \binom{N}{\ell} \sum_{\substack{0 \leq i \leq a, \\ j \geq \ell}} \binom{a}{i} \binom{b}{j} \binom{M}{i + \ell - j} W^j t^{i + \ell}, \quad (67)$$

where out-of-range binomial coefficients are zero. The condition $j \geq \ell$ is exactly the Hall-support feasibility condition after balancing the shores. Repeated application of Lemma 18.6 gives the full polynomial

$$P(t; T) = \sum_{\ell=0}^{\min(N, b)} P_\ell(t) (1 + Tt)^{K - N + \ell} (1 + W T t)^{N - \ell}. \quad (68)$$

Every factor and coefficient in this expression is explicit.

For $(a, b, N, M, W, T) = (20, 10, 14, 30, 21, 1)$, coefficient extraction gives

$$\begin{aligned} p_{308} &= 8131749153344342901404784768563550, \\ p_{309} &= 39894459936702682423206366171900, \\ p_{310} &= 97561086654439123859418490215. \end{aligned}$$

Consequently the endpoint margin is

$$309p_{309}^2 - 620p_{308}p_{310} = -726235 \cdot 4485169759863727935000 \cdot 21^{26} < 0.$$

This is an exact one-shore-weighted counterexample. Its full negative integer margin, full coefficient vector, edge list, matching, and cover are included in the verification data. [*Merge note:* Shipped as `data/73-one-shore-gadget_verification-one_shore_rank310_witness.json`, beside `code/73-one-shore-gadget_verification-*.py`; the gadget identity record is byte-identical to source 71's `data/71-unit-weight-gadget_identity_verification.json`.]

Keeping the same graph and allowing each center activity to be T , the stronger exact identity is

$$\begin{aligned} 309p_{309}(T)^2 - 620p_{308}(T)p_{310}(T) &= 4485169759863727935000 \cdot 21^{26} T^{558} \\ &\quad \cdot (-91777T^2 - 1459920T + 825462). \end{aligned}$$

At $T = 1$ the bracket is -726235 . This identity makes the finite activity explicit; no limiting value is used in the counterexample.

Verification. The gadget identity was checked against independent weighted endpoint-set enumeration on 150 small bipartite graphs, with a fixed seed. A separate standard-library script expands (68) in full, checks the three displayed coefficients and exact margin, and verifies the explicit rank certificates. An independent audit recomputed the three endpoint coefficients directly from Hall subset feasibility and the top coefficients of each gadget factor. The finite checks supplement the counting proofs above.

19 A sharp rank-38 boundary for the last Hall inequality, one common right-core activity (source 55)

Abstract of source 55, as delivered. For full Hall graphs with one common activity on the right core and unit activities elsewhere, the first possible failure of the last matching-rank-normalized Newton inequality occurs at rank 38. We prove this for all finite exterior populations and every positive core activity, rather than only in a limiting regime. An explicit positive identity rules out a negative finite-field window whenever the leading quadratic coefficient is nonnegative. A cubic then gives an exact parameter criterion, and a finite integer certificate proves the rank threshold. Precisely two core shapes occur at rank 38. The smaller of their minimal populations gives an 864-vertex example with an exact integer certificate.

Scope. The activity is common across one specified core block. The theorem concerns the *last* Newton inequality. It does not determine the first failure among arbitrary individual one-shore activities or among all gaps. General finite-activity rank-six one-shore ultra-log-concavity remains open. [*Merge note:* At rank six, the first gap holds for every positively weighted graph (*preorder-root-polytopes*, Theorem 180.2), and the second and last gaps for every one-shore graph (sources 05 and 13, Part III); the third and fourth stay open in general.] These are ordinary mathematical proofs with exact certificates, not formal kernel verification or external peer review; no global priority claim is made.

19.1 The family and the sharp threshold

Let $H_{a,b;n,m}$ have shores $A \sqcup L$ and $B \sqcup R$, with respective block sizes a, n, b, m , and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Here $a, b \geq 1$ and $n, m \geq 0$ are integers. Every vertex has activity one except the vertices of B , whose common activity is $w > 0$. Write

$$P(t) = \sum_{k=0}^d p_k t^k,$$

where p_k is the total activity of pairs of k -element endpoint sets admitting a perfect matching. Each pair is counted once, regardless of its number of matchings. The degree d is the matching rank. The last rank-normalized Newton inequality is

$$\Delta_{d-1} := (d-1)p_{d-1}^2 - 2dp_{d-2}p_d \geq 0 \quad (d \geq 2). \quad (69)$$

Theorem 19.1 (Sharp common-core-field endpoint boundary). *Within this family, the smallest matching rank for which (69) fails at some $w > 0$ is exactly 38. At rank 38, the only triples*

$$(a, b, h), \quad h = n - b + 1,$$

permitting failure for some finite m, w are

$$(7, 31, 3) \quad \text{and} \quad (8, 30, 3).$$

Their least possible m are 923 and 794, respectively. Thus the fewest vertices at rank 38 in this family with a failing last gap is 864, attained by $H_{8,30;32,794}$. For that graph the first failing positive integer activity is $w = 292602646$.

[Merge note: Source 47 (Theorem 20.1) shows that in this family no gap at all fails below matching number 38, so the least failure rank of full ULC is also 38. This is the one-shore analogue of Research question 88 of *preorder-root-polytopes*; that question itself, with activity on both cores, is not answered here.]

The qualifier “at rank 38” in the vertex statement matters: a previously constructed rank-50 example has only 125 vertices [16]. The new conclusion is a sharp rank threshold for the stated family and gap.

19.1.1 Deficient exterior populations

The actual matching rank is

$$d = \min\{a + b, a + n, b + m\}. \quad (70)$$

For a direct construction, first match $\min(a, m)$ pairs from A to R and $\min(b, n)$ pairs from L to B , then match the remaining core vertices to each other. This attains (70); the three displayed quantities are upper bounds from the core cover and the two shores.

If $n < b$ or $m < a$, this rank is the smaller shore size, and the established weighted smaller-shore theorem gives the entire actual-rank ULC sequence. [Merge note: That theorem is *preorder-root-polytopes*’s Theorem 70.1 (Part V), Lemma 3.1 here; the specialization recalled next is its proof.] Here is the weighted specialization explicitly. For any bipartite graph $G = (X, Y)$, introduce a private element $\bar{x}$ for each $x \in X$. Match these private elements and Y into X , with $\bar{x}$ adjacent only to x . The resulting transversal matroid has bases

$$\{\bar{x} : x \notin I\} \sqcup J$$

for exactly the feasible support pairs (I, J) . Its Lorentzian basis polynomial $\mathcal{B}$, specialized by

$$\left(\prod_{x \in X} u_x \right) \mathcal{B}((s/u_x)_{x \in X}, (tv_y)_{y \in Y}) = \sum_k p_k s^{|X|-k} t^k,$$

gives $\text{ULC}_{|X|}$ by the bivariate characterization and closure under nonnegative linear substitutions [1]. Transpose to obtain the smaller order. This is the weighted form of the relation/bimatroid argument in [7]. Consequently deficient populations cannot fail (69). We henceforth assume $n \geq b$ and $m \geq a$, and write $r = a + b = d$.

19.2 The exact endpoint quadratic

A size- k support selecting i core-left and j core-right vertices is feasible exactly when $i + j \geq k$. Indeed, the selected exterior-left vertices must fit into the selected B vertices; after matching the exterior vertices, the remaining core vertices can be matched because the core is complete. Thus

$$p_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{n}{k-i} \binom{b}{j} \binom{m}{k-j} w^j, \quad (71)$$

where out-of-range binomial coefficients are zero.

Set

$$h = n - b + 1, \quad x = \frac{b}{h}, \quad y = \frac{b(b-1)}{h(h+1)},$$

$$R = \frac{m-a+1}{a}, \quad \rho = \frac{a-1}{m-a+2}, \quad J = \binom{a}{2} + ax + y, \quad D = \binom{n}{b} \binom{m}{a-1} > 0.$$

Enumerating total core deficits zero, one and two in (71) gives

$$p_r = DRw^b,$$

$$p_{r-1} = D((a+x)w^b + bxRw^{b-1}),$$

$$p_{r-2} = D \left(J\rho w^b + b(ax+y)w^{b-1} + \binom{b}{2}yRw^{b-2} \right).$$

When $b = 1$, the last term is zero. Substitution in (69) yields

$$\Delta_{r-1} = D^2 w^{2b-2} (\alpha_m w^2 + 2bRBw + R^2 C), \quad (72)$$

where

$$\alpha_m = (r-1)(a+x)^2 - 2rR\rho J, \quad (73)$$

$$B = (r-1)x(a+x) - r(ax+y) = \frac{b[b(r-1) - ah^2]}{h^2(h+1)}, \quad (74)$$

$$C = b(b(r-1)x^2 - r(b-1)y)$$

$$= \frac{b^2\{b^2(r-1) + h[r(2b-1) - b^2]\}}{h^2(h+1)} > 0. \quad (75)$$

The last positivity uses $r(2b-1) - b^2 = a(2b-1) + b(b-1) > 0$. Define

$$K = \frac{2r(a-1)}{a}J, \quad \alpha_\infty = (r-1)(a+x)^2 - K. \quad (76)$$

Then

$$\alpha_m = \alpha_\infty + \frac{K}{m-a+2} \geq \alpha_\infty. \quad (77)$$

19.3 A finite field cannot create a hidden negative window

Lemma 19.2 (Positive discriminant margin). *Put $H = ah^2 - b(r-1)$ and define*

$$Q_0 = a^2(b^2 + 2b - 1) + a(b-1)(2b^2 + 3b - 1) + b(b-1)^2,$$

$$Q_1 = (r-1)(2ab - a - b),$$

$$R_0 = a(b^2 + 2b - 1) + (b-1)(3b - 1),$$

$$R_1 = a(3b - 1) + (b-1).$$

For all positive integers a, b ,

$$\frac{ah^3(h+1)^2}{b^2r}(\alpha_\infty C - b^2 B^2) = H(Q_0 + Q_1 h) + br(r-1)(R_0 + R_1 h). \quad (78)$$

If $H \geq 0$, the right side is strictly positive.

Proof. Substitution of (74), (75), and (76), followed by collecting powers of h , proves the identity. Each displayed term in Q_0, R_0, R_1 is nonnegative and their first terms are positive. Also $2ab - a - b = a(b-1) + b(a-1) \geq 0$, so $Q_1 \geq 0$. Finally $br(r-1) > 0$ and $R_0 + R_1 h > 0$. $\square$

Theorem 19.3 (Exact finite-field and population criterion). *Fix a, b, n, m with $n \geq b, m \geq a$. The last inequality fails for some $w > 0$ if and only if $\alpha_m < 0$. If $\alpha_m \geq 0$, it is strictly satisfied for every $w > 0$. If $\alpha_m < 0$, failure occurs precisely above the unique positive root of the quadratic in (72).*

For fixed a, b, h , there exist finite m and $w > 0$ giving failure if and only if $\alpha_\infty < 0$. In that case the least eligible integer population is

$$m_{\min} = a - 1 + \left\lfloor \frac{K}{-\alpha_\infty} \right\rfloor. \quad (79)$$

Proof. If $B \geq 0$ and $\alpha_m \geq 0$, every term in the quadratic is nonnegative and its constant term is positive. If $B < 0$, then $H > 0$ by (74). Lemma 19.2 and (77) imply

$$\alpha_m C - b^2 B^2 \geq \alpha_\infty C - b^2 B^2 > 0.$$

In particular $\alpha_m > 0$, and the quadratic has negative discriminant and is positive on the entire real line. This rules out every putative negative finite- w window with nonnegative leading coefficient.

If $\alpha_m < 0$, the constant term is positive, so the two real roots have opposite signs and failure is exactly above the positive root. Now (77) shows that some finite m has $\alpha_m < 0$ exactly when $\alpha_\infty < 0$. In that case the necessary and sufficient condition is

$$m - a + 2 > \frac{K}{-\alpha_\infty},$$

whose least integer solution is (79). Since $K + \alpha_\infty = (r-1)(a+x)^2 > 0$, the quotient in (79) exceeds one, so $m_{\min} \geq a$. $\square$

19.4 A cubic classification and the finite rank certificate

Clearing the positive denominator in (76) gives

$$ah^2(h+1)\alpha_\infty = F_{a,b}(h) = A_3 h^3 + A_2 h^2 + A_1 h + A_0, \quad (80)$$

where

$$\begin{aligned} A_0 &= ab^2(r-1), \\ A_1 &= -b(r-1)(ab - 2a - 2b), \\ A_2 &= a(a^2 + 2ab - a + 2b^2 - b), \\ A_3 &= a(a^2 + 2ab - a - b). \end{aligned}$$

The coefficients A_0, A_2, A_3 are strictly positive. Only A_1 can be negative. In particular $a \leq 2$ or $b \leq 2$ forces $F_{a,b}(h) > 0$ for all $h \geq 1$.

This gives a complete test for any fixed core shape, at arbitrary rank. If $A_1 \geq 0$, failure is impossible. Otherwise the unique positive critical point is

$$h_* = \frac{-A_2 + \sqrt{A_2^2 - 3A_3 A_1}}{3A_3}.$$

Since $F''(h) = 6A_3h + 2A_2 > 0$ on $h \geq 0$, the integer minimum for $h \geq 1$ is attained at

$$\max\{1, \lfloor h_* \rfloor\}, \quad \max\{1, \lceil h_* \rceil\}.$$

Some exterior populations and a common core field fail the last gap exactly when at least one of the two corresponding cubic values is negative.

For the sharp finite-rank certificate we use an integer-only tail bound. Set

$$H_0 = \max \left\{ 1, \left\lceil \frac{-A_1}{A_2} \right\rceil \right\}. \quad (81)$$

For $h \geq H_0$, the sum $A_2h^2 + A_1h = h(A_2h + A_1)$ is nonnegative, so $F(h) > 0$. It therefore suffices to evaluate $1 \leq h \leq H_0$, with the boundary included redundantly.

The following complete integer procedure covers every rank from 4 through 38:

$$\begin{aligned} r &= 4, \dots, 38; & a &= 2, \dots, r-2; & b &= r-a; \\ h &= 1, \dots, H_0(a, b); & & & & \text{evaluate } F_{a,b}(h). \end{aligned}$$

It contains 630 core pairs and exactly 2237 integer evaluations; the largest cutoff is 10. Every value at rank at most 37 is positive. At rank 38, the only negative values are

$$F_{7,31}(3) = -3056, \quad F_{8,30}(3) = -4392. \quad (82)$$

No zeros occur. All omitted $a = 1$ or $b = 1$ cases have $A_1 \geq 0$, and (81) proves positivity for all omitted h . The integer verifier and complete evaluation list accompany this report. For ease of inspection, Table 5 records the minimum evaluated value at each rank. The tail argument, not a numerical extrapolation, makes the finite calculation exhaustive.

Table 5: Minimum of the cubic over the complete finite certificate at each rank.

r	Minimum	r	Minimum	r	Minimum
4	96	16	7680	28	30936
5	200	17	9248	29	31248
6	360	18	11016	30	30921
7	588	19	12996	31	30312
8	896	20	15200	32	29184
9	1296	21	17640	33	26976
10	1800	22	20328	34	24304
11	2420	23	23276	35	21088
12	3168	24	26337	36	16281
13	4056	25	28000	37	8424
14	5096	26	29200	38	-4392
15	6300	27	30132		

19.5 The two rank 38 shapes and the smaller witness

For $(a, b, h) = (8, 30, 3)$ one has

$$\alpha_\infty = -\frac{61}{4}, \quad \alpha_m = \frac{48379 - 61m}{4(m-6)}.$$

Thus $\alpha_m < 0$ exactly when $m \geq 794$. For $(7, 31, 3)$, the limiting coefficient is $-764/63$, and (79) gives $m_{\min} = 923$. These are the complete possibilities at rank 38 by (82).

For the first graph at its minimal population, put

$$D = \binom{32}{30} \binom{794}{7}, \quad q(w) = -22w^2 + 6437219280w + 11385894392295.$$

The exact endpoint factorization is

$$\Delta_{37} = \frac{5}{6304} D^2 w^{58} q(w). \quad (83)$$

Direct integer substitution gives

$$q(292602645) = 725475345 > 0, \quad q(292602646) = -5711821777 < 0.$$

The unique positive root proves the stated first failing integer. This graph has 864 vertices and 7552 edges. The other rank-38 shape has at least $38 + 33 + 923 = 994$ vertices, establishing the rank-38 vertex minimum in Theorem 19.1.

For completeness, at that second minimal population its primitive endpoint quadratic is

$$-2w^2 + 3479879022w + 8332557477297,$$

whose first failing positive integer is 1739941906. The preceding and failing values are 1715636157 and -1764252443 .

19.6 Verification and attribution

The top-three formula and the earlier rank-38 construction appeared in the companion [16]. [*Merge note:* Source 73: Lemma 18.3, Proposition 18.4 (whose α, β, γ are the α_m , $2bRB$, R^2C above) and Proposition 18.5.] The additional conclusions here are the exact finite-field criterion, exhaustive rank-38 endpoint classification, and minimal populations within that rank. The smaller-shore input is established Lorentzian theory [1, 7].

The standard-library script `verify_boundary.py` [*Merge note:* shipped as `code/55-rank38-endpoint-verify_boundary.py` and `code/55-rank38-endpoint-verify_symbolic.py`, outputs `data/55-rank38-endpoint-*.json`] regenerates all 2237 exact cubic values, verifies both rank-38 population thresholds, and checks the endpoint formula against the complete support sum on 160 deterministic rational-activity samples. It also checks the full support coefficients of both new minimal-population witnesses; in each, the last inequality is the only failing gap. The separate script `verify_symbolic.py` expands (78) and (80) exactly over $\mathbb{Q}(a, b, h)$. The complete certificate, scripts, and outputs are included. The general identities and the proven finite tail bound establish the result; sample regressions are additional checks.

The exact first-failure question for arbitrary individual one-shore activities is broader. Earlier Newton gaps in the common-core-field family are also outside this report's classification. [*Merge note:* The earlier gaps are settled by source 47 (Section 20).] Neither is settled by the last-gap criterion proved here.

20 The sharp rank-38 ULC boundary for full Hall graphs, one common right-core activity (source 47)

Abstract of source 47, as delivered. For full Hall graphs with a common positive activity on the right core and unit activities elsewhere, the first possible failure of matching-rank-normalized ultra-log-concavity occurs at rank 38. We close the earlier-gap question left open by the endpoint classification. Every intermediate Newton gap through rank 38

has positive coefficients in the common field, for every eligible pair of exterior populations. The proof expands each gap in two binomial population bases and checks a finite array of 22,452,529 positive integer coefficients. This is an exact all-population certificate, not a bounded population or activity scan. Together with the first-gap inequality and the previously established endpoint theorem, it gives the complete failure classification at rank 38.

Scope. The field is common across one specified core block. The result does not determine the first failure for arbitrary individual one-shore activities or for arbitrary bipartite graphs. General finite-activity rank-six one-shore ultra-log-concavity remains open. [*Merge note:* See the note in the scope paragraph of source 55 (Section 19).] This is a computer-assisted mathematical proof with reproducible integer certificates, not formal kernel verification or an external peer-review claim.

20.1 The family and the conclusion

Let $H_{a,b;n,m}$ have shores $A \sqcup L$ and $B \sqcup R$, with block sizes a, n, b, m , respectively, and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Here $a, b \geq 1$ and $n, m \geq 0$ are integers. Every activity is one except those of B , which all equal $w > 0$. The support polynomial is

$$P(t) = \sum_{k=0}^d p_k(w) t^k.$$

Each pair of equally sized endpoint sets admitting a perfect matching is counted once, regardless of its number of matchings. Its degree is

$$d = \min\{a + b, a + n, b + m\}.$$

For $1 \leq k < d$, put

$$\Delta_k = k(d - k)p_k^2 - (k + 1)(d - k + 1)p_{k-1}p_{k+1}.$$

The sequence is ULC_d precisely when every $\Delta_k \geq 0$, since all coefficients up to the matching rank are positive.

Theorem 20.1 (Sharp boundary for all gaps). *Every graph in this common-core-field family of matching rank at most 37 is ULC_d for every $w > 0$ and all exterior populations.*

At matching rank 38, every gap except possibly the last is strictly positive whenever $a, b \geq 3$ and $n \geq b, m \geq a$. A failure at some finite m, w is possible only for

$$(a, b, n - b + 1) = (7, 31, 3) \quad \text{or} \quad (8, 30, 3).$$

The least corresponding m are 923 and 794. The smallest graph at rank 38 in this family admitting any failed gap therefore has 864 vertices, namely $H_{8,30;32,794}$. Its first failing positive integer core activity is 292602646.

The new ingredient is the following stronger statement about intermediate gaps.

Theorem 20.2 (Positive common-field coefficients). *Suppose $a, b \geq 3$, $r = a + b \leq 38$, $n \geq b$, and $m \geq a$. For every $2 \leq k \leq r - 2$, all coefficients of*

$$w^{-\max(0, 2k-2a)} \Delta_k$$

from degree zero through degree $\min(2b, 2k) - \max(0, 2k - 2a)$ are strictly positive. In particular $\Delta_k > 0$ for every $w > 0$.

At larger rank the last gap does not control the earlier ones: the separate rank-49 example $H_{14,35;38,624}$ has its last gap positive for every $w > 0$ but a failing penultimate gap at $w = 40000000$ [18]. The rank restriction in Theorem 20.2 is therefore essential to the present certificate.

20.2 Counting and the finite population expansion

Write $\binom{x}{y} = 0$ when $y < 0$ or $y > x$, for nonnegative integer x . For $j, q \geq 0$, define

$$T_{j,q}(n) = \sum_{\ell=0}^j \binom{a}{q+\ell} \binom{n}{j-\ell}; \quad (84)$$

set $T_{j,q} = 0$ when either subscript is negative. A support selecting j vertices of B and q of R must select at least q vertices of A . If it selects $q + \ell$ of A , it selects $j - \ell$ of L . The complete core makes this necessary condition sufficient. Hence

$$p_k(w) = \sum_{j=0}^b \binom{b}{j} \binom{m}{k-j} T_{j,k-j}(n) w^j. \quad (85)$$

This is endpoint-support counting throughout.

Fix $2 \leq k \leq r - 2$. For a coefficient of w^s , write $L = 2k - s$. Its possible range is

$$\max(0, 2k - 2a) \leq s \leq \min(2b, 2k). \quad (86)$$

For $0 \leq p \leq \lfloor L/2 \rfloor$, define

$$\begin{aligned} G_{k,s,p}(n) &= \binom{L-p}{p} \sum_{j=\max(0, s-b)}^{\min(b, s)} \binom{b}{j} \binom{b}{s-j} (U_j - V_j), \\ U_j &= k(r-k) \binom{L-2p}{k-j-p} T_{j,k-j}(n) T_{s-j, k-s+j}(n), \\ V_j &= (k+1)(r-k+1) \binom{L-2p}{k-1-j-p} T_{j, k-1-j}(n) T_{s-j, k+1-s+j}(n). \end{aligned} \quad (87)$$

Lemma 20.3 (Exact finite population transfer). *For every nonnegative integer m ,*

$$[w^s] \Delta_k = \sum_{p=0}^{\lfloor L/2 \rfloor} \binom{m}{L-p} G_{k,s,p}(n). \quad (88)$$

Proof. Two subsets of an m -element set, of sizes q_1, q_2 and intersection size p , can be specified by their union, their intersection, and the first disjoint remainder. Thus

$$\binom{m}{q_1} \binom{m}{q_2} = \sum_p \binom{m}{q_1 + q_2 - p} \binom{q_1 + q_2 - p}{p} \binom{q_1 + q_2 - 2p}{q_1 - p}.$$

Both products in the coefficient of w^s have $q_1 + q_2 = L$. Applying this identity to (85) gives exactly (87)–(88). $\square$

This transfer keeps all finite-population terms, including intersections. The diffuse limit alone would retain only $p = 0$ and would not justify the finite- m conclusion.

20.3 An exhaustive finite integer certificate

The polynomial $T_{j,q}(n)$ has degree at most j , so $G_{k,s,p}(n)$ has degree at most s . Its Newton finite-difference expansion about $n = b$ is

$$G_{k,s,p}(b+h) = \sum_{e=0}^s c_{a,b,k,s,p,e} \binom{h}{e}, \quad (89)$$

$$c_{a,b,k,s,p,e} = \sum_{i=0}^e (-1)^{e-i} \binom{e}{i} G_{k,s,p}(b+i). \quad (90)$$

These are integer identities. Combining them with Lemma 20.3 gives

$$\Delta_k = \sum_s \sum_{p=0}^{\lfloor (2k-s)/2 \rfloor} \sum_{e=0}^s c_{a,b,k,s,p,e} w^s \binom{m}{2k-s-p} \binom{n-b}{e}. \quad (91)$$

Every basis factor is nonnegative at the populations and activities in question.

Proposition 20.4 (Verified coefficient array). *Every integer $c_{a,b,k,s,p,e}$ is strictly positive over the finite range*

$$\begin{aligned} 6 \leq r \leq 38, \quad 3 \leq a \leq r-3, \quad b = r-a, \quad 2 \leq k \leq r-2, \\ \max(0, 2k-2a) \leq s \leq \min(2b, 2k), \quad 0 \leq p \leq \lfloor (2k-s)/2 \rfloor, \quad 0 \leq e \leq s. \end{aligned}$$

There are 561 core pairs, 13,651 intermediate gaps, 1,583,703 polynomials G , and 22,452,529 checked coefficients.

Exact verification. The accompanying standard-library verifier evaluates (87) at $n = b, \dots, b+s$, forms successive difference rows, and checks their first entries against zero. Every arithmetic operation is an integer addition, multiplication, binomial evaluation, or comparison. It visits precisely the displayed finite range; there are no numerical tolerances, sampled fields, or omitted population tails. Per-gap coefficient counts and SHA-256 records are included, allowing all entries to be regenerated and compared. A separate implementation constructs the support counts by adding exterior vertices using Pascal recurrence and computes the coefficients by the explicit inverse Pascal matrix. It agrees with every per-gap hash and count. The smallest checked coefficient is 9; there are no zero entries. $\square$

Proof of Theorem 20.2. Equation (91) and Proposition 20.4 give coefficientwise nonnegativity in w for all $n \geq b$ and $m \geq a$. For each s in (86), take $p = \lfloor L/2 \rfloor$ and $e = 0$. Then

$$L-p = \lceil L/2 \rceil \leq a \leq m,$$

so the corresponding population binomial is positive. Its coefficient is strictly positive by Proposition 20.4. Thus every stated common-field coefficient is positive. $\square$

20.4 Closing the first and last gaps

20.4.1 The first gap

[Merge note: Source 47 proves here, by the Motzkin–Straus route, that $\Delta_1 \geq 0$ for every positively weighted bipartite graph; this is *preorder-root-polytopes*’s Theorem 180.2 (Part XI, `hb:thm:firstidentity`) and the inequality printed with this route in Sections 11 and 14. The argument is not reprinted; in it $M_2 \geq p_2$ is the weighted number of two-edge matchings and the Motzkin–Straus theorem gives $M_2 \leq \frac{d-1}{2d} p_1^2$. Source 47’s strictness step is kept verbatim:]

In the genuine full Hall cases with $a, b \geq 3$, two vertices from each core form a $K_{2,2}$. Its support contributes two positive matchings to M_2 but only once to p_2 . Hence $M_2 > p_2$ and $\Delta_1 \geq 2d(M_2 - p_2) > 0$. [*Merge note:* In *preorder-root-polytopes*'s identity this is the positive term B_4 .]

20.4.2 The endpoint classification

We use the exact endpoint theorem of [17]. [*Merge note:* That theorem is source 55, printed in Section 19: the quadratic (72), Lemma 19.2, Theorem 19.3, the cubic (80) and the two negative values (82). Source 47 recalls its formulas without proof; the recall is kept below, in source 47's notation, which agrees with source 55's.] Its essential formulas are recalled to make the scope and resulting boundary explicit. Assume $n \geq b, m \geq a$, and put

$$h = n - b + 1, \quad x = \frac{b}{h}, \quad y = \frac{b(b-1)}{h(h+1)}, \quad R = \frac{m-a+1}{a}, \quad \rho = \frac{a-1}{m-a+2},$$

$$J = \binom{a}{2} + ax + y, \quad D = \binom{n}{b} \binom{m}{a-1}.$$

Then

$$\Delta_{r-1} = D^2 w^{2b-2} (\alpha_m w^2 + 2bRBw + R^2 C),$$

where

$$\alpha_m = (r-1)(a+x)^2 - 2rR\rho J, \quad B = \frac{b[b(r-1) - ah^2]}{h^2(h+1)},$$

$$C = \frac{b^2\{b^2(r-1) + h[r(2b-1) - b^2]\}}{h^2(h+1)} > 0.$$

The endpoint theorem proves that this quadratic is positive for every $w > 0$ if $\alpha_m \geq 0$. If $\alpha_m < 0$, it has one positive root and is negative precisely above it. Thus finite fields introduce no hidden negative window.

Writing

$$K = \frac{2r(a-1)}{a} J, \quad \alpha_\infty = (r-1)(a+x)^2 - K$$

gives $\alpha_m = \alpha_\infty + K/(m-a+2)$. Some finite m, w fail exactly when $\alpha_\infty < 0$, and then

$$m_{\min} = a - 1 + \left\lfloor \frac{K}{-\alpha_\infty} \right\rfloor.$$

The sign of α_∞ is the sign of the cubic

$$F_{a,b}(h) = a(a^2 + 2ab - a - b)h^3 + a(a^2 + 2ab - a + 2b^2 - b)h^2 - b(r-1)(ab - 2a - 2b)h + ab^2(r-1).$$

Only its linear coefficient can be negative. The endpoint certificate uses the finite tail cutoff $\max(1, \lceil -A_1/A_2 \rceil)$ for these cubic coefficients. Its 2237 exact evaluations show positivity through rank 37, and exactly

$$F_{7,31}(3) = -3056, \quad F_{8,30}(3) = -4392$$

as the rank-38 negative cases. The tail bound makes this exhaustive in h .

20.4.3 The remaining cover and population cases

If $n < b$ or $m < a$, the actual rank is the smaller shore size. The standard weighted transversal-matroid specialization then gives actual-rank ULC [1, 7]. [*Merge note: preorder-root-polytopes's* Theorem 70.1; Lemma 3.1 here.] Explicitly, with a private element for each left vertex, the extended transversal basis polynomial specializes to

$$\sum_k p_k s^{|X|-k} t^k$$

by assigning a private variable s/u_x and a right variable tv_y , then multiplying by $\prod_x u_x$. Matroid basis polynomials are Lorentzian, so this is $\text{ULC}_{|X|}$; transpose for the smaller shore.

When $n \geq b, m \geq a$ and $\min(a, b) \leq 2$, the previously proved minimum-cover theorem [12] gives actual-rank ULC for arbitrary positive activities. [*Merge note: Theorem 2.1; also source 75's Theorem 16.1.*] All remaining cases are precisely those in Proposition 20.4. The first gap and the endpoint classification therefore complete the proof of Theorem 20.1.

20.5 The sharp witness and reproducibility

For $H_{8,30;32,794}$, let $D = \binom{32}{30} \binom{794}{7}$. The endpoint factorization is

$$\Delta_{37} = \frac{5}{6304} D^2 w^{58} (-22w^2 + 6437219280w + 11385894392295).$$

The quadratic takes the values 725475345 and -5711821777 at 292602645 and 292602646, respectively. The graph has 864 vertices and 7552 edges. The other rank-38 shape requires at least 994 vertices. [*Merge note: These are source 55's numbers, (83); restated by source 47.*] The new intermediate-gap theorem shows that these endpoint possibilities also exhaust every-gap failure at rank 38.

The new certificate and the endpoint certificate have different purposes. The former covers every intermediate gap and both unbounded exterior populations through binomial-basis identities. The latter covers the endpoint through a cubic tail bound and an exact quadratic criterion. The source package contains the generators, compact per-gap coefficient records, regression checks of the population identity, and instructions for replay. [*Merge note: Shipped as `code/47-rank38-ulc-*.py` and `data/47-rank38-ulc-*.json` (33 per-rank certificate files, ranks 6–38); the replay instructions were in the delivered README, which is not shipped (this report's README gives them).*] Numerical scans used during exploration are not part of the proof.

The new conclusion is restricted to a common core field on full Hall graphs. The minimum rank for failure with arbitrary individual one-shore activities remains between 6 and 38. [*Merge note: The lower end six rests on source 74's Corollary 2.2 (Part I); the upper end on sources 73 and 55.*] The rank-49 separation [18] prevents extrapolating endpoint control to all higher ranks.

21 The last Hall gap does not control ultra-log-concavity, one common right-core activity (source 51)

Abstract of source 51, as delivered. We give a full Hall graph of rank 49 with a common activity on one right-core block for which the last rank-normalized Newton inequality holds strictly at every positive activity, but the preceding inequality fails. The example has 711 vertices. Four explicit coefficient formulas give an exact quartic certificate. This separates the

last-gap criterion from full ultra-log-concavity without changing the previously proved sharp rank-38 endpoint threshold.

Let $H_{a,b;n,m}$ have left blocks A, L of sizes a, n , right blocks B, R of sizes b, m , and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Give every vertex activity one except the vertices of B , each of which has activity $w > 0$. The coefficient p_k of its support polynomial counts each pair of k -element endpoint sets admitting a perfect matching once, with the product of its vertex activities. For matching rank r , write

$$\Delta_k = k(r - k)p_k^2 - (k + 1)(r - k + 1)p_{k-1}p_{k+1}.$$

Rank-normalized ultra-log-concavity requires $\Delta_k \geq 0$ at every $1 \leq k < r$.

Theorem 21.1. *The graph $H_{14,35;38,624}$ has matching rank 49, 711 vertices and 10556 edges. Its last gap Δ_{48} is strictly positive for every $w > 0$. At $w = 40000000$, its preceding gap Δ_{47} is negative. The first positive integer w for which $\Delta_{47} < 0$ is 38189138.*

Proof. The core is a cover of size 49, attained by disjoint matchings $A \rightarrow R$ and $L \rightarrow B$. This proves the rank. The vertex and edge counts are

$$14 + 35 + 38 + 624 = 711, \quad 14 \cdot 35 + 38 \cdot 35 + 14 \cdot 624 = 10556.$$

For general full Hall parameters, a size- k support selecting i vertices of A and j of B is feasible exactly when $i + j \geq k$. Hence, with out-of-range binomial coefficients zero,

$$p_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{n}{k-i} \binom{b}{j} \binom{m}{k-j} w^j. \quad (92)$$

Set $D = \binom{38}{35} \binom{624}{12} > 0$. The four coefficients needed here are

$$\begin{aligned} \frac{p_{46}}{Dw^{32}} &= \frac{8801172765}{2} + 32499495w + \frac{136955}{2}w^2 + \frac{27846}{613}w^3, \\ \frac{p_{47}}{Dw^{33}} &= 72736965 + 299880w + 273w^2, \\ \frac{p_{48}}{Dw^{34}} &= \frac{1258425}{2} + 1071w, \\ \frac{p_{49}}{Dw^{35}} &= \frac{14382}{7}. \end{aligned} \quad (93)$$

These follow directly from (92): the core-deficit pairs $(14 - i, 35 - j)$ have total at most 3, 2, 1, 0, respectively. Thus the formulas require only ten, six, three and one terms.

Substituting (93) gives

$$\Delta_{47} = \frac{882}{613} D^2 w^{66} q(w), \quad (94)$$

$$\Delta_{48} = 89964 D^2 w^{68} (w^2 + 47940w + 48443370), \quad (95)$$

where

$$\begin{aligned} q(w) &= -13w^4 + 496441140w^3 + 673931488920w^2 \\ &\quad + 331783888454280w + 68528424519495225. \end{aligned}$$

Every factor in (95) is positive for $w > 0$. In contrast,

$$q(40000000) = -1506688736346303933404280504775 < 0,$$

proving the finite counterexample. The quartic has one sign change in descending coefficient order; Descartes' rule, together with $q(0) > 0$ and $q(w) \rightarrow -\infty$, gives exactly one positive root. Finally,

$$\begin{aligned} q(38189137) &= 358201675329747057329792 > 0, \\ q(38189138) &= -365864633482267706996743 < 0. \end{aligned}$$

This proves the first-integer claim. □

Scope and reproducibility. The sharp rank-38 theorem in [17] concerns the last gap in this common-core-activity family. It remains valid. The present example shows that its criterion does not characterize the full ULC sequence. It does not decide whether an earlier gap can fail below rank 38, or the first failure for arbitrary individual one-shore activities. [*Merge note:* The first is decided by source 47 (Theorem 20.1): no gap fails below rank 38 in this family. The second stays open; it lies in [6, 38].]

The accompanying standard-library script [*Merge note:* shipped as `code/51-last-gap-verify_counterexample.py`, output `data/51-last-gap-verification.json`] independently sums (92), verifies all four coefficient identities and both polynomial factorizations exactly, and checks the stated integer signs. At $w = 40000000$, it also finds that $k = 47$ is the only negative gap. No numerical root approximation is used. These are ordinary proofs with reproducible exact arithmetic, not formal kernel verification; global priority is not claimed.

Part III

Matching number six with one weighted shore at every finite activity

Contents of Part III. Part III prints seven sources on graphs of matching number six with *one weighted shore*: all left activities one, right activities arbitrary positive (Research question 85 of *preorder-root-polytopes*). By source 74 one may assume a genuine $3 + 3$ minimum cover. Source 60 (Section 22) lifts rank-two exterior column blocks of matrices and gives exact conditioning reductions; source 10 (Section 23) proves a Lorentzian three-vertex marginal and all five inequalities for star cores; source 23 (Section 24) proves rooted sector theorems and gives a 25-vertex graph whose all-right-core conditional cubic fails its unshifted Newton inequalities; source 13 (Section 25) proves the *last* inequality and source 05 (Section 26) the *second*, for every graph and every activity; source 69 (Section 27) proves all five for every core with complete exterior neighbourhoods, and source 63 (Section 28) for a complete core with nested exterior neighbourhoods, with a barrier and a conditional theorem. Sources 60 and 10 also allow weights on both shores in their main theorems; their section titles say so.

Status at matching number six, one weighted shore. Gap one holds for every positively weighted graph (*preorder-root-polytopes*, Theorem 180.2); gap two holds by source 05 and gap five by source 13, for every graph and every finite activity. Gaps three and four are open in general; they hold for star cores (source 10), for every core with complete exterior neighbourhoods (source 69) and for the nested complete-core family (source 63). Several sources, written before others, call more gaps open; their sentences are kept with notes.

Same results in several sources. Source 60's 9×9 rank-three example is *preorder-root-polytopes*'s Theorem 192.1 (Part XII, `mr:thm:matrix`); it is replaced by a pointer. Source 05's Lemma 26.3 is source 60's Theorem 22.1 at $q = 0$ (a second route, printed), and source 05's one-vertex compression is the case $h = 1$ of source 60's slack reductions. Source 05's quintic example is source 23's 25-vertex graph. Source 13's incidence lemma and sector identity are source 21's (Part IV), and the lemma is the case $b = 3$ of source 08s's two-plane lemma; it is printed here with source 13's proof. The first-gap arguments of sources 69 and 63 are *preorder-root-polytopes*'s Theorem 180.2 and become pointers. Gaps two and five of sources 69 and 63 also follow from sources 05 and 13, and *preorder-root-polytopes*'s unit $H_{3,n}$ result (Corollary 177.2) is a special case of source 69's theorem.

Non-claims of Part III. No source proves ULC_6 at every finite one-shore activity; source 60's conditioned and eventual conclusions do not assert ULC of the original polynomial at every finite activity; source 23's obstruction does not refute rank-six ULC or an eventual-scaling theorem; source 13's local lemma is graph-generic only; source 05's quartic route is a research direction and its quintic route is refuted, not scalar ULC_6 ; source 63's barrier rules out the full-marginal approach, not ULC_6 ; source 69's barrier coefficient obstructs a proof method, not ULC.

22 Rank-two matrix blocks and exterior conditioning, both shores weighted (source 60)

Abstract of source 60, as delivered. An explicit matroid lift proves ultra-log-concavity of order $q + 2$ for weighted regular-minor support polynomials of matrices whose columns consist of q ordinary columns and a block of rank two. The normalization is the actual matrix rank when that rank is $q + 2$. Exact conditioning reductions with Hall slack at most two then show that, in a genuine $a + q$ bipartite cover, forcing any matchable $a - 2$ exterior-right vertices leaves an actual-degree- $(q + 2)$ ULC polynomial after the forced factor is removed. Scaling those activities sufficiently far also makes the full original polynomial strictly rank-ULC, with an explicit field-dependent threshold. The compressed object may be a matrix rather than a graph; a positive rank-three matrix example shows why this distinction matters. The general finite-activity rank-six one-shore problem remains open.

22.1 A rank two exterior column block

Let M be a matrix over a field, with finite row set X and column set Y . For positive real activities u_i, v_j , define its regular-minor support polynomial by

$$P_M(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, \det M[I, J] \neq 0}} u_I v_J t^{|I|}, \quad u_I = \prod_{i \in I} u_i, \quad v_J = \prod_{j \in J} v_j.$$

The empty minor is one. Each row-column pair is counted once, and determinant magnitudes are not used. The degree is $r = \text{rank } M$. For $p(t) = \sum p_k t^k$, write

$$\Delta_k^{(D)}(p) = k(D - k)p_k^2 - (k + 1)(D - k + 1)p_{k-1}p_{k+1}.$$

The property ULC_D means these gaps are nonnegative and the coefficient sequence has no internal zeros. This is equivalently log-concavity after division by $\binom{D}{k}$.

Theorem 22.1. *Partition the columns as $Y = B \sqcup R$, where $|B| = q$ and $\text{rank } M[X, R] = 2$. For every positive right activity vector, the polynomial*

$$F_M(s, z) = \sum_{\substack{I, J: |I|=|J| \\ \det M[I, J] \neq 0}} z_I v_J s^{q+2-|I|} \tag{96}$$

is Lorentzian. Consequently P_M is ULC_{q+2} for all positive activities on both sides. This is actual-rank normalization if $\text{rank } M = q + 2$.

If the rank is smaller, the theorem asserts order $q + 2$, not the stronger normalization by that smaller rank. The activity parameters are positive real numbers, independently of the field containing the matrix entries.

Proof. Factor the exterior block and write

$$M = [C, DE], \quad D \text{ has two columns, } E \text{ has two rows and rank two.}$$

For the column E_j indexed by $j \in R$, put

$$Q = \sum_{\substack{i < j \text{ in } R \\ \det(E_i, E_j) \neq 0}} v_i v_j > 0, \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Construct a representable matroid in dimension $q + 2$ with three kinds of elements:

$$a_i = (C_i^\top, D_i^\top) \ (i \in X), \quad e_b \ (b \in B), \quad p_j = (0, J_0 E_j) \ (j \in R).$$

The private elements e_b are the first q coordinate vectors. Together with any independent plane pair p_i, p_j , they form a basis, so the matroid has rank $q + 2$.

In its basis polynomial substitute

$$a_i \mapsto z_i, \quad e_b \mapsto s/v_b, \quad p_j \mapsto sv_j/Q, \quad (97)$$

and multiply by $Q \prod_{b \in B} v_b$. We claim the result is (96).

Fix a selected original row set I and write the private set as $B \setminus J_B$. The basis-size equation forces the number of plane elements to be

$$\ell = 2 + |J_B| - |I|.$$

Only $\ell = 0, 1, 2$ are possible. Eliminate the private coordinates and consider the remaining determinant.

If $\ell = 2$, it factors as the ordinary minor $C[I, J_B]$ times the plane-pair determinant, up to sign. The total plane-pair activity is Q/Q^2 , which cancels the outside factor Q . This records selection of no exterior column.

If $\ell = 1$, the plane element p_j yields the nonzero-minor criterion for $M[I, J_B \cup \{j\}]$. The elementary identity behind this is

$$\det(d, J_0 e) = d^\top e.$$

Its activity v_j/Q cancels the outside factor Q , recording that one exterior column.

If $\ell = 0$, the augmented minor uses the two columns of D . An original minor with exterior columns i, j equals this augmented minor times $\det(E_i, E_j)$, up to sign. Thus either no exterior pair works or exactly all independent pairs work; their total activity is Q , as supplied by the outside factor. Larger exterior selections are dependent.

In all three cases the private activities supply v_{J_B} , and the degree in s is $q + 2 - |I|$. This proves the claimed polynomial identity without a determinant multiplicity. Matroid basis polynomials are Lorentzian, and nonnegative linear substitutions preserve the property [1]. Setting $z_i = u_i t$ proves the ULC conclusion. $\square$

[Merge note: Theorem 22.1 answers in part Research question 101 of *preorder-root-polytopes* (`mr:q:bimatroid`, “Which representable bimatroids satisfy actual-rank ULC despite the dense rank-three obstruction?”): every regular-minor matrix whose columns are q ordinary columns and a block of rank two, when its rank is $q + 2$. At $q = 0$ it is source 05’s Lemma 26.3, proved there by a different route.]

The construction generalizes the two-vertex-shore graph lift: there the exterior columns are supported on two rows. [Merge note: Source 74’s lift, Section 5.] Theorem 22.1 requires only a rank-two exterior column space, not a particular zero pattern. The normalization can be smaller than the ambient-shore normalization in the general bimatroid theorem [7].

22.1.1 The rank two hypothesis cannot simply be increased

[Merge note: Source 60 considers here the positive 9×9 matrix $M_{ij} = 1 + ij + (ij)^2$ of rank three, all of whose minors of orders one to three are positive, so that at unit activities $P_M(t) = 1 + 81t + 1296t^2 + 7056t^3$ and $\Delta_2^{(3)}(P_M) = -69984 < 0$. This is *preorder-root-polytopes*’s Theorem 192.1 (Part XII, `mr:thm:matrix`) at $m = n = 9$, with the same Cauchy–Binet and

Vandermonde proof; source 60 does not cite it. Its value -69984 is twice Part XII's defect $1296^2 - 3 \cdot 81 \cdot 7056 = -34992$, because $\Delta_2^{(3)} = 2 \cdot 1296^2 - 6 \cdot 81 \cdot 7056$ carries the factor $k(D-k) = 2$. The statement and proof are not reprinted; source 60's comments on the example follow.] The rank-two case is covered by Theorem 22.1 with $q = 0$, whereas rank one is trivial. Thus arbitrary regular-minor matrices can first fail at rank three. This elementary example is used as a scope check, not as a novelty claim. Its entry-support graph is complete bipartite and has matching rank nine; that graph's endpoint polynomial is a different polynomial.

22.2 Exact conditioning when Hall slack is at most two

For a weighted bipartite graph $G = (X, Y)$ and a right set F , let

$$P_G^{(F)}(t) = \sum_{\substack{(I,J) \text{ feasible} \\ F \subseteq J}} u_I v_J t^{|I|}.$$

This is the coefficient of $\lambda^{|F|}$ when the activities in F are multiplied by λ . Suppose F is matchable, $|F| = h \geq 1$, and let $N = N_G(F)$. The Hall slack here means $|N| - h$.

Proposition 22.2 (Slack zero and one). *If $|N| = h$, then*

$$P_G^{(F)} = v_F u_N t^h P_{G-N-F}.$$

If $|N| = h + 1$, put

$$E_h = \sum_{\substack{I \subseteq N, |I|=h \\ G[I,F] \text{ matchable}}} u_I > 0.$$

Delete N, F and add one left vertex of activity u_N/E_h . Its neighborhood is the union of the unforced-right neighborhoods of those $i \in N$ for which $N \setminus \{i\}$ matches F . Call the resulting graph H . Then

$$P_G^{(F)} = v_F E_h t^h P_H. \quad (98)$$

The remaining graph has matching rank $\nu(G) - h$ in either case.

Proof. In slack zero all of N is consumed by F . In slack one, supports selecting exactly h members of N likewise consume them all; summing their weights gives E_h , with the virtual vertex absent. A support selecting all $h + 1$ neighbors has the fixed product u_N . Its one available vertex can be matched outside F exactly through the stated union neighborhood. This is a Boolean existence test, so different successful choices do not multiply a support's weight. The virtual activity makes $E_h(u_N/E_h) = u_N$.

A matchable right set extends to a maximum matchable right set. Thus $P_G^{(F)}$ has degree $\nu(G)$; the exact identities prove the rank statements. $\square$

For example, conditioning on a right vertex of degree two merges its neighbors a, b into a union-neighborhood vertex of activity $u_a u_b / (u_a + u_b)$, with outside factor $v_F(u_a + u_b)t$.

Theorem 22.3 (Slack two kernel reduction). *If $|N| = h + 2$, the conditioned polynomial has an exact representation*

$$P_G^{(F)} = v_F E_h t^h P_{M'}, \quad (99)$$

where M' is an explicit regular-minor matrix with positive row activities and rank $\nu(G) - h$. In general M' need not represent a graph endpoint polynomial.

Proof. Represent G by its generic matrix M , putting an algebraically independent indeterminate on every edge and zero elsewhere. Work over the rational-function field. A square minor is nonzero exactly when its endpoint graph has a perfect matching, since distinct matchings give distinct determinant monomials.

Let $\Gamma = M[N, F]$, of rank h , and define E_h by summing u_I over its nonsingular h -row submatrices. The coefficient space

$$K = \ker(\Gamma^\top)$$

has dimension two. For every $i \in N$ such that $\Gamma[N \setminus \{i\}, F]$ has rank h , choose a nonzero vector k_i spanning the line

$$K \cap \{x_i = 0\}.$$

If this intersection has dimension two, omit the index i : the corresponding $h + 1$ rows cannot cover all the forced columns. Define the virtual row and its activity by

$$r'_i = k_i^\top M[N, Y \setminus F], \quad w_i = u_{N \setminus \{i\}} / E_h. \quad (100)$$

Retain the original rows outside N , their activities, and the columns outside F . Together with the virtual rows these form M' . Nonzero rescaling of a virtual row does not change any minor's zero/nonzero status.

For retained i, j , the kernel lines k_i, k_j are independent exactly when

$$\det \Gamma[N \setminus \{i, j\}, F] \neq 0. \quad (101)$$

Indeed, their intersection is the kernel supported on that h -row complement. It is zero exactly when the complement is nonsingular. A nonsingular h -row set exists, so the retained lines span K .

Partition supports by their selected neighbors in N . Exactly h selected neighbors contribute E_h times the unchanged outside support polynomial. Exactly $h + 1$ neighbors, $N \setminus \{i\}$, reduce after eliminating the forced columns to the one virtual row r'_i . Their activity is recovered because $E_h w_i = u_{N \setminus \{i\}}$.

When all $h + 2$ neighbors are selected, any independent kernel pair is a basis of K . Changing that basis multiplies a residual determinant by a nonzero scalar. All such pairs therefore have the same residual nonzero-minor profile, including the possibility that this profile is identically zero. Their total assigned activity is

$$\begin{aligned} \sum_{\substack{i < j \\ k_i, k_j \text{ independent}}} w_i w_j &= \frac{u_N}{E_h^2} \sum_{\substack{i < j \\ \det \Gamma[N \setminus \{i, j\}, F] \neq 0}} u_{N \setminus \{i, j\}} \\ &= \frac{u_N}{E_h}. \end{aligned} \quad (102)$$

Thus the prefactor E_h recovers u_N exactly once. Dependent kernel pairs and all triples of virtual rows contribute zero. This proves (99).

The independence in (102) is independence of the kernel vectors, not necessarily of their image rows. Independent kernel vectors can have dependent images in a rank-deficient residual matrix; then their common determinant profile is zero, and the identity remains valid.

Finally, projecting the original row space onto the F coordinates has rank h . Its kernel is spanned by the outside rows and the combinations from K . The retained virtual lines span K , so deleting the zero F coordinates leaves rank $\text{rank } M - h = \nu(G) - h$. $\square$

For a single degree-three forced vertex, with forced-column entries $\gamma_1, \gamma_2, \gamma_3$ at its neighbors, the virtual rows can be written as

$$\gamma_2 r_1 - \gamma_1 r_2, \quad \gamma_3 r_1 - \gamma_1 r_3, \quad \gamma_3 r_2 - \gamma_2 r_3.$$

Their respective activities are $u_1u_2/E_1, u_1u_3/E_1, u_2u_3/E_1$, with $E_1 = u_1 + u_2 + u_3$. Their pair-activity sum is $u_1u_2u_3/E_1$, giving a concrete instance of (102).

22.3 Forcing exterior vertices in a genuine cover

Write a bipartite graph as $X = A \sqcup L, Y = B \sqcup R$, with $A \cup B$ a minimum vertex cover, $|A| = a$ and $|B| = q$. Call the displayed cover genuine when $\nu(G) = a + q$. There are no L – R edges. Any maximum matching then consists of an a -matching $A \rightarrow R$ and a disjoint q -matching $L \rightarrow B$: an edge inside the cover would consume two cover vertices and prevent attaining its size.

Theorem 22.4. *Suppose $a \geq 3$ and the displayed $a + q$ cover is genuine. Let $F \subseteq R$ be any matchable set of $h = a - 2$ vertices. For arbitrary positive activities on both shores,*

$$P_G^{(F)}(t) = C_F t^h Q_F(t), \quad (103)$$

where $C_F > 0$ and Q_F has actual degree $q + 2$ and is ULC_{q+2} . Consequently the entire conditioned sector is ULC_{a+q} .

Proof. The neighborhood $N(F)$ is contained in A and has size $h, h + 1$, or $h + 2$. Slack zero deletes h core-left vertices and leaves a genuine $2 + q$ cover. Slack one leaves one unchanged core-left vertex and one virtual union vertex, again giving a genuine $2 + q$ cover. A generic matrix of either graph has a rank-two exterior-column block and total rank $q + 2$, so Theorem 22.1 applies.

In slack two, $N(F) = A$. The generic exterior block originally has rank a , and its h forced columns are independent. Eliminating those columns reduces its rank to $a - h = 2$. The total compressed rank is $a + q - h = q + 2$. Thus Theorems 22.3 and 22.1 give (103), with the compressed object interpreted as a regular-minor matrix.

Multiplication by t^h weakens the required Newton ratios. At coefficient index j of a degree- d polynomial, $d = q + 2$, the required ratio changes from

$$\frac{(j+1)(d-j+1)}{j(d-j)} \quad \text{to} \quad \frac{(j+h+1)(d-j+1)}{(j+h)(d-j)}.$$

The latter is smaller. This proves ULC at order $d + h = a + q$ for the shifted sector. $\square$

[Merge note: Compare source 23's Theorem 24.1, which forces all of B and exterior-left vertices instead.] Further forcing is a coefficientwise limit at the same fixed normalization. In particular, for a genuine $3 + 3$ cover, conditioning on any exterior-right vertex leaves t times a degree-five ULC polynomial, for arbitrary fields on both shores. Every further conditioned sector containing such a vertex is safe. Among three-right forced sets, only the all- B set escapes this reduction. This does not prove the original polynomial is ULC at arbitrary finite activities.

22.4 Eventual strictness and an explicit threshold

Theorem 22.5. *Under the assumptions of Theorem 22.4, multiply just the activities in F by a common $\lambda > 0$, keeping all other activities fixed. The full, unconditioned polynomial is strictly ULC_{a+q} for all sufficiently large λ . The threshold depends on the fixed graph and activities.*

Proof. Put $h = |F| \geq 1$, $d = q + 2$, and $r = h + d$. Write $p_k(\lambda)$ for the full coefficient. For $0 \leq k \leq h$,

$$p_k(\lambda) = b_k \lambda^k + O(\lambda^{k-1}),$$

where (b_k) is the coefficient sequence of the graph restricted to right shore F . It has rank h and is ULC_h . One can see this directly from the usual basis matroid on its original row vectors and private coordinate elements in dimension h ; positive specialization gives its degree- h homogenization. For $0 \leq j \leq d$,

$$p_{h+j}(\lambda) = c_j \lambda^h + O(\lambda^{h-1}),$$

where (c_j) is a positive scalar multiple of the ULC_d sequence in Theorem 22.4. All b_k, c_j are positive, and $b_h = c_0$.

For $k < h$, the coefficient of λ^{2k} in the order- r gap is at least

$$\frac{(k+1)d}{h-k} b_{k-1} b_{k+1} > 0,$$

by the stronger ULC_h inequality. At $k = h$, the square has order λ^{2h} while the adjacent product has at most order λ^{2h-1} ; the leading coefficient is $hdc_0^2 > 0$. Above it, at $k = h + j$ with $1 \leq j < d$, ULC_d gives a leading coefficient at least

$$\frac{h(d-j+1)}{j} c_{j-1} c_{j+1} > 0$$

at order λ^{2h} . Every gap polynomial therefore has a positive leading coefficient. There are finitely many gaps, proving eventual strictness. $\square$

For an explicit coefficient bound, write each gap as

$$G_k(\lambda) = L_k \lambda^{m_k} + \sum_{j < m_k} g_{kj} \lambda^j, \quad L_k > 0,$$

and set

$$\Lambda = 1 + \max_k \frac{\sum_{j < m_k} \max\{-g_{kj}, 0\}}{L_k}. \quad (104)$$

For every $\lambda \geq \Lambda$, the leading term dominates all negative terms, since $\lambda \geq 1$ and

$$G_k(\lambda) \geq \lambda^{m_k-1} \left(L_k \lambda - \sum_{j < m_k} \max\{-g_{kj}, 0\} \right) > 0.$$

This is a field-dependent bound, not a uniform numerical threshold. With exact numerical activities, it is directly computable from the finitely many coefficient polynomials.

22.5 Verification and scope

Run `python3 checks/run_all.py`. [*Merge note:* Shipped as `code/60-rank-two-blocks-run_all.py` and `code/60-rank-two-blocks-check_*.py`, outputs `data/60-rank-two-blocks-*`; rerun on a copy of the delivered layout (README).] The package independently checks 32 rank-two-block basis identities, 32 degree-three contractions, 50 general kernel compressions with $h = 1, 2, 3, 4$, and 200 graph conditioning identities. The kernel cases include omitted and parallel kernel lines. Further exact checks verify all 8,433 positive minors in the rank-three scope example and the leading coefficients and explicit threshold on 36 genuine covers with positive integer activities. The universal proofs are the determinant-sector, kernel, and leading-coefficient arguments above; finite examples are regression checks.

The regular-minor matrix after slack-two compression must not be replaced by its entry-support graph. Its linear dependencies are part of the construction. Likewise, order $q + 2$

is actual-rank ULC only when the total matrix rank is $q + 2$. The conditioned and eventual conclusions do not assert ULC at every finite choice of positive activities on the original graph. In particular, the general rank-six one-shore question, including its all-core-right cubic limit, remains unresolved here. [*Merge note:* Within the batch, at matching number six with one weighted shore, gap one holds by *preorder-root-polytopes*'s Theorem 180.2, gap two by source 05 (Section 26) and gap five by source 13 (Section 25), at every activity; gaps three and four are open in general. The all-core-right cubic limit is source 21's theorem (Part IV).]

The note uses the matroid Lorentzian theorem and the regular-minor viewpoint of bimatroids [1, 7]; it makes no exhaustive literature-priority claim. The arguments and exact verifiers received independent mathematical review. They are not formal theorem-prover verification or external peer review.

23 Lorentzian marginals on three vertices and star cores, both shores weighted (source 10)

Abstract of source 10, as delivered. For a bipartite graph with three vertices on one shore, fix arbitrary positive activities on the other shore. The degree-three homogenization of its selected-vertex marginal is Lorentzian. The proof combines rank-three matroid Rayleigh inequalities with an exact positive-coefficient certificate for one Hessian determinant. An endpoint-set vertex-sum identity then proves weighted rank-normalized ultra log-concavity for every genuine $3 + q$ cover whose core is a star centered on the three-vertex shore. In particular, both star orientations of a $3 + 3$ cover satisfy all five rank-six inequalities for arbitrary positive activities on both shores. No general rank-three-block recombination theorem is asserted.

23.1 The selected-vertex marginal

Let $G = (X, Y)$ be bipartite, with $X = \{1, 2, 3\}$, and give each $y \in Y$ an activity $v_y > 0$. For $I \subseteq X$, define

$$c_I = \sum_{\substack{J \subseteq Y, |J|=|I| \\ G[I, J] \text{ perfectly matchable}}} \prod_{y \in J} v_y, \quad c_\emptyset = 1.$$

Each pair of endpoint sets is counted once, regardless of the number of perfect matchings realizing it. Write c_i, c_{ij}, c_{123} for the corresponding coefficients.

Theorem 23.1. *The homogeneous cubic*

$$H(s, z_1, z_2, z_3) = s^3 + \sum_i c_i z_i s^2 + \sum_{i < j} c_{ij} z_i z_j s + c_{123} z_1 z_2 z_3 \quad (105)$$

is Lorentzian.

For a homogeneous polynomial with nonnegative coefficients and M -convex support, the Lorentzian criterion is that each quadratic coordinate derivative has a Hessian with at most one positive eigenvalue [1, Theorem 2.25]. For a cubic these are its first coordinate derivatives. We verify them directly. Positive left activities can subsequently be introduced by nonnegative linear substitution.

23.2 Three Rayleigh inequalities

First suppose that every c_I is positive. Represent the right vertices by columns whose entries at graph edges are algebraically independent, and adjoin the three private coordinate columns

e_1, e_2, e_3 . This gives a rank-three transversal matroid. A generic minor is nonzero exactly when its endpoints are matchable, since distinct matchings give distinct determinant monomials.

Every rank-three matroid is Rayleigh [19, Theorem 1.1]. Set the private variable at i to zero, retain the other two private variables, and specialize all right variables to their activities. For $\{i, j, k\} = \{1, 2, 3\}$, its basis polynomial is

$$c_{123} + x_j c_{ik} + x_k c_{ij} + x_j x_k c_i.$$

The Rayleigh inequality, extended to zero specializations by continuity, gives

$$c_{ij} c_{ik} \geq c_i c_{123}. \quad (106)$$

The nonzero Hessian block of $\partial_{z_i} H$ is

$$\begin{pmatrix} 2c_i & c_{ij} & c_{ik} \\ c_{ij} & 0 & c_{123} \\ c_{ik} & c_{123} & 0 \end{pmatrix}.$$

Its determinant is $2c_{123}(c_{ij}c_{ik} - c_i c_{123}) \geq 0$. Its lower principal block has one positive and one negative eigenvalue. Interlacing and the determinant therefore imply that the full block has at most one positive eigenvalue.

23.3 The remaining Hessian

The Hessian of $\partial_s H$ is

$$\begin{pmatrix} 6 & 2c_1 & 2c_2 & 2c_3 \\ 2c_1 & 0 & c_{12} & c_{13} \\ 2c_2 & c_{12} & 0 & c_{23} \\ 2c_3 & c_{13} & c_{23} & 0 \end{pmatrix}.$$

Eliminating its positive first pivot reduces the assertion to $Q \succeq 0$, where

$$Q_{ii} = \frac{2}{3}c_i^2, \quad Q_{ij} = \frac{2}{3}c_i c_j - c_{ij} \quad (i \neq j). \quad (107)$$

The one- and two-dimensional principal minors are nonnegative: $0 \leq c_{ij} \leq c_i c_j$ gives $-\frac{1}{3}c_i c_j \leq Q_{ij} \leq \frac{2}{3}c_i c_j$. We prove the remaining determinant inequality by an exact finite identity.

Group right vertices by their nonempty neighborhoods $S \subseteq \{1, 2, 3\}$. Let

$$m_S = \sum_{y: N(y)=S} v_y, \quad h_S = \sum_{y: N(y)=S} v_y^2.$$

For distinct i, j , put

$$s_{ij} = \sum_{S \supseteq \{i, j\}} m_S, \quad h_{ij} = \sum_{S \supseteq \{i, j\}} h_S, \quad c_i = \sum_{S \ni i} m_S.$$

Exact endpoint counting yields

$$c_{ij} = c_i c_j - \frac{s_{ij}^2 + h_{ij}}{2}. \quad (108)$$

Indeed, $c_i c_j$ includes the forbidden repeated use of each common neighbor, and counts each distinct pair of common neighbors twice. Its excess is the sum of their squared activities plus their second elementary symmetric sum.

Encode neighborhoods by binary masks $1, \dots, 7$. For the four nonsingleton classes set

$$y_S = \sqrt{h_S}, \quad z_S = m_S - y_S \geq 0. \quad (109)$$

Then $m_S = y_S + z_S$ and $h_S = y_S^2$ exactly. The singleton classes enter only through their masses x_1, x_2, x_4 . Thus $M = 6Q$ has entries

$$M_{ii} = 4c_i^2, \quad M_{ij} = -2c_i c_j + 3(s_{ij}^2 + h_{ij}), \quad (110)$$

as polynomials in the eleven nonnegative variables

$$(x_1, x_2, x_4, y_3, z_3, y_5, z_5, y_6, z_6, y_7, z_7).$$

Lemma 23.2. *After the substitutions above, $\det M$ has exactly 5339 nonzero monomials. Every coefficient is a positive integer; the least is 36 and the greatest is 10656.*

Proof. This is the exact coefficient identity in the accompanying `data/determinant.json`. The generator constructs (110) from the neighborhood sums and expands the six permutation terms using sparse integer arithmetic. An independently written checker uses explicit neighborhood forms and, for a symmetric matrix $\begin{pmatrix} a & d & e \\ d & b & f \\ e & f & c \end{pmatrix}$, expands

$$abc + 2def - af^2 - be^2 - cd^2.$$

Both reconstructions agree coefficient by coefficient and check that every listed coefficient is positive and every exponent vector has total degree six. The complete identity, rather than a numerical evaluation or residual tolerance, is the certificate. Section 23.5 specifies its replay. $\square$

Consequently $\det Q \geq 0$. All principal minors of Q are nonnegative, so $Q \succeq 0$. This proves the required Hessian condition. If all $c_I > 0$, the support of (105) is the degree-three homogenization of the independent sets of a free matroid on three elements; it is M -convex. The cubic criterion proves Theorem 23.1 in this case. For an arbitrary graph, add at each left vertex a private right leaf of activity $\varepsilon > 0$. All eight coefficients become positive. Let $\varepsilon \downarrow 0$ and use closure of the Lorentzian cone to obtain the theorem, including isolated vertices and deficient matching rank.

23.4 Gluing at one left vertex

For a polynomial $p(t) = \sum_{k=0}^D p_k t^k$, write $p \in \text{ULC}_D$ when it has no internal zeros and

$$k(D-k)p_k^2 \geq (k+1)(D-k+1)p_{k-1}p_{k+1}, \quad 1 \leq k < D.$$

Equivalently, its degree- D bivariate homogenization is Lorentzian [1, Example 2.26].

Theorem 23.3. *Suppose G has a vertex cover $A \sqcup B$ with $|A| = 3$, $|B| = q$, and all edges of $G[A, B]$ are incident to one vertex $v \in A$. For arbitrary positive activities on both shores, $P_G \in \text{ULC}_{q+3}$. If the cover is minimum, this is ULC at the actual matching rank $q + 3$.*

Proof. Write the shores as $A \sqcup L$ and $B \sqcup R$. Since $A \sqcup B$ is a cover, there are no L - R edges. Split G into G_1 on $(L \sqcup \{v\}, B)$ and G_2 on (A, R) ; these meet only at v . Write

$$p_1 = A(t) + u_v t B(t), \quad p_2 = C(t) + u_v t D(t).$$

Then

$$P_G = AC + u_v t(BC + AD). \quad (111)$$

If v is omitted, both selected components are balanced. If v is selected, the component using it is determined uniquely by its imbalance away from v : its opposite shore has one extra selected vertex. Thus (111) counts each feasible endpoint support once.

For $q \geq 1$, the right-complement transversal basis polynomial of G_1 gives the degree- q Lorentzian polynomial

$$F(s, t, x) = A_h(s, t) + xB_h(s, t), \quad A_h = s^q A(t/s), \quad B_h = s^{q-1} B(t/s).$$

Explicitly, take the transversal matroid on the left vertices and one private element for each $b \in B$, with the B vertices as its q slots. Substitute s/v_b for the private element, tu_ℓ for each $\ell \in L$, and x for v , and multiply by $\prod_{b \in B} v_b$. This is Lorentzian because every matroid basis polynomial is Lorentzian, and nonnegative substitutions preserve the class [1, Theorems 3.10 and 2.10].

[Merge note: Identity (111) is the vertex-sum identity of source 74 (display (13)) and of *preorder-root-polytopes*'s Part VI; source 74's star transfer (Section 10) treats $2 + q$ covers in the same way.] Theorem 23.1, with the other two left variables specialized to tu_i and the root variable to x , gives

$$K(s, t, x) = C_h(s, t) + xD_h(s, t), \quad C_h = s^3 C(t/s), \quad D_h = s^2 D(t/s).$$

The product FK is Lorentzian [1, Corollary 2.32]. Retain only terms of x -degree at most one, then set $x = u_v t$. The result is

$$A_h C_h + u_v t (B_h C_h + A_h D_h) = s^{q+3} P_G(t/s),$$

which is Lorentzian. The case $q = 0$ is Theorem 23.1 itself. $\square$

For completeness, the partial truncation used here preserves the Lorentzian property directly. Its support is the intersection of an M -convex set with the coordinate bound $\deg_x \leq 1$, which is again M -convex when nonempty. In a quadratic coordinate derivative, a differentiation using no x simply removes the nonnegative x^2 coefficient from the original quadratic; this subtracts a positive semidefinite diagonal Hessian entry and cannot increase the positive index. A derivative using x once is the corresponding derivative of $\partial_x(FK)$ specialized at $x = 0$. A derivative using x twice or more vanishes. The Hessian criterion gives the assertion.

Corollary 23.4. *Every genuine $3 + 3$ -cover graph whose core has matching rank at most one has $P_G \in \text{ULC}_6$ for arbitrary positive activities on both shores.*

Proof. A nonempty bipartite graph of matching rank one is a star. A row star is Theorem 23.3; a column star follows after interchanging the shores. For an empty core, the polynomial is a product of two smaller-shore ULC_3 polynomials, so product closure applies. $\square$

The proof also retains every non-root left activity as its own variable: the selected-left marginal of the row-star cover, homogenized to degree $q + 3$, is Lorentzian. The scalar conclusion then allows arbitrary left activities. This vertex-sum result does not provide a rank-three exterior-block lift for general cores.

23.5 Exact verification and scope

The source package contains the manuscript, the complete determinant coefficient list and three standard-library Python checks. [Merge note: Shipped as `code/10-three-vertex-*.py` and `data/10-three-vertex-*.json`; the determinant list is `data/10-three-vertex-determinant.json`.] From its root run

```
python -O code/replay_all.py.
```

The two independent polynomial reconstructions produce the same lexicographically ordered coefficient list. Its canonical SHA-256 is

The third checker independently enumerates endpoint sets in 240 weighted three-left graphs and 160 weighted star-core graphs, including 38 genuine minimum covers. It verifies the overlap identity, Rayleigh specialization, rooted gluing and all fixed-cover Newton inequalities. These finite regressions supplement the proof; only the complete determinant expansion is used as a finite certificate for a universal assertion.

The result concerns all fields in the stated star-core class. The general finite-activity one-shore rank-six problem remains open at its three middle Newton gaps. [*Merge note:* After source 05 only two middle gaps remain: at matching number six with one weighted shore, gap one holds by *preorder-root-polytopes*'s Theorem 180.2, gap two by source 05 (Section 26) and gap five by source 13 (Section 25), at every activity; gaps three and four are open in general.] No universal real-rootedness or arbitrary-core Lorentzian recombination is claimed. The standard Lorentzian and Rayleigh theorems are attributed below; no exhaustive priority claim is made.

24 Rooted Lorentzian Hall sectors and an unrooted cubic obstruction (source 23)

Abstract of source 23, as delivered. For a genuine $a + q$ bipartite cover, condition on all q right-core vertices and on any matchable $q - 2$ exterior-left vertices. The remaining degree- a exterior-right marginal is Lorentzian, for arbitrary positive activities on both shores. This follows from exact weighted Hall-slack compression and the rank-two exterior-block matroid lift. The roots cannot simply be omitted: a connected 25-vertex rank-six graph, with unit left activities and a unique $3 + 3$ minimum cover, has an all-right-core conditional cubic failing both degree-three Newton inequalities. Its shifted degree-six inequalities remain strict, and its full polynomial is strictly ULC_6 for every common right-core activity. This identifies a normalization and recombination barrier without refuting rank-six ULC .

24.1 Matching supports and the rooted theorem

For a bipartite graph with left and right activities $u_x, v_y > 0$, its matching-support polynomial is

$$P_G(t) = \sum_{\substack{|I|=|J| \\ G[I,J] \text{ has a perfect matching}}} u_I v_J t^{|I|}.$$

Each feasible endpoint pair is counted once. Call a displayed cover $A \sqcup B$ genuine when

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = a, \quad |B| = q, \quad \nu(G) = a + q.$$

There are no L – R edges. Every maximum matching consists of disjoint exterior matchings of sizes a and q .

Theorem 24.1. *Let $a \geq 1$, $q \geq 2$, and let $F \subseteq L$ be any matchable set of $q - 2$ vertices. Condition simultaneously on selecting all B and all F . After removing the size factor t^q and fixed positive activity factors, the remaining polynomial has degree a and is ULC_a .*

More strongly, keep the exterior-right activities as separate variables. The degree- a homogenization

$$H_F(s, z_R) = \sum_{J \subseteq R} c_F(J) z_J s^{a-|J|}$$

is Lorentzian, where $c_F(J)$ sums the fixed left activities of feasible endpoint sets containing F and matched to $B \cup J$.

The theorem permits arbitrary positive activities on both shores. The fixed positive normalization does not affect the Lorentzian or Newton properties. In particular, in a genuine $3+3$ cover, forcing all three B vertices and one specified exterior-left vertex leaves a degree-three Lorentzian marginal.

24.1.1 Exact inputs

We use the following previously proved weighted results [21] [*Merge note*: source 60: Theorem 22.1, Proposition 22.2 and Theorem 22.3; identity (112) is source 60's (102). They are restated here as source 23 states them]; the compression is a regular-minor construction and need not produce an ordinary graph.

First, if a matrix consists of a ordinary columns and an exterior column block of rank two, with total rank $a + 2$, its degree- $(a + 2)$ left marginal of nonzero square minors is Lorentzian after fixing positive column activities.

Second, a matchable forced set of size h with h , $h + 1$ or $h + 2$ neighbors admits exact weighted compression. Slack zero deletes the forced set and its neighbors. Slack one replaces the neighbors by a weighted union-profile element. Slack two replaces them by a two-dimensional kernel-line family. The compressed matrix has rank reduced by h , and its support polynomial recovers the conditioned polynomial after a positive factor and t^h .

For precision, in the slack-two case let Γ be the forced-column matrix on its $h+2$ neighboring rows, write α_i for their activities, and put

$$E_h = \sum_{\substack{|I|=h \\ \det \Gamma[I] \neq 0}} \alpha_I.$$

Each eligible omitted index i gives a kernel line in $\ker(\Gamma^\top) \cap \{x_i = 0\}$, with virtual activity $\alpha_{N \setminus \{i\}}/E_h$. Independent kernel pairs have a common residual determinant profile, and their total activity is

$$\sum_{\text{independent kernel pairs } i < j} \frac{\alpha_{N \setminus \{i\}} \alpha_{N \setminus \{j\}}}{E_h^2} = \frac{\alpha_N}{E_h}. \quad (112)$$

This identity counts the full neighbor set once. Independence is in the kernel, not necessarily in the image rows; zero image profiles contribute zero. These exact weighted identities are essential.

24.1.2 Proof by transposed compression

Proof of Theorem 24.1. For $q = 2$, use the rank-two-block marginal theorem directly after transposition, with the a original A elements as ordinary columns. Extracting both B rows gives degree a .

For $q \geq 3$, transpose the graph and force the right set F , of size $h = q - 2$. Its neighbor set N lies in the original B , so $h \leq |N| \leq h + 2$. Apply the exact compression above. The residual matrix has rank $a + 2$.

In the transposed residual matrix, the original A elements remain exactly a ordinary columns. All other columns come from $L \setminus F$. Their span originally had rank q and was supported only on B . Eliminating the h independent forced directions reduces this exterior-column rank to $q - h = 2$. The kernel family spans the residual directions. Thus the

rank-two-block theorem makes the residual degree- $(a + 2)$ left marginal Lorentzian. Its left variables are the retained or virtual original-right variables.

Selecting all original B now has a uniform description:

- If $|N| = h$, exactly two original B rows remain, and both must be selected.
- If $|N| = h + 1$, one original B row and one union-profile virtual row must be selected.
- If $|N| = h + 2 = q$, an independent pair from the kernel-line family must be selected. Equation (112) recovers the original full- N activity exactly.

Call these distinguished original or virtual rows D . Give each its assigned positive activity times a common variable y , while keeping all z_R separate. Nonnegative linear substitution preserves the Lorentzian property [1]. The y -degree is at most two: there are two distinguished elements in the first cases, and the virtual family has rank two in the last.

Apply $\frac{1}{2}\partial_y^2$ and set $y = 0$. Derivatives and nonnegative boundary specialization preserve Lorentzianity. The result is the coefficient of y^2 , homogeneous of degree a , and the exact compression identities identify it, up to a positive factor, with H_F . The total size shift is $h + 2 = q$, and the power of s is $(a + 2) - 2 - |J| = a - |J|$.

Finally, F extends to a size- q exterior-left basis. Together with an a -matching from A to R , this gives a maximum matching containing F and all B . The top coefficient is positive; the same left basis gives a positive constant coefficient. Thus the actual residual degree is a , as claimed. $\square$

24.2 Why the roots cannot be omitted

[*Merge note:* This 25-vertex graph is also the example of source 05's last section (Section 26), which uses it, without citing source 23, against a quintic Lorentzian route; it is printed once, here.] We construct a connected graph whose unrooted all- B cubic fails ULC_3 , although Theorem 24.1 applies after any one exterior-left vertex is additionally selected.

Take $A = \{A_1, A_2, A_3\}$ and $B = \{B_1, B_2, B_3\}$. The core edges are

$$A_1B_2, \quad A_2B_1.$$

There are eight exterior-left vertices, each adjacent to all of B . All left activities and the three B activities are one. The exterior-right vertices are:

neighborhood in A	population	activity per vertex
$\{A_1\}$	1	15
$\{A_2\}$	1	15
$\{A_1, A_2\}$	7	1
$\{A_3\}$	1	15
$\{A_1, A_3\}$	1	1/10

The graph has 25 vertices and 45 edges. Three common exterior-left vertices match all B , and the three private exterior-right vertices match all A , so its matching rank is six.

Its minimum cover is uniquely $A \cup B$. Omitting any B vertex would require all eight common left neighbors, impossible in a six-vertex cover. Once all B are selected, omitting either A_1 or A_2 would require all seven common right neighbors. The remaining one-vertex budget must include A_3 , because its private neighbor and the hybrid neighbor are distinct. The hybrid joins the former A_3 star to the other component, so the graph is connected.

24.2.1 Exact conditional coefficients

First give the hybrid vertex activity ε . Removing it leaves an isolated A_3 star and a quadratic factor. Direct support counting gives

$$C_0(t) = (1 + 15t)(120 + 3500t + 25536t^2). \quad (113)$$

Indeed, with $n = 8$, $p = \binom{8}{2} = 28$, $q = \binom{8}{3} = 56$, the constant quadratic coefficient is $q + 2p + n = 120$. Its linear coefficient is $2(q + p)15 + (2q + p)7 = 3500$, and its quadratic coefficient is $q(15^2 + 30 \cdot 7 + \binom{7}{2}) = 25536$.

The hybrid contributes

$$C_\varepsilon(t) = C_0(t) + \varepsilon(204t + 5992t^2 + 44016t^3). \quad (114)$$

For the first increment, $2q + 3p + n = 204$. Pairing the hybrid with the four old right classes contributes

$$15(q + p) + 15(2q + p) + 7(3q + p) + 15(q + p) = 5992.$$

The top increment is $q(2 \cdot 15^2 + 3 \cdot 15 \cdot 7 + \binom{7}{2}) = 44016$. These are endpoint-support counts, not counts of matching allocations.

At $\varepsilon = 1/10$, the conditioned polynomial is $t^3 C(t)$, with

$$C(t) = 120 + \frac{26602}{5}t + \frac{393176}{5}t^2 + \frac{1937208}{5}t^3. \quad (115)$$

Multiply by the harmless positive factor five, and write

$$\hat{C}(t) = 600 + 26602t + 393176t^2 + 1937208t^3.$$

Its degree-three Newton gaps are

$$26602^2 - 3 \cdot 600 \cdot 393176 = -50396,$$

$$393176^2 - 3 \cdot 26602 \cdot 1937208 = -13454672.$$

Both fail strictly. Its cubic discriminant is -9199780352 , so scalar real-rootedness fails as well.

By contrast, the two nontrivial shifted degree-six gaps are

$$8 \cdot 26602^2 - 15 \cdot 600 \cdot 393176 = 2122747232 > 0,$$

$$5 \cdot 393176^2 - 12 \cdot 26602 \cdot 1937208 = 154533548288 > 0.$$

Thus this is not a counterexample to the required rank-six normalization. [*Merge note:* The tempting false reading: this cubic does not contradict source 21's all-core bounds (Part IV). They are bounds for the *shifted* degree-six inequalities, with constants $2n/(n-1)$ and $21/8$, both below three; on this graph the two ratios are $2.99979 \geq 16/7$ and $2.99974 \geq 21/8$ (recomputed at placement). Only the *unshifted* degree-three inequalities, with constant three, fail.]

24.2.2 The recombination obstruction

Forcing a specified common exterior-left vertex replaces each factor $\binom{8}{j}$ counting selected common left vertices by $\binom{7}{j-1}$. Summing over the eight possible roots therefore counts a support j times when it uses j exterior-left vertices:

$$8 \binom{7}{j-1} = j \binom{8}{j}.$$

All eight rooted marginals are Lorentzian by Theorem 24.1; they do not recover the unrooted marginal with its required multiplicity one. This example shows that the normalization loss cannot be repaired by simply dropping the roots.

24.3 The original graph is safe under every common core field

There is a stronger check for this particular graph. Keep the exterior-right activities as above, and give every B vertex a common activity $w > 0$. Direct endpoint enumeration gives the full support coefficients

k	$p_k(w)$
0	1
1	$296/5 + 26w$
2	$11241/10 + (14951/10)w + 117w^2$
3	$34593/5 + (276443/10)w + (61207/10)w^2 + 120w^3$
4	$(830232/5)w + (521166/5)w^2 + (26602/5)w^3$
5	$(2905812/5)w^2 + (393176/5)w^3$
6	$(1937208/5)w^3$

Put $G_k = k(6 - k)p_k^2 - (k + 1)(7 - k)p_{k-1}p_{k+1}$. The third and fourth gaps have nonnegative coefficients and are nonzero. The remaining gaps have the exact forms

$$\begin{aligned}
5G_1 &= 20170 - 12746w + 9880w^2, \\
25G_2 &= 99127242 - 8903046w + 94261217w^2 + 7629855w^3 + 1567800w^4, \\
25G_5 &= w^4(22918732029648 - 690327945216w + 154533548288w^2).
\end{aligned}$$

The displayed quadratic parts have positive leading coefficients and respective discriminants

$$-634657884, \quad -37296153647015940, \quad -13690299259267221125283840.$$

They are positive for every real w . Hence every $G_k(w) > 0$ for $w > 0$. The original graph is strictly ULC_6 throughout this common-core-field family, while its all- B cubic fails ULC_3 . No statement for arbitrary independent B fields is inferred from this check.

24.4 Verification and remaining questions

The package [*Merge note:* shipped as `code/23-rooted-sectors-*.py` and `data/23-rooted-sectors-*.json`] includes a Hall-subset enumeration of the conditional coefficients, all four hybrid increments, the full common-field coefficient table and its exact gap certificates. A separate endpoint-state dynamic enumeration verifies the original 24-vertex construction. An exhaustive enumeration of all 2048 left-cover masks confirms the connected graph's unique minimum cover and rank.

The Lorentzian theorem depends on the exact weighted matrix/kernel compression inputs [21]. The residual object must not be replaced by its entry-support graph, since its linear dependencies matter. The counterexample uses unit left activities, positive right activities, a genuine $3 + 3$ cover and no degenerating zero field.

The general finite-activity rank-six one-shore ULC problem remains open. The present obstruction also does not refute an all-core eventual-scaling theorem at the weaker rank-six normalization. [*Merge note:* Such a theorem is source 21's Corollary (Part IV). On the open problem: at matching number six with one weighted shore, gap one holds by *preorder-root-polytopes*'s Theorem 180.2, gap two by source 05 (Section 26) and gap five by source 13 (Section 25), at every activity; gaps three and four are open in general.] It identifies why the blanket stronger scalar- ULC_3 and real-rootedness routes cannot prove that result. No global minimal-vertex or priority claim is made.

25 The last Newton inequality at matching number six, one weighted shore (source 13)

Abstract of source 13, as delivered. For every bipartite graph of matching rank six, with unit activities on one shore and arbitrary positive activities on the other, we prove the last matching-rank-normalized Newton inequality. The three core activities reduce the gap to a positive-definite quadratic in their reciprocals. Two matroid Hessians and a weighted plane-incidence inequality reduce its Schur complement to a quantitative two-point Rayleigh bound. In the graph-generic setting, that bound has ten support profiles; exact moment aggregation and 858 rational weighted-square terms certify all ten. The conclusion holds at every finite positive activity. The full rank-six ULC question remains open at the three middle gaps.

25.1 Statement and support convention

For a bipartite graph $G = (X, Y)$, define

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, J] \text{ perfectly matchable}}} u_I v_J t^{|I|} = \sum_k p_k t^k.$$

Each feasible pair of endpoint sets is counted once, regardless of how many perfect matchings realize it. Its degree is the matching rank.

Theorem 25.1. *If G has matching rank six, every $u_x = 1$, and every $v_y > 0$, then*

$$5p_5^2 \geq 12p_4p_6. \quad (116)$$

This is the last ULC_6 inequality. It concerns arbitrary finite positive activities. The first gap is already universally nonnegative by the edge-compatibility Motzkin–Straus argument [6]; the present result leaves only gaps 2, 3, 4 as possible failures at rank six. We do not claim full ULC_6 . [*Merge note:* The first gap is *preorder-root-polytopes*’s Theorem 180.2 (Part XI), printed with the Motzkin–Straus route in Section 11. Gap two is settled by source 05 (Section 26); only gaps three and four remain open in general.]

We use the following precise earlier result: if a bipartite graph has a minimum vertex cover with at most two vertices on either shore, its support polynomial is ULC_r at its actual matching rank r , for arbitrary positive activities on both shores. The two-vertex case is [12, Theorem 1.1]; the pure and singleton cases are proved in Section 2 of that report. [*Merge note:* Theorem 2.1 and Section 3 here; the pure and singleton cases are *preorder-root-polytopes*’s Theorem 206.2.] Thus, after choosing a minimum vertex cover, it suffices to treat a genuine $3 + 3$ cover

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = |B| = 3, \quad \nu(G) = 6.$$

There are no L – R edges. Each maximum matching avoids core edges, so both exterior blocks have rank three. Isolated vertices may be removed.

We use two established inputs: matroid basis polynomials are Lorentzian, with the usual derivative and nonnegative-specialization closures [1]; and every rank-three matroid is Rayleigh [19]. The new finite input is the ten exact identities in Section 25.6.

25.2 The exact reciprocal activity quadratic

Represent the graph by a matrix whose nonzero edge entries are algebraically independent. A square minor is nonzero exactly when its endpoint sets are matchable: distinct matchings give

distinct determinant monomials. Write K for the 3×3 core, ℓ_i for the exterior-left rows, and r_j for the exterior-right columns. Matrix indeterminates are separate from vertex activities.

Let q count independent triples in L , let p_i count pairs matching $B \setminus \{i\}$, and let n_i count left neighbors of B_i . All are positive. In the right rank-three matroid on R and the private coordinate basis $E = \{e_1, e_2, e_3\}$, let U, a, T be the weighted basis counts with one, two and three R elements, respectively. Its degree-three Newton inequality gives

$$a^2 \geq 3TU. \quad (117)$$

For each independent left pair P , choose a nonzero annihilator column v_P and introduce a virtual right vector $w_P = Kv_P$. Retain every pair label, including zero or parallel virtual vectors. Let P_B count weighted bases consisting of one virtual vector and two R columns, and let P_V count bases with one virtual vector, one R column and one private coordinate vector.

For a left row ℓ , let β_ℓ be the total activity of those r_j for which

$$\begin{pmatrix} K & r_j \\ \ell & 0 \end{pmatrix} \quad (118)$$

is nonsingular, and put $N = \sum_{\ell \in L} \beta_\ell$. Forcing all B and removing their activity product and t^3 gives a residual cubic with

$$C_1 = qU + P_V + N, \quad C_2 = qa + P_B, \quad C_3 = qT. \quad (119)$$

The support partition uses three, two or one selected left-exterior vertices. [*Merge note:* The same sector decomposition is source 21's (Part IV).] Eliminating a selected independent pair gives its virtual column, and complementary private coordinate vectors record the selected A rows once.

For each independent right pair J , choose a row normal b_J and introduce a virtual left row $z_J = b_J K$, with activity v_J . Let B_i count weighted bases (ℓ, e_i, z_J) . If exactly $B \setminus \{i\}$ is selected, its residual t^2 coefficient is

$$E_i = p_i a + B_i. \quad (120)$$

This is the same determinant elimination after transposition. Let D count weighted bases (ℓ, z_J, z_H) for distinct pair labels J, H . Their activity products may share original R factors.

Write $w_i > 0$ for the three B activities, $W = w_1 w_2 w_3$ and $\tau_i = 1/w_i$. Exact endpoint counting gives

$$\begin{aligned} p_6/W &= qT, \\ p_5/W &= C_2 + T \sum_i p_i \tau_i, \\ p_4/W &= C_1 + \sum_i E_i \tau_i + T \sum_{i < j} n_k \tau_i \tau_j, \quad \{i, j, k\} = \{1, 2, 3\}. \end{aligned} \quad (121)$$

For example, the top coefficient with exactly two B vertices uses all A and two L vertices, giving $p_i T$. With exactly one B vertex it gives $n_i T$.

Set $x_i = T \tau_i$. The last gap is

$$\frac{5p_5^2 - 12p_4 p_6}{W^2} = A_0 + 2c^\top x + x^\top Q x, \quad (122)$$

where

$$A_0 = 5C_2^2 - 12qTC_1, \quad c_i = 5C_2 p_i - 6qE_i,$$

and

$$Q_{ii} = 5p_i^2, \quad Q_{ij} = 5p_i p_j - 6q n_k \quad (i \neq j). \quad (123)$$

25.3 Two Schur bounds

Let $\mathcal{N}$ have zero diagonal and $\mathcal{N}_{ij} = n_k/2$. The rank-three left matroid on L and the private coordinates has basis polynomial

$$qt^3 + \sum_i p_i x_i t^2 + \sum_{i < j} n_k x_i x_j t + x_1 x_2 x_3.$$

Its derivative Hessian, with positive pivot $6q$, gives

$$H = pp^\top - 3q\mathcal{N} \succeq 0. \quad (124)$$

Thus $Q = pp^\top + 4H$ is positive definite. Indeed, a vector in its kernel is orthogonal to p and killed by H , hence by $\mathcal{N}$, whereas

$$\det \mathcal{N} = \frac{n_1 n_2 n_3}{4} > 0.$$

No simplicity, coloop or nonsingular-core assumption is needed.

Adjoin all virtual left rows z_J to this matroid, with a common variable multiplying their activities. The derivative Hessian in the original-left direction gives

$$\begin{pmatrix} H & d \\ d^\top & P_B^2 - 3qD \end{pmatrix} \succeq 0, \quad d = pP_B - \frac{3q}{2}B. \quad (125)$$

To check the factors, before taking its Schur complement the entries at the original-left pivot are $6q$, $2p_i$ and $2P_B$. The private block is $2\mathcal{N}$, its mixed virtual entries are B_i , and the virtual diagonal is $2D$. Eliminate $6q$ and multiply the resulting negative matrix by $-3q/2$.

From (120),

$$c = (P_B - qa)p + 4d.$$

The Cauchy–Schwarz inequality applied to (125) yields

$$c^\top Q^{-1}c \leq (P_B - qa)^2 + 4P_B^2 - 12qD. \quad (126)$$

For a direct verification, use $|d^\top y|^2 \leq (P_B^2 - 3qD)y^\top Hy$, then apply the two-term Cauchy–Schwarz inequality to $c^\top y$. The denominator is $y^\top Qy = (p^\top y)^2 + 4y^\top Hy$.

Put $\mathcal{R} = aP_B - TP_V$. Subtraction in (126) and (119) gives

$$A_0 - c^\top Q^{-1}c \geq 4q[q(a^2 - 3TU) + 3(\mathcal{R} - TN + D)]. \quad (127)$$

It remains to establish

$$D \geq \frac{3}{4}TN, \quad \mathcal{R} \geq \frac{1}{4}TN. \quad (128)$$

25.4 The weighted plane incidence bound

We recall the rank-three incidence lemma from the all-core conditional argument [22], including its short proof. [*Merge note:* Source 21’s Lemma (Part IV); it is the case $b = 3$ of source 08s’s two-plane incidence lemma (Part IV). It is printed with its proof here, where it is first used; Part IV refers back.] For a weighted rank-three configuration and a plane H_0 , let β be the mass outside H_0 , q_0 the weighted basis sum, and h_δ the pair mass whose span meets H_0 in the line δ . Put $h = \sum_\delta h_\delta$. Then

$$h^2 - \sum_\delta h_\delta^2 \geq \frac{3}{2}q_0\beta. \quad (129)$$

Adjoin diffuse generic plane points of total activity y and one point x_δ on each line. The limiting matroid basis cubic is

$$q_0 t^3 + h y t^2 + \frac{\beta}{2} y^2 t + \sum_{\delta} x_{\delta} ((h - h_{\delta}) t^2 + \beta y t) + \beta t \sum_{\delta < \eta} x_{\delta} x_{\eta}.$$

Its ∂_t Hessian has at most one positive eigenvalue. Eliminating its positive yy pivot β leaves line block $-\beta I$, cross entries $-2h_{\delta}$ and tt entry $6q_0 - 4h^2/\beta$. The remaining Schur complement is nonpositive, which is (129). Parallel elements and arbitrary positive weights are allowed.

Apply (129) to the weighted R configuration for each fixed ℓ . If K is invertible, the effective plane is $\ker(\ell K^{-1})$. If $\text{rank } K = 2$ and ℓ is nonzero on $\ker K$, it is $\text{im } K$. In the remaining rank-two cases, and when $\text{rank } K \leq 1$, $\beta_{\ell} = 0$.

In each nonzero case, independent mapped pairs of right-pair normals correspond exactly to distinct intersection lines with this plane. This follows by restricting the normals to the plane: their projective restrictions agree exactly when their kernels there agree. At rank two the quotient by the kernel of the transpose identifies with the image plane. Consequently the ordered virtual-left pair count at ℓ is at least $3T\beta_{\ell}/2$. The definition of D uses unordered pairs. Summing over the unit left rows proves the first inequality in (128).

25.5 The quantitative local lemma

Lemma 25.2. *Let E be the coordinate basis in dimension three. Let R consist of columns generic independently on specified nonempty coordinate supports, with positive activities. Let v_1, v_2 be two independent vectors over the field of the other graph entries, so they are independent of the R indeterminates.*

Let β be the total R activity outside $\text{span}(v_1, v_2)$. Let P count weighted bases with two R columns and one of v_1, v_2 , and let B count bases with one R column, one coordinate element and one of v_1, v_2 . Then

$$aP - TB \geq \frac{1}{2}T\beta. \quad (130)$$

The graph-generic hypothesis is explicit. The analogous quantitative assertion for arbitrary numerical configurations is not needed and is not proved here.

Write $S_1 = \text{supp}(v_1)$, $S_2 = \text{supp}(v_2)$ and $S_0 = \text{supp}(v_1 \times v_2)$. A determinant involving one virtual vector and one or two generic R columns is nonzero exactly when the coordinate masks permit a matching. Distinct monomials in the independent R entries cannot cancel. The outside-plane indicator for a column of type S is $\mathbf{1}_{S \cap S_0 \neq \emptyset}$. Thus the actual normal support is retained; the plane is not replaced by a generic plane.

Up to coordinate permutations and swapping the two virtual vectors, there are ten profiles, listed in Table 6. We use masks 1, 2, 4 for the three coordinates. If $|S_0| = 1$, the plane is a coordinate plane and gives the first three types. If $|S_0| = 2$, it is spanned by the remaining coordinate axis and a line supported on the other two coordinates, giving four types. If $|S_0| = 3$, every nonzero point has support of size at least two; equal size-two supports would give dependent vectors, leaving three types.

Equivalently, on common support coordinates, a zero minor means equality of the ratios $(v_1)_i/(v_2)_i$. Set partitions of these ratios, with the one-product minors forced nonzero, give exactly 55 labeled nonparallel profiles and ten orbits. This is an exhaustive classification, rather than a vector sample.

25.6 Moments and exact finite identities

In one coordinate class, with positive activities u_j , set

$$s = \sum_j u_j, \quad y = \sqrt{\sum_j u_j^2}, \quad z = s - y \geq 0.$$

Its elementary symmetric moments satisfy

$$e_1 = y + z, \quad e_2 = yz + \frac{z^2}{2}, \quad e_3 \leq \frac{yz^2}{2} + \frac{z^3}{6}. \quad (131)$$

The last inequality follows from $\sum_j u_j^3 \leq (\sum_j u_j^2)^{3/2}$. Only e_1 matters for a singleton coordinate class, only e_1, e_2 for a size-two class, and only the full-support class can contribute e_3 to T .

Replace the full-support e_3 by its upper bound. This lowers $2aP - T(2B + \beta)$, since its multiplier is nonnegative. The other used moments are exact. Denote the resulting quartic by F_{S_1, S_2, S_0} , in the eleven nonnegative variables

$$X = (x_1, x_2, x_4, y_3, z_3, y_5, z_5, y_6, z_6, y_7, z_7).$$

The first three are singleton-class totals.

For complete specification, let $\chi(M_1, \dots, M_k)$ be the indicator that the masks admit distinct coordinate representatives. For a multiset I of right types, put $e(I) = \prod_S e_{\text{mult}_I(S)}(S)$, using (131). Then

$$\begin{aligned} a &= \sum_{|I|=2} e(I) \sum_{e \in \{1,2,4\}} \chi(I, e), \\ P &= \sum_{|I|=2} e(I) (\chi(I, S_1) + \chi(I, S_2)), \\ T &= \sum_{|I|=3} e(I) \chi(I), \\ B &= \sum_S e_1(S) \sum_{e \in \{1,2,4\}} (\chi(S, e, S_1) + \chi(S, e, S_2)), \\ \beta &= \sum_{S \cap S_0 \neq \emptyset} e_1(S), \quad F = 2aP - T(2B + \beta). \end{aligned}$$

Multisets and elementary symmetric moments count distinct original column labels exactly once.

For every profile the supplied rational data give the exact identity

$$F = \sum_j \frac{\lambda_j X^{\mu_j}}{2r_j} (r_j X^{m_j} - X^{n_j})^2 + \sum_\alpha d_\alpha X^\alpha. \quad (132)$$

All exponents are nonnegative integers; all λ_j, r_j, d_α are positive rationals. Every multiplier X^{μ_j} is therefore nonnegative on the entire moment domain. There are no residual tolerances in (132). This proves $F \geq 0$, hence Lemma 25.2.

The accompanying standard-library checker independently reconstructs all ten targets from the matching rule and expands every rational term of (132). It checks 8680 target monomials. A numerical linear program found the decompositions; exact rational reconstruction and exact coefficient equality are the proof-relevant steps. Two active-basis solutions were solved again over the rationals, and every final residual is exactly zero.

Table 6: The exhaustive virtual profiles and exact certificate sizes.

S_1	S_2	S_0	Weighted squares	Positive monomials
1	2	4	64	716
1	3	4	61	718
3	3	4	54	728
4	3	3	74	839
4	7	3	122	792
3	7	3	90	821
7	7	3	93	798
3	5	7	95	807
3	7	7	107	807
7	7	7	98	775
Total			858	7801

25.7 Summing the local bound

For each ℓ with $\beta_\ell > 0$, choose two other left rows completing it to a basis. The normals of the two pairs containing ℓ span $\ker \ell$. Their images under K are independent and span the effective plane whose outside R activity is β_ℓ . Indeed, nonsingularity of (118) for some r_j forces the image of $\ker \ell$ to have dimension two; the rank cases were identified above. Apply Lemma 25.2 to these two virtual points.

For any individual virtual point w_P , the ordinary rank-three Rayleigh theorem gives

$$aP_B(w_P) - TP_V(w_P) \geq 0.$$

This follows by summing the Rayleigh inequalities between w_P and the three private coordinate elements, and setting their external activities to zero by continuity. It includes zero and parallel virtual points.

Each labeled left pair can be chosen at most twice, once at each endpoint. Summing the local inequalities and using nonnegativity of every unused individual slack gives

$$2(aP_B - TP_V) \geq \frac{1}{2}T \sum_{\ell} \beta_{\ell}.$$

This is the second bound in (128). Now (117), (127) and (128) imply $A_0 - c^\top Q^{-1}c \geq 0$. Finally,

$$A_0 + 2c^\top x + x^\top Qx = (x + Q^{-1}c)^\top Q(x + Q^{-1}c) + A_0 - c^\top Q^{-1}c \geq 0.$$

Equations (122) and (116) prove Theorem 25.1. In fact the reciprocal-core quadratic is nonnegative without restricting x to the positive orthant.

25.8 Reproduction and remaining scope

The package [*Merge note*: shipped as `code/13-last-newton-*.py` and `data/13-last-newton-*.json`] includes the complete ten-profile rational certificate, an independent standard-library replay, the exhaustive ratio-partition orbit checker, and the endpoint-set dynamic program. The latter checks the top-three coefficient formulas, positive definiteness and 1536 full quadratic identities on 512 deterministic graph shapes. These finite graph checks are supplementary; the universal proof is the reduction and exact identities above.

The theorem concerns the last gap at every finite positive one-shore activity. It strengthens the earlier eventual-scaling conclusion in this particular gap. The three middle rank-six

inequalities remain open. [*Merge note:* The earlier eventual-scaling conclusion is source 21's (Part IV). Of the three middle inequalities, the second is settled by source 05; the third and fourth remain open.] The unrestricted-vector quantitative local lemma, optimal constants, equality classification for the new last-gap theorem and global novelty priority are not claimed. These are ordinary mathematical proofs with independently checked exact certificates, not a formal-kernel proof or an externally refereed publication.

26 The second Newton inequality at matching number six, one weighted shore (source 05)

Abstract of source 05, as delivered. For every bipartite graph of matching rank six, with unit activities on one shore and arbitrary positive activities on the other, we prove the second matching-rank-normalized Newton inequality. For a genuine $3 + 3$ minimum cover, we prove the stronger Lorentzian property of a degree-three truncation in the individual right variables. A neighbor-feature Gram identity and exact kernel compression settle its homogenizing and exterior derivatives. Each remaining core derivative reduces to a two-dimensional Schur complement. The empty core column has a direct proof; the other columns have an exact certificate on 61 rooted cores and 51 exterior-basis cones, comprising 50 037 694 nonnegative coefficient checks. The third and fourth rank-six gaps remain open. An explicit quadratic derivative rules out the analogous universal quintic Lorentzian approach to the fourth gap.

26.1 The inequality and its cubic strengthening

For a bipartite graph $G = (X, Y)$ with positive activities, let

$$P_G(t) = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|, G[I, J] \text{ perfectly matchable}}} u_I v_J t^{|I|} = \sum_k p_k t^k.$$

Every feasible endpoint pair is counted once. We never count its matching allocations. The degree is the matching rank.

Theorem 26.1. *If G has matching rank six, $u_x = 1$ for every $x \in X$, and $v_y > 0$ for every $y \in Y$, then*

$$8p_2^2 \geq 15p_1p_3. \quad (133)$$

This is the second ULC_6 inequality. Together with the universal first gap and the earlier last-gap theorem, it leaves only gaps 3, 4 open at rank six. [*Merge note:* The universal first gap is *preorder-root-polytopes*'s Theorem 180.2; the earlier last-gap theorem is source 13, Theorem 25.1, which source 05 does not cite.] It holds at every finite positive activity.

The earlier two-vertex-cover theorem [12] proves all-field actual-rank ULC whenever a minimum cover has at most two vertices on one shore. We may therefore assume

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = |B| = 3, \quad \nu(G) = 6,$$

with $A \sqcup B$ a vertex cover. Both exterior blocks have rank three, since every maximum matching avoids core edges.

For a fixed right set J , let a_J be the number of left sets that match it. Put $d_j = a_{\{j\}}$ and $p_k(\mathbf{z}) = \sum_{|J|=k} a_J z_J$.

Theorem 26.2. *For every such genuine $3 + 3$ cover, the homogeneous polynomial*

$$F(s, \mathbf{z}) = 20s^3 + 10p_1(\mathbf{z})s^2 + 4p_2(\mathbf{z})s + p_3(\mathbf{z}) \quad (134)$$

is Lorentzian.

Its support is M -convex: it is the homogenized independent-set support of the rank-three truncation of the right transversal matroid. Thus it suffices to show that every first derivative has a Hessian with at most one positive eigenvalue [1, Theorem 2.25]. After $z_j = tv_j$, the degree-three Newton inequality gives $(4p_2)^2 \geq 3(10p_1)p_3$, which is (133).

Two precise companion inputs are bundled with this report. The selected-left marginal of a graph with three left vertices, homogenized to degree three, is Lorentzian [20]. [*Merge note:* Source 10's Theorem 23.1.] For a displayed $2 + q$ cover, the two-vertex lift gives a degree- $(q + 2)$ Lorentzian marginal in *all individual variables on the two-core-vertex shore*, with the opposite-shore activities fixed [12, Section 3]. [*Merge note:* Source 74's Theorem 2.1, proved in Section 5 (the retained-variable statement at its end).] We use this retained-variable statement after transposing a $3 + 2$ cover.

26.2 The homogenizing derivative

We first allow arbitrary positive left activities. Let d_i be the total activity adjacent to right vertex i and let a_{ij} be its weighted two-endpoint coefficient. Set

$$G_{ii} = d_i^2, \quad G_{ij} = d_i d_j - a_{ij} \quad (i \neq j).$$

For common-neighborhood mass s_{ij} and squared-activity sum h_{ij} , endpoint counting gives $G_{ij} = (s_{ij}^2 + h_{ij})/2$. [*Merge note:* This is source 10's overlap identity (108).]

Take one feature for each left vertex, weighted by u_ℓ^2 , and one for each unordered left pair, weighted by $u_\ell u_m$. A right vertex belongs to a feature if it meets every left vertex of that feature. Their Gram matrix K has exactly the off-diagonal entries of G , and

$$G = K + \text{diag} \left(\sum_{\ell < m: \ell, m \sim i} u_\ell u_m \right). \quad (135)$$

Split $d = \alpha + n$, according to contributions from A and L . Retain the three singleton A features; by Cauchy their Gram matrix dominates $\alpha\alpha^\top/3$. Retain all singleton and pair L features and their diagonal corrections. Applied to the transpose of B versus L , the three-vertex marginal theorem gives the bound $nn^\top/3$. Every discarded term in (135) is positive semidefinite. Hence

$$G \succeq \frac{\alpha\alpha^\top + nn^\top}{3} \succeq \frac{dd^\top}{6}. \quad (136)$$

The Hessian of $\partial_s F$ has entries $ss = 120$, $sz_i = 20d_i$ and $z_i z_j = 4a_{ij}$, with zero z_i^2 entries. Its positive s pivot leaves the negative of $4(G - dd^\top/6)$. This settles $\partial_s F$.

26.3 Exterior derivatives and rank two compression

The Hessian of $\partial_{z_j} F$ has s pivot $20d_j$, cross entries $4a_{ji}$, and off-diagonal entries a_{jik} . Its required Schur matrix is

$$Q^{(j)} = \frac{4}{5d_j} \mathbf{a}_j \mathbf{a}_j^\top - B^{(j)}, \quad (\mathbf{a}_j)_i = a_{ji}, \quad B_{ik}^{(j)} = a_{jik} \quad (i \neq k), \quad B_{ii}^{(j)} = 0. \quad (137)$$

[*Merge note:* Lemma 26.3 is source 60's Theorem 22.1 at $q = 0$, transposed; source 05's direct Hessian proof is printed as a second route.]

Lemma 26.3. *For any matrix of rank at most two, with arbitrary positive row activities, its degree-two selected-column regular-minor marginal is Lorentzian.*

Proof. Ranks zero and one are immediate. Factor a rank-two matrix through a two-dimensional space and group its nonzero rows by projective direction, with total activities w_h . Put $W = \sum_h w_h$ and $Q = \sum_{h < k} w_h w_k > 0$. Compress parallel column classes by adding their variables, and let m be the number of distinct nonzero column directions. A column annihilates either one row direction or none. Its singleton coefficient is $d_i = W - w_i$, where the nonzero w_i are distinct members of the row-class list, and other w_i are zero. Every pair of distinct column directions has coefficient Q . The quadratic Hessian condition is

$$QI + \frac{1}{2}dd^\top - Q\mathbf{1}\mathbf{1}^\top \succeq 0.$$

The rank-one Schur criterion reduces this to $(m-1)(2Q + \sum_i d_i^2) \leq (\sum_i d_i)^2$. The difference is exactly

$$\left(W - \sum_i w_i\right)^2 + (m-1)\left(\sum_h w_h^2 - \sum_i w_i^2\right) \geq 0.$$

The support is the degree-two homogenization of a rank-two column matroid's independent sets. Restoring parallel variables preserves the result. $\square$

Consider a row-block matrix $(U; L)$ in which U has rank at most two and L is graph-generic on three columns B , with entries independent over the field of U . Let α, n be the singleton coefficient vectors of the two blocks, and let S_U, S_L be their weighted common-support Gram matrices. The mixed two-row determinant is nonzero exactly when one diagonal product is nonzero, by independence of the L entries. Consequently, with the definition of G above,

$$G = G_U + G_L + S_U \circ S_L + \text{diag}(\alpha_i n_i).$$

The last two terms are positive semidefinite. Lemma 26.3 and the three-vertex marginal give

$$G \succeq \frac{1}{2}\alpha\alpha^\top + \frac{1}{3}nn^\top \succeq \frac{1}{5}(\alpha + n)(\alpha + n)^\top. \quad (138)$$

Now fix $j \in R$. Its neighborhood lies in the three A vertices. Forcing j , dividing by its activity and by t , gives exactly E times a residual regular-minor polynomial, where $E = \sum_{a \sim j} u_a$. Here is the needed compression explicitly. [*Merge note:* It is the case $h = 1$ of source 60's Proposition 22.2 and Theorem 22.3, including its degree-three example; source 05 does not cite source 60.] One neighbor is deleted. Two neighbors are replaced by their kernel row of activity $u_1 u_2 / E$. Three neighbors are replaced by the three pair-kernel rows, of activities $u_1 u_2 / E, u_1 u_3 / E, u_2 u_3 / E$. Each single kernel row has the endpoint profile of its original pair. Every pair has the same three-row minor profile, and its total activity is

$$\frac{u_1 u_2 u_3 (u_1 + u_2 + u_3)}{E^2} = \frac{u_1 u_2 u_3}{E}.$$

Thus all support coefficients have the correct weights, including dependent images. Kernel rows are obtained by eliminating the forced column; these identities concern regular minors, not ordinary graph contraction.

The remaining and virtual A rows span a space of dimension at most two. The unchanged L block is graph-generic on B and independent of them. Its residual singleton and pair coefficients are a_{ji}/E and a_{jik}/E . Equation (138), multiplied by $E = d_j$, is precisely (137). This proves every exterior derivative, even with arbitrary positive original left activities.

26.4 Exact core profiles

We now use unit left activities and force a core column B_0 . Write $J, C_1, C_2 \subseteq A$ for the core neighborhoods of B_0, B_1, B_2 . Let N_S be the exterior-left population of nonempty B -mask S . Define

$$n = |N_L(B_0)|, \quad m_i = |N_L(B_i)|, \quad t_i = |N_L(B_0) \cap N_L(B_i)|, \quad p_i = nm_i - \binom{t_i + 1}{2}, \quad E = n + |J|.$$

The exterior rank gives $n \geq 1$. Let $h(U, V)$ count two-element A sets matching columns with supports U, V , and let $\chi(U, V, W)$ indicate matchability to all three A vertices. For every nonempty exterior-right mask S , put

$$\begin{aligned} r_S &= n|S| + h(J, S), \\ x_i &= p_i + n|C_i| + m_i|J| - t_i|J \cap C_i| + h(J, C_i), \\ y_i(S) &= p_i|S| + nh(C_i, S) + m_i h(J, S) \\ &\quad - t_i[h(J, S) + h(C_i, S) - h(J \cup C_i, S)] + \chi(J, C_i, S). \end{aligned} \tag{139}$$

These are exactly $a_{B_0 R_S}, a_{B_0 B_i}, a_{B_0 B_i R_S}$. For y_i , partition by two, one or zero selected L rows. If the one L row meets both core columns, the remaining A pair uses either core column with R_S . Its Boolean union is $J \cup C_i$; the displayed subtraction removes the duplicate allocation.

Let q_{all} count bases of all three B columns. To compute it, add the three core-row mask populations to N_S , obtaining M_S , and ignore zero rows. Set $M = \sum_S M_S$ and $d'_i = \sum_{S \ni i} M_S$. Then

$$\begin{aligned} q_{\text{all}} &= \binom{M}{3} - \sum_i \binom{M - d'_i}{3} + \sum_i \binom{M_{\{i\}}}{3} \\ &\quad - \sum_i \binom{M_{\{i\}}}{2} \sum_{S \supseteq B \setminus \{i\}} M_S. \end{aligned} \tag{140}$$

Inclusion-exclusion first excludes a missed column. Every remaining Hall failure has just one selected row meeting a chosen pair of columns, and the other two rows meet only the third. The three failure types are disjoint.

The class-limit matrix has seven R coordinates and the two other B coordinates. Its R block and cross columns are

$$\begin{aligned} Q_R(S, T) &= \frac{4r_S r_T}{5E} - nh(S, T) - \chi(J, S, T), \\ v_i(S) &= \frac{4x_i r_S}{5E} - y_i(S). \end{aligned} \tag{141}$$

The remaining block has diagonal $4x_i^2/(5E)$ and off-diagonal $4x_1 x_2/(5E) - q_{\text{all}}$. This matrix suffices for all finite class populations: if X_S is the sum of arbitrary real test coordinates in class S , the actual quadratic form is the class-limit form in $(B_1, B_2, \mathbf{X})$ plus

$$\sum_S [nh(S, S) + \chi(J, S, S)] \sum_{r: N(r)=S} z_r^2 \geq 0.$$

Absent classes set $X_S = 0$. This identity does not assume equal right activities.

26.5 The two dimensional Schur complement

For $|J| \leq 1$, all profile functions in a right mask are combinations of the six containment features 1, 2, 3, 12, 13, 23. The mask-7 row and column equal the sum of those for masks 3, 5, 6

minus those for masks 1, 2, 4, including the cross terms. We therefore retain masks 1, ..., 6. For $|J| = 2, 3$ retain all seven.

Every principal matrix containing R and only one other B_i is positive semidefinite. Indeed, delete the other core vertex. The graph has a genuine $3 + 2$ cover. Transpose and apply the full marginal statement of [12]: its selected-right marginal is Lorentzian of degree five. Differentiate in B_0 and then twice in its homogenizing variable. The quadratic Schur coefficient is $3/(4E)$, whereas (137) has $4/(5E)$. The difference is the positive rank-one matrix $\mathbf{a}\mathbf{a}^\top/(20E)$. Using arbitrarily many twins with equal test coordinates divided by their population proves the same assertion for the class-limit principal matrix.

For $J = 1, 3, 7$ in binary-mask notation, direct exact calculation gives the following determinants for the retained block $A_R = 5EQ_R$:

J	$\det A_R$
1	$6250n^4(n+1)^8$
3	$15625n^4(n+2)^7(2n^2+4n+3)$
7	$31250n^3(n+2)^2(n+3)^6(2n^2+7n+9)$.

They are positive for $n \geq 1$. The retained block is therefore positive definite. After eliminating it, both diagonal entries of the remaining 2×2 matrix are nonnegative by the preceding principal-matrix argument. Only its determinant remains to be proved nonnegative.

The case $J = \emptyset$ has a direct proof. Let q count L bases, let O count L pairs matchable both to $\{B_0, B_1\}$ and to $\{B_0, B_2\}$, and put $e_i = |C_i|$, $c = |C_1 \cap C_2|$. The Schur complement is exactly

$$D_i = \frac{p_i^2}{2n} + 2e_i p_i + n \binom{e_i}{2}, \quad L_{12} = \frac{p_1 p_2}{2n} - q + cO. \quad (142)$$

To verify this identity, use the six containment features above, put $u = (1, 1, 1, 0, 0, 0)^\top$ and let f_i be the feature vector of C_i . The R block is $nF^\top(I - uu^\top/5)F$, the cross vector in feature coordinates is $nf_i - (p_i + ne_i)u/5$, and the feature inverse is $I + uu^\top/2$. Substitution cancels the mixed core terms in (142).

Rank-three Rayleigh gives $nq \leq p_1 p_2$ [19]. Also $O \leq \min(p_1, p_2)$ and $c \leq \min(e_1, e_2)$. With $b = p_1 p_2 / (2n)$ we have $-b \leq L_{12} \leq b + cO$, while

$$D_1 D_2 - (b + cO)^2 \geq \frac{c}{n} p_1 p_2 (p_1 + p_2 - O) + c^2 (4p_1 p_2 - O^2) \geq 0.$$

Thus the empty-column Schur complement is positive semidefinite.

26.6 The complete finite certificate

For the nonempty column, permute A so J is 1, 3 or 7. Permutations stabilizing J , together with exchange of B_1, B_2 , leave 24, 24, 13 rooted core orbits respectively. The two shores are never transposed in this enumeration.

Choose any exterior L basis. Its neighborhood multiset is one of exactly 51 Hall-matchable triples among the 84 triples with repetition from seven masks. If b_S records this basis, write $N_S = b_S + X_S$ with $X_S \geq 0$. Every genuine exterior block has such a decomposition.

For each core, form the exact rational 2×2 Schur complement from (139)–(141). Clear a positive common denominator and then a positive scalar denominator in its determinant. For every basis choice, translate $N = b + X$. The complete certificate states that every coefficient of every resulting ordinary polynomial in the seven X_S is nonnegative. The exact totals are

$ J $	Core orbits	Basis cones	Nonzero coefficient entries
1	24	1224	10 025 635
2	24	1224	22 628 373
3	13	663	17 383 686
Total	61	3111	50 037 694

Every listed nonzero coefficient is a positive integer; the minimum is one. Absent monomials have coefficient zero. The cleared determinants have degree 10, 12 or 14. Their possible common denominators before squaring are

$$2n^2, \quad n(2n^2 + 4n + 3), \quad 2n(2n^2 + 7n + 9), \quad 2n(n + 2)(2n^2 + 7n + 9),$$

all strictly positive on the required domain.

The complete per-core records contain the denominator, scalar multiplier, degree, coefficient counts and the SHA-256 of each lexicographically ordered coefficient list. The source reconstructs every coefficient with exact rational arithmetic and checks its sign. An independent reconstruction uses labeled Hall tests for the endpoint coefficients, verifies each symbolic R inverse and uses a different population translation. The full replay status and commands are recorded in the package’s verification document. [*Merge note:* Shipped as `05-second-newton-VERIFICATION.md`; the scripts as `code/05-second-newton-*.py`, the per-core records as `data/05-second-newton-core_schur_*.json` and the independent replays as `data/05-second-newton-independent-*.json`.]

These finitely many identities prove every remaining determinant inequality on every nonnegative real cone $X \geq 0$, hence on all original integer populations. With both diagonal Schur entries nonnegative, each Schur matrix is positive semidefinite. Sections 2–5 [*Merge note:* Subsections 26.2–26.5 here] now settle every derivative of (134). Its M -convex support completes Theorem 26.2, and the minimum-cover reduction proves Theorem 26.1.

26.7 Scope and the next truncations

The next possible truncation is

$$F_4 = 15s^4 + 10p_1s^3 + 6p_2s^2 + 3p_3s + p_4, \quad \partial_s F_4 = 3F.$$

Its Lorentzian property would prove $9p_3^2 \geq 16p_2p_4$. Only the two-distinct-right-variable quadratic derivatives are new. This is a research direction, not a conclusion of the present theorem.

The analogous quintic

$$F_5 = 6s^5 + 5p_1s^4 + 4p_2s^3 + 3p_3s^2 + 2p_4s + p_5$$

is *not* universally Lorentzian. An explicit graph has core row masks $(2, 1, 0)$, eight exterior-left vertices of mask 7, and exterior-right types 1, 2, 3, 4, 5 with populations 1, 1, 7, 1, 1 and activities 15, 15, 1, 15, 1/10. All left activities are one. It is connected and has matching rank six. [*Merge note:* This is source 23’s 25-vertex graph (Section 24.2); source 05 does not cite source 23. Its cubic below is source 23’s (115).] Its all- B conditional cubic has coefficients

$$(C_0, C_1, C_2, C_3) = \left(120, \frac{26602}{5}, \frac{393176}{5}, \frac{1937208}{5}\right).$$

Differentiate F_5 in the three B variables and specialize the R variables. The resulting quadratic is $3C_0s^2 + 2C_1st + C_2t^2$, but

$$C_1^2 - 3C_0C_2 = -\frac{50396}{25}.$$

Its Hessian has positive diagonal and determinant 201584/25, hence two positive eigenvalues. The source package directly verifies these counts. This refutes the stronger quintic approach, not scalar ULC₆. The fourth gap requires a weaker argument.

The present result is an ordinary mathematical proof with an exact finite coefficient certificate. It is not proof-assistant formalization or external refereeing, and no exhaustive priority claim is made.

27 Matching number six with common exterior neighbourhoods, one weighted shore (source 69)

Abstract of source 69, as delivered. We prove matching-rank ultra-log-concavity for one-shore activities on a rank-six family with an arbitrary 3×3 core and complete exterior neighborhoods. The proof reduces the last four Newton gaps to 8,964 explicit univariate coefficient polynomials, all nonnegative coefficientwise; the first gap has a general elementary proof. A small exact verifier and the complete integer certificate accompany the note. We also give a cover-changing reduction that handles certain adjacent exterior classes, and an explicit negative coefficient showing why the main certificate does not extend unchanged to arbitrary exterior neighborhoods. The general rank-six one-shore problem remains unresolved.

27.1 The theorem and its scope

For a bipartite graph $G = (X, Y; E)$ with activities $u_x, v_y > 0$, put

$$P_G(t) = \sum_{(I,J) \text{ feasible}} u_I v_J t^{|I|} = \sum_{k=0}^d p_k t^k.$$

Feasibility means $I \subseteq X$, $J \subseteq Y$, $|I| = |J|$, and $G[I, J]$ has a perfect matching. Each endpoint pair is counted once. Thus P_G is the weighted perfectly matchable set polynomial, not the matching polynomial; see Davis–Kohl [5]. Its degree is $d = \nu(G)$. Define

$$\Delta_k^{(d)}(P) = k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1}. \quad (143)$$

The property ULC_d means all these gaps are nonnegative, with no internal zero coefficients.

Theorem 27.1. *Let $X = A \sqcup L$, $Y = B \sqcup R$, where $|A| = |B| = 3$, $|L| = n \geq 3$ and $|R| = m \geq 3$. Let*

$$E = (L \times B) \cup (A \times R) \cup K, \quad K \subseteq A \times B.$$

Give every vertex in X activity one and every vertex in Y an arbitrary positive activity. Then P_G has degree six and is ULC₆. The same conclusion holds with the two shores interchanged.

The core K may be any of the 512 labeled 3×3 bipartite graphs. The exterior-right activities need not agree. A common positive activity on the entire unit-weight shore can also be allowed, by rescaling t .

This is a partial rank-six result. Arbitrary positive two-shore fields are known to satisfy rank-normalized ULC through rank five, whereas a one-shore counterexample of rank 38 is available in companion notes [12, 16]. The present theorem excludes this entire common-exterior rank-six family from being a smaller one-shore counterexample. [Merge note: Its gaps two and five now also follow from sources 05 and 13 for every one-shore graph of matching number six; what is new here for this family is gaps three and four. Taking K complete and all activities one gives *preorder-root-polytopes*'s unit $H_{3,n}$, so the unit-weight assertion of its

Corollary 177.2 (`hb:cor:asymptoticweight`) is a special case of Theorem 27.1.] It does not settle arbitrary $3 + 3$ covers or claim exhaustive literature priority. In particular, the ambient-shore normalization of Röhrle–Ulirsch [7] need not equal six here.

27.2 Exact endpoint counting and the deleted-core baseline

Write b_1, b_2, b_3 for the B activities, $r_1, \dots, r_m$ for the R activities, and $e_l(R)$ for their elementary symmetric polynomials. For $J \subseteq B$, let $b_J = \prod_{j \in J} b_j$. Out-of-range binomial coefficients are zero.

Lemma 27.2. *For $0 \leq k \leq 6$,*

$$p_k = \sum_{\substack{J \subseteq B, 0 \leq l \leq 3 \\ |J| + l = k}} T_{J,l}(n) b_J e_l(R), \quad (144)$$

where

$$T_{J,l}(n) = \sum_{\substack{I \subseteq A, |I| \geq l \\ \nu(K[I,J]) \geq |I| - l}} \binom{n}{|J| + l - |I|}. \quad (145)$$

Proof. Fix I, J and l selected exterior-right vertices. A support of this type has $h = |J| + l - |I|$ exterior-left vertices. A perfect matching must use l edges from I to those exterior-right vertices and h edges from the selected exterior-left vertices into J . The remaining $|I| - l$ edges are core edges. Hence the stated core matching condition is necessary. Conversely, take that many disjoint core edges, match the remaining l vertices of I to the selected exterior-right vertices, and match the remaining h vertices of J to the selected exterior-left vertices. Completeness of both exterior neighborhoods permits these choices. This proves existence without introducing a multiplicity for different realizing matchings. There are $\binom{n}{h}$ possible exterior-left endpoint sets, giving (145) and (144). $\square$

Delete all core edges and denote the resulting polynomial by Q . Then

$$Q(t) = \left(\sum_{j=0}^3 \binom{n}{j} e_j(B) t^j \right) \left(\sum_{l=0}^3 \binom{3}{l} e_l(R) t^l \right), \quad T_{J,l}^0(n) = \binom{3}{l} \binom{n}{|J|}. \quad (146)$$

Both factors are ULC_3 . For the first, the normalized coefficient is $\binom{n}{j} e_j(B) / \binom{3}{j}$, a product of two log-concave sequences by Newton's inequalities. For the second, it is $e_l(R)$, again log-concave. Their degree-three homogenizations are Lorentzian; the product is Lorentzian and therefore Q is ULC_6 [1, Example 2.26 and Corollary 2.32].

27.3 The finite nonnegative coefficient certificate

Set $z = n - 3 \geq 0$. Every right-activity variable occurs with exponent at most two in a Newton gap. A monomial is specified, up to permuting R , by its core exponent vector $v \in \{0, 1, 2\}^3$, the number ρ of exterior variables occurring once, and the number τ occurring twice. Necessarily

$$|v| + \rho + 2\tau = 2k.$$

For any coefficient table T , define

$$C_{a,b}(T; v, \rho, \tau) = \sum_{\substack{J, H \subseteq B \\ \mathbf{1}_J + \mathbf{1}_H = v}} \binom{\rho}{a - |J| - \tau} T_{J, a - |J|}(z + 3) T_{H, b - |H|}(z + 3). \quad (147)$$

A summand is zero unless both exterior indices lie in $\{0, 1, 2, 3\}$ and their sum is $\rho + 2\tau$. Formula (147) is the coefficient of the specified monomial in $p_a p_b$: every squared exterior variable lies in both supports, and the binomial factor chooses the singleton variables belonging to the first support.

Let $D_k(T) = k(6 - k)C_{k,k}(T) - (k + 1)(7 - k)C_{k-1,k+1}(T)$, with the monomial parameters suppressed. The certificate consists of the coefficient vectors in increasing powers of z of

$$36(D_k(T) - D_k(T^0)), \quad k = 2, 3, 4, 5. \quad (148)$$

They are integer vectors because T is a sum of $\binom{z+3}{h}$ for $0 \leq h \leq 3$, and

$$\begin{aligned} 6\binom{z+3}{0} &= 6, & 6\binom{z+3}{1} &= 18 + 6z, \\ 6\binom{z+3}{2} &= 18 + 15z + 3z^2, & 6\binom{z+3}{3} &= 6 + 11z + 6z^2 + z^3. \end{aligned}$$

There are 36 core orbits under separate row and column permutations. The binary mask of K is $\sum_{(i,j) \in K} 2^{3i+j}$ with $0 \leq i, j \leq 2$. The orbit representatives are

$$\begin{aligned} &0, 1, 3, 7, 9, 10, 11, 14, 15, 27, 29, 31, 63, 73, 74, 75, 78, 79, \\ &83, 84, 85, 87, 91, 93, 94, 95, 118, 119, 127, 219, 221, 223, 238, 239, 255, 511. \end{aligned}$$

The verifier constructs their full orbits and checks that they partition all 512 masks. It then enumerates $v \in \{0, 1, 2\}^3$, $0 \leq \tau \leq 3$, $0 \leq \rho \leq 6 - \tau$, subject to the displayed total-degree equation, and evaluates (148) by integer polynomial addition and multiplication.

Lemma 27.3 (Exact certificate). *Every integer in every vector (148) is nonnegative. The complete finite accounting is*

Gap k	Vectors checked	Nonzero vectors	Nonzero scalar entries
2	1,224	1,001	2,475
3	2,196	1,878	5,247
4	2,952	1,618	6,105
5	2,592	420	2,205
Total	8,964	4,917	16,032

The file `data/exact_certificate.json` lists every vector, including zeros, and `checks/verify_certificate.py` independently reconstructs each one from (145)–(148) and checks all signs and values.

This is a finite exact certificate, not sampling over activity values. For example, at the complete core, gap five and monomial type $(v, \rho, \tau) = ((2, 2, 1), 5, 0)$, the gap-increment coefficient before multiplication by 36 is

$$5(z + 2)(z + 3)^2(z + 6),$$

which is nonnegative. The complete certificate has the same coefficientwise property, without needing to factor each entry.

Proof of Theorem 27.1. For $k = 2, 3, 4, 5$, Lemma 27.3 gives coefficientwise nonnegativity of $\Delta_k^{(6)}(P) - \Delta_k^{(6)}(Q)$ in z and all right activities. Since Q is ULC_6 , these four gaps of P are nonnegative.

For the first gap, $5p_1^2 \geq 12p_2$, which is the first gap since $p_0 = 1$. [*Merge note:* This is *preorder-root-polytopes*'s Theorem 180.2 (Part XI) at $r = 6$. Source 69 proves it by the weighted Motzkin–Straus route on the edge-disjointness graph, the route printed in Section 11; that argument is not reprinted.]

Finally, $A \sqcup B$ is a six-vertex cover, and disjoint three-matchings $L \rightarrow B$ and $A \rightarrow R$ exist. Thus the degree is exactly six. Subsets of a maximum matching and positivity of all activities give positive coefficients at every degree from zero through six. $\square$

27.4 Adjacent exterior classes and a precise barrier

The following reduction applies to arbitrary exterior neighborhoods and arbitrary positive fields on both shores.

Proposition 27.4. *Suppose a graph of matching rank six has a displayed cover $A \sqcup B$ with $|A| = |B| = 3$. If some $a \in A$ has only private leaves as its exterior neighbors, then its support polynomial is ULC_6 . The analogous statement holds on the other shore.*

Proof. All those exterior neighbors have neighborhood exactly $\{a\}$. A feasible support uses at most one of them, so replace them by one leaf whose activity is their sum. This preserves every support coefficient and the matching rank. [*Merge note:* Weighted leaf compression, *preorder-root-polytopes*'s Lemma 206.1.] There is at least one such leaf, since a size-six matching forces all of A to match to exterior-right vertices. In the compressed graph, replace a in the cover by its single leaf. The resulting six-vertex minimum cover has split $2 + 4$. The rank-normalized theorem for a minimum cover meeting a shore in at most two vertices [12] gives the result. [*Merge note:* Theorem 2.1.] $\square$

For example, this includes arbitrary cores and arbitrary exterior-left neighborhoods when exterior-right neighborhoods are singletons together with the one possible pair class $\{A_1, A_2\}$. Each exterior block has matching rank and minimum-cover size three. If the exterior block between A and R has a minimum cover using at most two vertices of A , combining it with B directly gives the same reduction. Thus a genuinely unhandled case must survive both this cover test and private-leaf compression on each shore.

The coefficientwise proof of the main theorem cannot simply be applied after allowing arbitrary exterior neighborhoods. Take complete core $K_{3,3}$ and n unit exterior-left vertices adjacent to all of B . Include exterior-right vertices r_1, r_2, r_3 with respective neighborhoods $\{A_1\}, \{A_2\}, \{A_3\}$ and variable activities. If Q again deletes the core, then

$$[b_2 b_3 r_1 r_2](\Delta_2^{(6)}(P) - \Delta_2^{(6)}(Q)) = 56 - 4n. \quad (149)$$

This is negative for $n \geq 15$. The verifier `verify_barrier.py` derives this identity exactly by enumerating left endpoint sets for the relevant right subsets. Yet this graph is ULC_6 : its right shore has six vertices, or one can use Proposition 27.4. Thus (149) obstructs a proof method, not ULC itself.

27.5 Reproducibility and remaining question

Run `python3 checks/run_all.py` [*Merge note:* shipped as `code/69-common-exterior-run_all.py`; the files `data/exact_certificate.json`, `checks/verify_certificate.py` and `verify_barrier.py` named above are shipped as `data/69-common-exterior-exact_certificate.json`, `code/69-common-exterior-verify_certificate.py` and `code/69-common-exterior-verify_barrier.py`] to regenerate the endpoint checks, compare the complete coefficient certificate, and verify (149). The core certificate verifier uses only

Python’s standard library, integer arithmetic and Hall’s subset criterion. It also compares the support criterion with a separate injection-based core matching computation on 100,352 support types across all 512 labeled cores. The mathematical formula and the certificate were independently reconstructed before packaging.

The next unresolved step is a uniform treatment of genuine $3 + 3$ minimum-cover graphs whose exterior blocks have arbitrary neighborhood classes and survive the cover-changing reductions above. The finite positivity certificate here does not resolve that step. Neither numerical minimization nor a negative coefficient of a gap increment would by itself settle the full one-shore question.

28 Nested families and conditional barriers at matching number six, one weighted shore (source 63)

Abstract of source 63, as delivered. We extend the rank-six one-shore ULC result to a complete core with nested exterior neighborhoods. An exact finite certificate proves the result for arbitrary positive activities on the weighted shore. We then give a 20-vertex graph with a unique $3 + 3$ minimum cover whose full right marginal is not Lorentzian, ruling out that stronger approach to the general problem. A conditional theorem excludes every quadratic sector obtained by forcing four right vertices in a genuine $3 + 3$ cover. The general rank-six one-shore ULC question remains open.

28.1 A nested exterior theorem

For a positively weighted bipartite graph $G = (X, Y)$, let

$$P_G(t) = \sum_{(I,J) \text{ feasible}} u_I v_J t^{|I|} = \sum_{k=0}^d p_k t^k,$$

where a feasible pair has $|I| = |J|$ and $G[I, J]$ admits a perfect matching. Each endpoint pair is counted once. Its degree is the matching rank d . We use

$$\Delta_k^{(d)}(P) = k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1}$$

for the rank-normalized Newton gap. Positive support polynomials have no internal zeros, so ULC_d is equivalent to nonnegativity of these gaps.

Theorem 28.1. *Let $A = \{A_1, A_2, A_3\}$ and $B = \{B_1, B_2, B_3\}$. Take the complete core $A \times B$, together with $n \geq 3$ exterior-left vertices L , each adjacent to all of B . Let every exterior-right vertex in R have neighborhood in the chain*

$$\{A_1\} \subset \{A_1, A_2\} \subset A.$$

Assume A can be matched into R . Give all vertices of $A \cup L$ activity one and the vertices of $B \cup R$ arbitrary positive activities. Then P_G is ULC_6 .

If the three exterior class sizes are m_1, m_2, m_3 , the rank assumption is exactly $m_3 \geq 1$, $m_2 + m_3 \geq 2$, and $m_1 + m_2 + m_3 \geq 3$. The complete-core hypothesis is part of this theorem. The earlier common-exterior result allowed an arbitrary core, but that larger core scope is not claimed here. [Merge note: The earlier result is source 69 [23], Theorem 27.1; neither family contains the other. As for source 69, gaps two and five of Theorem 28.1 also follow from sources 05 and 13, and its new content is gaps three and four for this family.]

For $I \subseteq A$ and a selected exterior multiset with class counts $c = (c_1, c_2, c_3)$, define $H(c, I)$ by

$$c_1 \leq |I \cap \{A_1\}|, \quad c_1 + c_2 \leq |I \cap \{A_1, A_2\}|, \quad |c| \leq |I|. \quad (150)$$

These are precisely the nested Hall conditions for matching those right vertices into I . Fix a subset $J \subseteq B$ of size j and an actual set of exterior-right vertices with counts c . Its coefficient as a monomial in the individual right activities is

$$T_{j,c}(n) = \sum_{\substack{I \subseteq A: H(c,I) \\ 0 \leq j+|c|-|I| \leq j}} \binom{n}{j+|c|-|I|}. \quad (151)$$

Indeed, the selected exterior-left set has size $j + |c| - |I|$. Once the selected exterior-right vertices are matched, completeness of the core and of $L \times B$ completes the support exactly when the displayed bounds hold. This counts endpoint sets rather than their realizing matchings.

Delete the core edges and call the resulting polynomial Q . Its coefficient $T_{j,c}^0$ is obtained by keeping only $|I| = |c|$ in (151). The graph for Q is a disjoint union of two graphs with shores of size three. The standard transversal-matroid basis specialization makes both factors ULC_3 ; their degree-three Lorentzian homogenizations have a Lorentzian product. Thus Q is ULC_6 [1, 12].

Lemma 28.2 (Finite coefficient certificate). *For $k = 2, 3, 4, 5$, every coefficient of $\Delta_k^{(6)}(P) - \Delta_k^{(6)}(Q)$, expanded in the individual right activities and in $z = n - 3$, is nonnegative.*

Here is the complete finite reduction used by the certificate. A right-activity monomial has core exponent vector $v \in \{0, 1, 2\}^3$. For each of the three exterior classes let o_i count distinct variables occurring once and d_i count variables occurring twice. Thus

$$|v| + |o| + 2|d| = 2k, \quad |d| \leq 3, \quad |o| + 2|d| \leq 6.$$

Put $a = \#\{i : v_i = 2\}$ and $b = \#\{i : v_i = 1\}$. Distribute the singly occurring exterior variables by a vector e with $0 \leq e_i \leq o_i$, and the b core singleton variables by an integer $h \in [0, b]$. The two factors have exterior counts $d + e, d + o - e$ and core counts $a + h, a + b - h$, with multiplicity

$$\binom{b}{h} \prod_{i=1}^3 \binom{o_i}{e_i}.$$

Use (151) for the two support coefficients, and retain the distributions of first-factor degree k with multiplier $k(6 - k)$ and those of degree $k - 1$ with multiplier $-(k + 1)(7 - k)$. Subtract the corresponding T^0 product. This gives the coefficient in Lemma 28.2 with no activity specialization.

The file `data/exact_certificate.json` contains all 3,391 coefficient vectors, scaled by 36 and listed in ascending powers of z . They are integer vectors because each T is a sum of $\binom{z+3}{h}$ with $h \leq 3$. Every entry is nonnegative; the numbers of nonzero vectors at gaps 2, 3, 4, 5 are respectively 132, 495, 532, 152. The standard-library verifier reconstructs all vectors from (150)–(151), compares them to the stored certificate, and checks their signs. An independent literal matching enumeration checks 80 small weighted graphs.

Proof of Theorem 28.1. Lemma 28.2 and ULC_6 of Q prove gaps two through five. The first gap holds for every positive weighted bipartite support polynomial; here it gives $5p_1^2 \geq 12p_2$. [*Merge note:* This is *preorder-root-polytopes*'s Theorem 180.2; source 63's one-sentence Motzkin–Straus justification (edge-disjointness graph, clique number the matching rank, two-matchings at least size-two supports) is the route printed in Section 11.]

The cover $A \cup B$ has size six. The assumed three-matching $A \rightarrow R$ and a disjoint three-matching $L \rightarrow B$ attain that size. The degree is therefore six, as required. $\square$

28.2 An obstruction to the full marginal approach

With unit activities on X , define the homogeneous right marginal of degree six by

$$F_G(s, z) = \sum_{\substack{J \subseteq Y \\ |J| \leq 6}} \#\{I \subseteq X : (I, J) \text{ feasible}\} z_J s^{6-|J|}.$$

If F_G were Lorentzian, all positive one-shore specializations would be ULC_6 . The converse is not asserted, and the stronger property already fails in the irreducible $3 + 3$ class.

Proposition 28.3. *There is a 20-vertex graph of matching rank six with a unique minimum vertex cover, of split $3 + 3$, for which F_G is not Lorentzian.*

Proof. Use three-bit masks 1, 2, 4 for the first, second and third core vertices. The left-core neighborhoods in B are $(2, 0, 5)$. The seven exterior-left neighborhoods in B are

$$(3, 6, 5, 4, 3, 1, 4),$$

and the seven exterior-right neighborhoods in A are

$$(2, 5, 5, 5, 1, 4, 7).$$

There are no other edges. Each exterior block satisfies $|N(S)| > |S|$ for every nonempty subset of its three core vertices, so it has a three-matching. Also, omitting any nonempty set of core vertices from a cover requires more exterior vertices than were omitted. Thus $A \cup B$ is the unique minimum cover, and the matching rank is six.

Differentiate F_G in all three B variables and the variable of the first exterior-right vertex, whose neighborhood is $\{A_2\}$. This forces A_2 into the support. Of the six remaining right variables, four are then common neighbors of A_1, A_3 , and two are private neighbors, one at each. Their support coefficients are exactly

$$\text{constant } 64, \quad \text{each common singleton } 74, \quad \text{each private singleton } 42, \quad \text{each pair } 27.$$

These integers can be obtained by enumerating left subsets of sizes four, five and six and applying Hall's criterion. The accompanying verifier independently obtains them by dynamic enumeration of matchable left endpoint sets; it also enumerates all size-six covers.

Set the four common variables equal to x and both private variables equal to y . The resulting quadratic is

$$64s^2 + (296x + 84y)s + 27(6x^2 + 8xy + y^2). \tag{152}$$

At $x = t, y = 19t$, this becomes

$$64s^2 + 1892st + 14013t^2, \quad 1892^2 - 4 \cdot 64 \cdot 14013 = -7664 < 0.$$

A nonnegative bivariate quadratic is Lorentzian exactly when its middle coefficient squared is at least four times the product of its endpoint coefficients. Derivatives and nonnegative linear substitutions preserve Lorentzian polynomials [1]. Hence F_G cannot be Lorentzian. $\square$

This is not a scalar ULC_6 counterexample. A quadratic sector forced to degrees 4, 5, 6 needs the weaker ratio $12/5$, whereas the displayed sector's ratio is close to four. The result specifically rules out universal full-marginal Lorentzianity as a proof route.

28.3 Every four right vertex sector is safe

Theorem 28.4. *Let G have matching rank six and a displayed minimum cover $A \cup B$ with $|A| = |B| = 3$. Give all left vertices activity one and all right vertices arbitrary positive activities. Force four designated right vertices by scaling their activities together and taking the leading coefficient of degree four in the scaling parameter. Suppose the resulting sector is*

$$t^4(c_0 + c_1t + c_2t^2), \quad c_0c_2 > 0.$$

Divide out the fixed positive product of the four forced right activities. Then $c_1^2 \geq 3c_0c_2$. Thus its last rank-six Newton gap is strictly positive, since the required constant is $12/5$.

The theorem concerns these conditioned sectors. It does not assert ULC at arbitrary finite activities, and it does not address cubic sectors obtained by forcing only three right vertices.

28.3.1 Hall tight subfamilies

If the forced set contains S with $T = N(S)$, $|T| = |S| = h \geq 1$, and a perfect matching between T and S , every support containing S contains all of T , matched to S . Deleting both endpoint sets gives a coefficient-preserving correspondence apart from the fixed factor $u_T v_S t^h$.

The remaining graph has rank $6 - h$. Indeed, replacing all edges incident to T in a maximum matching by a matching from T to S cannot decrease size. The weighted rank-at-most-five theorem applies to the remaining graph [12]. [*Merge note:* Corollary 2.2.] Force the other $4 - h$ right vertices and take a coefficientwise limit at this fixed normalization. Its last gap gives

$$c_1^2 \geq \frac{2(6-h)}{5-h} c_0 c_2 \geq \frac{5}{2} c_0 c_2, \quad 1 \leq h \leq 4. \quad (153)$$

This includes a forced exterior vertex with a singleton neighborhood, two forced exterior vertices with a two-vertex combined neighborhood, and any three matchable exterior-right vertices.

28.3.2 Two core vertices and two exterior vertices

This case permits arbitrary positive activities on both shores. Let $J \subseteq B$ be the forced pair, and let z be the missing- B activity. Let E be the sum of the activity products of the A -pairs that match the two forced exterior-right vertices, and put $a = \prod_{i=1}^3 u_{A_i}$. Let S be the union of the core neighborhoods of the possible unmatched A rows, namely those whose complementary A -pair matches the two forced exterior vertices. Define weighted exterior-left counts

$$\begin{aligned} b &= \#\{L\text{-pairs matching } J\}, & c &= \#\{L\text{-triples matching } B\}, \\ V &= \#\{L\text{-singletons completing } J \text{ with row } S\}, & U &= \#\{L\text{-pairs completing } B \text{ with row } S\}. \end{aligned}$$

Here each count includes the product of the selected left activities. Let W be the total activity of remaining exterior-right vertices that complete the forced exterior pair to a matching of all of A . Partitioning supports by whether they select two or three vertices of A gives

$$c_0 = Eb + aV, \quad c_1 = (Ec + aU)z + abW, \quad c_2 = aczW.$$

In the three- A case the common product a makes the virtual union row exact: an endpoint support is counted once even if several unmatched rows work. Eligibility in W depends only on the exterior-right triple.

Take the rank-three transversal matroid on the exterior-left elements, the virtual row S , and a private element at the missing- B slot. After the exterior-left activity substitution, its

basis polynomial in the last two variables is $c + Ux + by + Vxy$. Every rank-three matroid is Rayleigh [19, Theorem 1.1], so $bU \geq cV$. Therefore

$$c_1^2 - 4c_0c_2 = [(Ec + aU)z - abW]^2 + 4a^2zW(bU - cV) \geq 0. \quad (154)$$

This stronger bound settles the two-core/two-exterior case.

28.3.3 A useful quadratic inequality

Suppose remaining right elements have parallel-class activity totals $x_1, \dots, x_m$ and a total g of other elements, each forming an independent pair with every other distinct element. Put $w = g + \sum x_i$. Suppose their linear costs are $q - \delta_i$ on the parallel classes and q on the other elements, where $\delta_i \geq 0$ and $\sum \delta_i \leq q$. If a pair of independent right elements has coefficient $T > 0$, then

$$c_1 = qw - \sum_i \delta_i x_i, \quad c_2 \leq \frac{T}{2} \left(w^2 - \sum_i x_i^2 \right).$$

The missing term in the second formula is the nonnegative sum of squared individual activities in the generic group. Set $D = q^2 - \sum_i \delta_i^2$. Ordinary Cauchy–Schwarz, or the reverse inequality for the quadratic form $z_0^2 - \sum z_i^2$, gives

$$c_1^2 \geq D \left(w^2 - \sum_i x_i^2 \right) \geq \frac{2D}{T} c_2. \quad (155)$$

For a direct check, replace $\sum \delta_i x_i$ by $\|\delta\| \|x\|$. Subtracting $D(w^2 - \|x\|^2)$ from $(qw - \|\delta\| \|x\|)^2$ leaves $(q\|x\| - \|\delta\|w)^2$.

28.3.4 All three core vertices and a degree three exterior vertex

All left activities are now one. Let the forced exterior vertex have neighborhood A . Remove isolated exterior-left vertices and let N be the remaining number. Write T for the number of exterior-left triples matching B ; $T \geq 1$. For each core row A_i , let $\mathcal{P}_i$ be the family of exterior-left pairs completed to B by A_i . Put

$$W = |\mathcal{P}_1 \cup \mathcal{P}_2 \cup \mathcal{P}_3|, \quad a_i = |\mathcal{P}_i \setminus (\mathcal{P}_j \cup \mathcal{P}_k)|, \quad A_0 = \sum_i a_i \leq W.$$

Let V count exterior-left singletons completed to B by some two core rows. Exact endpoint counting gives

$$C := c_0 = 3T + 3W - A_0 + V.$$

The three singleton-neighborhood classes of remaining right vertices have costs $2T + W - a_i$. Every right vertex with neighborhood size at least two has cost $q = 3T + W$. Any pair completes the forced exterior vertex to a matching of A , except two vertices in the same singleton class. Thus (155) applies with $\delta_i = T + a_i$, since $\sum \delta_i \leq q$.

We show $D \geq (3/2)CT$. The exterior-left rank-three matroid has N nonloops. A fixed basis and one distinct replacement basis for every other element show $T \geq N - 2$. Since $V \leq N$, we have $V \leq T + 2$. Expanding and using $A_0 \leq W$ yields

$$\begin{aligned} D - \frac{3}{2}CT &= \frac{3}{2}T^2 + \frac{3}{2}TW - \frac{1}{2}TA_0 + W^2 - \sum_i a_i^2 - \frac{3}{2}TV \\ &\geq T(W - 3) + W^2 - \sum_i a_i^2 \geq 0 \quad (W \geq 3). \end{aligned}$$

For the small- W boundary, represent the transversal matroid generically and let F be the span of the three core-row vectors. If $\dim F \leq 1$, then $V = 0$, and

$$D - 2CT = W^2 - \sum_i a_i^2 - 2TV \geq 0.$$

If $\dim F \geq 2$, choose a generic vector $e \in F$, avoiding all N exterior-left directions. Contraction by e gives a rank-two matroid on these N nonloops, with exactly W bases: a pair extends e exactly when at least one core row extends it. Hence $W \geq N - 1$. If $W \leq 2$, necessarily $N = 3, T = 1, W = 2$. The sole exterior-left pair not extending e spans F , which is two-dimensional. Exactly one exterior-left vector is outside F , so $V = 1$. Consequently

$$D - \frac{3}{2}CT = 7 - A_0/2 - \sum_i a_i^2 \geq 7 - 1 - 4 > 0.$$

Together with (155), this proves $c_1^2 \geq 3c_0c_2$.

28.3.5 All three core vertices and a lower degree exterior vertex

Relabel its neighborhood as $\{A_1, A_2\}$. Let U count exterior-left pairs completed by a virtual row with neighborhood $N(A_1) \cup N(A_2)$, let P count those completed by A_3 , and let W count those completed by at least one of these two rows. Put

$$O = U + P - W, \quad u = U - O, \quad p = P - O.$$

Let V count exterior-left singletons completed by those two rows. With T as before,

$$C = c_0 = 2T + U + 2P + V = 2T + 3W - 2u - p + V.$$

The remaining right vertices have two parallel classes: neighborhoods contained in $\{A_1, A_2\}$, of cost $T + P$, and neighborhood $\{A_3\}$, of cost $2T + U$. All other nonempty neighborhoods have cost $q = 3T + W$. Apply (155) with $\delta_1 = 2T + u$, $\delta_2 = T + p$. Expansion gives

$$D = 2CT + (2UP - O^2) - 2TV.$$

The previously proved sharp rank-three two-row inequality is

$$4UP - 2O^2 \geq 3TV;$$

apply it to the virtual union row and A_3 [12]. [*Merge note:* Source 74's inequality (9) (Theorem 9.2).] This inequality allows arbitrary positive exterior-left activities. It gives

$$D \geq 2CT - TV/2 \geq \frac{3}{2}CT,$$

because $C \geq V$. Again $c_1^2 \geq 3c_0c_2$ follows.

If the forced exterior vertex has degree one, remove it and its forced left neighbor. Two unit core-left rows remain. With the same two-row counts T, U, P, V, O , put $W = U + P - O$, $u = U - O$, and $p = P - O$. This time

$$C = T + U + P + V, \quad q = 2T + W, \quad \delta_1 = T + u, \quad \delta_2 = T + p.$$

The exact axis costs are $T + P$ and $T + U$, and the generic cost is q . The same expansion again gives

$$D = 2CT + (2UP - O^2) - 2TV \geq \frac{3}{2}CT.$$

Thus the constant three holds in this case as well.

Completion of Theorem 28.4. A size-six matching uses all of $A \cup B$ and no core edge. Four feasible forced right vertices contain one, two or three exterior-right vertices. Three are Hall-tight with $h = 3$, giving constant three in (153); two are handled by (154); one has degree one, two or three and is covered by the corresponding argument above. These cases exhaust the claim. $\square$

Verification. Run `python3 checks/run_all.py`. [Merge note: Shipped as `code/63-nested-exterior-run_all.py`; `data/exact_certificate.json` above is `data/63-nested-exterior-exact_certificate.json`.] The package uses only Python’s standard library and contains the complete nested certificate, independent endpoint checks, and an exact verification of Proposition 28.3, and weighted checks of (154). The proof of Theorem 28.4 is analytic; the finite experiments are not its justification. It contains no numerical searches or internal review reports. These are computer-assisted mathematical proofs, not formal theorem-prover verification or claims of exhaustive priority.

A Provenance of this report

A.1 Sources

All sources arrived in ProveIt’s incoming directory in commit 1512ef835 (batch 72, 79 archives) and were placed in commit 8bb543f0f. Each is dated 1 October 2026. Their delivered PDFs, manuscripts, READMEs, checksum ledgers and the identical helper `build_local.sh` are not shipped; their code, data, audit and provenance files are, under the prefixes listed (README of this report). “Pin” is a ProveIt commit that the manuscript itself cites; sources without a pin cite their siblings by archive name only. At this stage sixteen sources are printed.

No.	Archive and title	Pin	Labels	Printed as
74	ProveIt_Weighted_Rank_Five_and_Two_Shore_Covers.zip: <i>Weighted Rank Five and Covers with a Two Vertex Shore</i> (16 pages, A4)	1b3960d8a, a866ff9a2 (archives cited)	mrn:w:rf:	Sections 2–13 (base)
72	strictness_equality_package.zip: <i>Strictness and equality for weighted matching support polynomials</i> (8 pages, Letter; “AI-assisted mathematical research”)	none	mrn:w:se:	Section 14
76	ProveIt_Weighted_Rank_Four_Stability.zip: <i>Stability of Two by Two Covers and Weighted Rank Four</i> (8 pages)	1b3960d8a, a866ff9a2	mrn:w:sb:	Section 15
75	ProveIt_Sharp_Full_Hall_Classification.zip: <i>A Sharp Weighted Classification of Full Hall Bottleneck Graphs</i> (9 pages)	4a3b6991d	mrn:w:fh:	Section 16
71	ProveIt_Unit_Weight_Rank_Normalization_Counterexample.zip: <i>Unit Weight Failure of Matching Rank Normalization</i> (5 pages)	a866ff9a2	mrn:u:uw:	Section 17
73	ProveIt_One_Shore_Counterexamples.zip: <i>One Shore Activities Can Destroy Matching Rank Ultra Log Concavity</i> (6 pages, with pendant_appendix.tex)	4a3b6991d	mrn:u:os:	Section 18

No.	Archive and title	Pin	Labels	Printed as
55	ProveIt_Sharp_Rank_38_Hall_E ndpoint.zip: <i>A Sharp Rank 38 Boundary for the Last Hall Inequality</i> (6 pages)	none	mrn:u:ep:	Section 19
47	ProveIt_Sharp_Rank_38_Full_H all_ULC.zip: <i>The Sharp Rank 38 ULC Boundary for Full Hall Graphs</i> (6 pages)	none	mrn:u:ag:	Section 20
51	ProveIt_Last_Hall_Gap_Separa tion.zip: <i>The Last Hall Gap Does Not Control Ultra Log Concavity</i> (2 pages)	none	mrn:u:sp:	Section 21
60	ProveIt_Rank_Two_Matrix_Bloc ks_and_Conditioning.zip: <i>Rank Two Matrix Blocks and Exterior Conditioning</i> (7 pages)	none	mrn:s:rb:	Section 22
10	ProveIt_Three_Vertex_Margina ls_and_Star_Cores.zip: <i>Lorentzian Marginals on Three Vertices</i> (5 pages)	none	mrn:s:tv:	Section 23
23	ProveIt_Rooted_Hall_Sectors_ and_Cubic_Obstruction.zip: <i>Rooted Lorentzian Hall Sectors and an Unrooted Cubic Obstruction</i> (5 pages)	none	mrn:s:rs:	Section 24
13	ProveIt_Rank_Six_Finite_Last _Gap.zip: <i>The Last Newton Inequality at Rank Six</i> (7 pages)	none	mrn:s:lg:	Section 25
05	ProveIt_Rank_Six_Second_Newt on_Inequality.zip: <i>The Second Newton Inequality at Rank Six</i> (7 pages)	none	mrn:s:sn:	Section 26
69	ProveIt_Rank_Six_Common_Ext erior_Theorem.zip: <i>Rank Six With Common Exterior Neighborhoods</i> (5 pages)	none	mrn:s:ce:	Section 27
63	ProveIt_Rank_Six_Nested_and_ Conditional_Results.zip: <i>Nested Rank Six Families and Conditional Barriers</i> (7 pages, with <code>conditioning.tex</code>)	none	mrn:s:ne:	Section 28

The full commit hashes are 1b3960d8acdabd983147b7c609bd9dfb2d67a0a0 (the batch-70 archive of the manuscript now printed as Part XII of *preorder-root-polytopes*), a866ff9a2 (the batch-68 archive of the Hall-bottleneck manuscript, now Part XI) and 4a3b6991d0cc12cc09b884e22728e2128a54c6e3 (the commit that wrote Part XI). Source 72 cites source 74 by title, and its shipped 72-strictness-SOURCES.md records a hash of source 74's delivered PDF; that file is kept as delivered.

A.2 What each source contributed

- **74** (base): the $2 + q$ transversal lift (Theorem 2.1), weighted normalization through matching number five and the least weighted failure rank six (Corollary 2.2), the cover obstruction (Corollary 5.1), the forced-sector and scaling refinements, the negative family at every rank ≥ 6 , the Ehrhart corollary, the generic-line matroid model, the sharp all-core constants $2q/(q - 1)$ and three, the star-core transfer proofs, a second route to the first-gap equality case, and quantitative strictness thresholds.
- **72**: equality in any one gap only for balanced star forests, and strict ULC_r otherwise, under the cover hypothesis of source 74 (Theorem 14.1); a weighted Chan–Pak contraction criterion (Theorem 14.3); a second route to the first-gap equality case.
- **76**: real stability of the basis polynomial, hence real-rootedness of p_G , for every graph with a vertex cover of two vertices on each shore (Theorem 15.1); a second route to weighted rank four; the sharp stability criterion for complete $2 \times s$ mixed cores (Theorem 15.5) and its 17-vertex boundary example.
- **75**: the full Hall classification (Theorem 16.1), with a second route to its positive half, strictness, real-rootedness for a singleton core, explicit asymmetric failures, and a strictly ultra-log-concave rank-five example with nonreal zeros.
- **71**: a connected all-unit graph with 6916 vertices and matching number 3450 that is not ULC_{3450} (Theorem 17.1); a deficiency-one example and infinitely many such; failure of the two Davis–Kohl h^* -polynomials.
- **73**: one-shore counterexamples of matching number 50 (125 vertices) and 38; the endpoint quadratic and a construction criterion; the weighted pendant-arm identity (which contains source 71’s) and a rank-310 transfer example.
- **55**: for one common right-core activity on full Hall graphs, the least matching number with a failing last gap is exactly 38, with the two shapes and minimal populations (864 vertices).
- **47**: in the same family no intermediate gap fails through matching number 38, so 38 is the least failure rank of full ULC.
- **51**: a rank-49 full Hall graph whose last gap holds for every activity while the penultimate gap fails.
- **60**: Lorentzian lift for matrices with a rank-two exterior column block; exact Hall-slack conditioning (slack zero, one, two); forcing $a - 2$ exterior-right vertices of a genuine $a + q$ cover leaves a ULC_{q+2} factor; eventual strictness with an explicit threshold.
- **10**: the degree-three left marginal of any graph with three left vertices is Lorentzian (a 5339-monomial determinant certificate); star cores on the three-vertex shore give ULC_{q+3} for all activities on both shores, hence all five rank-six inequalities for both star orientations.
- **23**: rooted sector marginals are Lorentzian; the 25-vertex graph whose all- B conditional cubic fails both unshifted Newton inequalities although the graph is strictly ULC_6 for every common core activity.
- **13**: the last rank-six inequality for every one-shore graph and every finite activity (858 weighted squares, ten profiles).

- **05**: the second rank-six inequality likewise, through a Lorentzian cubic truncation (a certificate of 50,037,694 coefficients); the quintic route is refuted.
- **69**: ULC_6 for every 3×3 core with complete exterior neighbourhoods (8,964 certificate vectors); a private-leaf reduction; a barrier coefficient $56 - 4n$.
- **63**: ULC_6 for the complete core with nested exterior neighbourhoods; a 20-vertex graph whose full right marginal is not Lorentzian; forcing four right vertices gives $c_1^2 \geq 3c_0c_2$.

A.3 Where the merge chose

- *Placement and base.* The batch's manuscripts continue Parts XI–XIII of *preorder-root-polytopes* and answer several of its research questions, but they form a self-contained programme of 26 archives; they were placed as a new report in **log-concavity-and-unimodality**, with source 74 as the base because it proves the most general theorem and the others cite it. *preorder-root-polytopes* is cited, not reprinted, and receives dated pointers to this report.
- *Source 74's layout.* Its eight section files were inlined into **article.tex** and removed; its appendices are ordinary sections of Part I. Its two short lemmas that restate *preorder-root-polytopes* (Lemmas 3.1 and 3.2) keep their statements and receive pointer proofs. Table 3 was added from its recorded verifier output.
- *Re-proofs of preorder-root-polytopes.* Sources 74 and 72 re-prove Part XI's first-gap inequality and its equality case by the Motzkin–Straus route: both proofs are printed, marked as second routes. Source 74's rank-six witness is Part XI's $H_{3,33}$ and its $(4, 4)$ shape Part XI's $H_{4,11}$: printed with notes. The leaf compression of sources 72, 74 and 76 is Part XIII's; source 76's inversion identity, gluing lemma, expected-determinant lemma and $K_{2,k}$ stability are Part VI's: proofs of those are pointers.
- *Same theorem in the batch.* Source 76's weighted rank four and the positive half of source 75's classification are cases of source 74's theorem; both are printed in full as second routes, because their proofs (real stability; a comparison polynomial) differ from source 74's and give more (real-rootedness, strictness).
- *Stale sentences.* Statements of sources 75 and 76 that the first weighted failure rank is five or six, or that rank five is open, are kept with notes: source 74 settles them.
- *Part II.* Source 71's pendant lemma is printed once, in source 73's weighted form (source 71's statement is kept with a pointer proof). Source 47's first-gap proof is a pointer to Part XI of *preorder-root-polytopes* (its strictness step is kept); its recall of source 55's endpoint theorem is kept with pointers to source 55, which is printed before it. Source 73's rank-310 example is kept as a remark; it is one-shore, not unit, weighted. Statements of sources 55, 47 and 51 that call rank-six one-shore ULC open or that leave earlier gaps open are kept with notes.
- *Part III.* Source 60's restatement of Part XII's rank-three matrix is a pointer. The 25-vertex graph is printed once, in source 23; source 05's description of it is kept with a note. Source 13's incidence lemma, which is also source 21's (Part IV) and a case of source 08s's, is printed with source 13's proof, where it is first used. The first-gap arguments of sources 69 and 63 are pointers to *preorder-root-polytopes*'s Theorem 180.2. Source 05's direct proof of its rank-two lemma is kept as a second route of source 60's theorem. Sentences of sources 10, 13, 23, 60, 63 and 69 that call three middle gaps, or the whole rank-six one-shore problem, open are kept with notes giving the status above.

References

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- [8] *A Reflexive Root-Polytope Model for Preorder h -Polynomials*, ProveIt research report, SetTheory/Cardinals/docs/reports/enumerative-combinatorics/preorder-root-polytopes (“preorder-root-polytopes” in this report), in thirteen Parts. Theorem numbers cited here are those of its `article.pdf` at the placement commit `8bb543f0f`. [Merge note: Source 76 cites its Part VI (independent-column expectations, complete bipartite stability and block locality), inspected 1 October 2026.]
- [9] *Hall Bottlenecks and the Limits of Rank Normalization*, research manuscript prepared for Vladimir Reshetnikov with ChatGPT, 30 September 2026; now Part XI of *preorder-root-polytopes*. Cited as the incoming archive `ProveIt_Hall_Bottlenecks.zip` at commit `a866ff9a2` by sources 74 (“The rank-six witness is restated and verified here”) and 76, and as Part XI of *preorder-root-polytopes* pinned at commit `4a3b6991d0cc12cc09b884e22728e2128a54c6e3` by source 75. Source 71 cites the archive at `a866ff9a2`, its section “Further research questions”, question “Unit weights”, and notes that it “gives two-shore-weighted counterexamples and explicitly leaves the unit-weight question open”; source 73 cites Part XI at `4a3b6991d`.
- [10] *Rank-Three Ultra-Log-Concavity and Exact Kernels for Matching Supports*, research manuscript prepared for Vladimir Reshetnikov with ChatGPT, 30 September 2026; now Part XII of *preorder-root-polytopes*. Cited by sources 74 and 76 as the incoming archive `ProveIt_Matching_Rank_Research.zip` at commit `1b3960d8acdabd983147b7c609bd9dfb2d67a0a0`; source 76 refers to its question “Finish rank four” (Research question 93 of *preorder-root-polytopes*).
- [11] *Leaf Compression and Weighted Rank Three for Matching Support Polynomials*, research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026, delivered as `ProveIt_Weighted_Rank_Three_Leaf_Compression.zip` in batch 71 and again (byte-identical) as archive 78 of batch 72; now Part XIII of *preorder-root-polytopes*. Source 74: “The compression proof is repeated in” Lemma 3.2 (here a pointer to Part XIII).
- [12] Batch-72 source 74, *Weighted Rank Five and Covers with a Two Vertex Shore*, research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026; printed here as Sections 2–13. Its Theorem 1.1 is Theorem 2.1 here, and its Sections 2 and 3 are Sections 3 and 4. Source 72 cites it as an “AI-assisted ProveIt research report” containing “the rank- $(q+2)$ transversal lift

and weighted rank-five theorem; the lift’s exact interface is recalled in” Section 14.3. Later sources cite it as *Weighted Rank Five and Two Shore Covers*.

- [13] Batch-72 source 75, *A Sharp Weighted Classification of Full Hall Bottleneck Graphs*, printed here as Section 16. Source 74 cites it as *Sharp Full Hall Classification*, a companion report: “Its general negative-shape construction is reproduced in” Section 6.
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- [17] Batch-72 source 55, *A Sharp Rank 38 Boundary for the Last Hall Inequality*, printed here as Section 19.
- [18] Batch-72 source 51, *The Last Hall Gap Does Not Control Ultra Log Concavity*, printed here as Section 21. Source 47 cites it under the wrong title *The Last Hall Gap Does Not Control All Earlier Gaps*.
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- [20] Batch-72 source 10, *Lorentzian Marginals on Three Vertices*, printed here as Section 23. Source 05 embeds an exact copy of its archive as a prerequisite.
- [21] Batch-72 source 60, *Rank Two Matrix Blocks and Exterior Conditioning*, printed here as Section 22. Source 23: “Exact weighted Hall-slack compression and the rank-two-block marginal theorem”.
- [22] Batch-72 source 21, *Rank Six Conditioning for Every Core*, printed in Part IV. Source 13 recalls its incidence lemma with proof.
- [23] Batch-72 source 69, *Rank Six With Common Exterior Neighborhoods*, printed here as Section 27. Source 63 lists it in its bibliography as a companion report without citing it in the text; its “earlier common-exterior result” is this source.